Estimating Tax Elasticities along the Income Distribution

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Preliminary and incomplete. The latest version is available here.

Abstract

The elasticity of earnings with respect to taxation governs the trade-off between redistribution and efficiency, but existing estimates are local to an income threshold (bunching) or to the bracket a reform affects (reform designs). This paper proposes a strategy to estimate the elasticity along the income distribution. It builds on the observation that, under a tax system that taxes spousal income jointly and gives tax advantages for children, individuals with the same earnings are differentially likely to see a tax change when the composition of the tax unit changes (a child leaves, a couple marries). The strategy formalises such comparisons and isolates the variation that stems from partner income. In a panel built from the universe of French tax returns, it reveals that the earnings elasticity declines from 0.45 in the bottom decile to 0.22 in the middle deciles and below 0.1 in the top decile. The profile is shifted up for women and at older ages, and the lowest earners' participation response comes from those with higher-earning partners. Accounting for this heterogeneity flattens optimal progressivity at the lower end of the distribution and alters the predicted effects of reforms towards individualisation.

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1 Introduction

How much revenue a tax raises and how much it harms economic activity depend on how elastic (responsive) earnings are with respect to taxation. This elasticity may be heterogeneous, but existing estimates are local. Bunching designs use a threshold of the tax schedule, and tax-reform designs study the bracket a reform affects, e.g., top incomes for a cut to top rates. An estimate at one threshold or for one bracket cannot recover how the elasticity varies along the income distribution, and because it is unclear how it varies (Gruber & Saez, 2002; Saez et al., 2012), it is often assumed not to (Diamond & Saez, 2011; Saez, 2001). Yet the predicted effects of reforms, the calibration of macroeconomic models and the assessment of the trade-off between redistribution and efficiency rest on this assumption and are biased if it is wrong. With a declining profile of elasticities, the standard model of optimal taxation (Saez, 2001) recommends slower withdrawal of transfers at the bottom and steeper progressivity further up, with a rising profile, the opposite.

This paper proposes a strategy to estimate the tax elasticity of earnings along the income distribution. The main identification challenge is simultaneity: tax rates are a function of income, and the effort to earn income a function of how much it is taxed, implying that observed changes in tax rates are endogenous and need instrumentation. The instrument proposed in this paper builds on the observation that, under a tax system that taxes spousal income jointly and gives tax advantages for children,[1] otherwise similar individuals are differentially likely to see their marginal tax rate change at "events" that change the composition of the tax unit. Consider two women with the same earnings but different partner incomes. When a child reaches the age at which it can no longer be counted in the family tax unit (the event), the woman with the higher-earning partner may be pushed into the bracket above while the other remains in hers. The strategy formalises such comparisons, building on the simulated-IV strategy that Gruber and Saez (2002) developed for reforms. Among individuals with similar own income, we simulate from pre-event incomes who would cross a threshold and who would not, and the simulated change in the marginal tax rate is the instrument for the realised one. Whereas a reform supplies one change in tax rates to instrument with, the events supply many, at every level of income. Each tax change a child's exit can induce, at

[1] Under joint taxation such as in France, the couple's income is divided by a quotient that rises with the number of children (2 for a married couple, 2.5 with one child, 3 with two) before the tax schedule applies.

each level of own income, is analysed on its own,[2] and the earnings growth comparisons, one per threshold and income level, give the elasticity along the income distribution.

We apply the strategy to the universe of French tax returns, reshaped, with open-source code, from a panel of tax units into an individual-level panel that follows each person across tax units. With around fifty million individuals a year and 21 years (2003 to 2023), it can be split particularly finely by own and partner income and by the other dimensions of heterogeneity we study.

We find the earnings elasticity[3] to decline with income, most steeply at the bottom. It falls from 0.45 (standard error 0.11) in the bottom decile of the income distribution to 0.23 (0.03) in the middle deciles and to 0.07 (0.05) in the top decile.[4] Averaged over the French taxpayer distribution, the earnings elasticity is 0.23 (standard error 0.03) and the participation elasticity 0.25 (0.02). The decline holds for both genders, at a higher level for women: their earnings elasticity falls from 0.45 in the bottom decile to 0.14 in the top decile, men's from 0.29 in the deciles below the median to 0.07 in the top decile, and the averages over French taxpayers are 0.29 and 0.17.[5] Partner income shows which households drive the large elasticities at low incomes. Among the lowest-earning women, the participation response comes from those with higher-earning partners, with elasticities of 0.69 and 0.35 in the middle and top partner-income terciles, against 0.02 in the bottom one (standard errors 0.08 to 0.11).

The heterogeneity we document enters both normative models of how taxes should be set and positive predictions of what reforms do. The normative models set the marginal tax rate at each income by trading off redistribution against efficiency (Mirrlees, 1971). With the

[2]Analysing each on its own also avoids the negative weights that pooling changes of different strength produces when responses differ across individuals (Graber et al., 2026).

[3]The elasticity of own earnings (work income) with respect to the household net-of-tax rate, one minus the marginal tax rate the household faces at the individual's income.

[4]Deciles of the 2023 work-income distribution of all earners in France, men and women together (Appendix F.15). The estimates are for bins of own income €10,000 wide, in 2015 prices, with men's and women's estimates pooled (Figure 6). The lowest bin, below €10,000, holds the bottom 18 percent of earners. The two bins around the median, from €10,000 to €30,000, have estimates of 0.23 (0.05) and 0.22 (0.04). The bins from €40,000 to €70,000 cover the top decile up to the 98th percentile, with estimates of 0.04 (0.06), 0.06 (0.09) and 0.18, the last one imprecise (standard error 0.13).

[5]Standard errors: women 0.11 and 0.06, men 0.07 and 0.06, averages 0.05 and 0.03. Deciles of each gender's own 2023 work-income distribution (Appendix F.15); a decile's value averages the €10,000 bins it spans, weighted by their shares of the decile, with standard errors that treat the bins as independent. The top decile is covered up to the 98th percentile for women and the 97th for men, the profiles' cap of €70,000; the women's bin above €60,000 rests on a single cell and is left out. Below €10,000, only one of the men's comparisons meets the precision screens, so the men's lower range runs from the 16th to the 50th percentile. The averages are those of Table 4, not averages of the decile values.

estimated profile of elasticities (estimated on parents whose child leaves the tax unit) in place of a single one, the optimal rates flatten at the bottom of the income distribution, falling by about 14 percentage points on average below €10,000 (Brewer et al., 2010; Saez, 2001). With our profiles over own and partner income in place of one elasticity per spouse, the evaluation of reforms towards more individual taxation (Bierbrauer et al., 2024) raises the assessed welfare gains and relocates them, concentrating them where both partners earn little and lowering them over much of the remaining distribution.

The assumptions are relatively weak. The strategy imposes no structure and does not require the event to be exogenous. Its main requirement is that, without the schedule change, earnings growth would not vary with partner income (or vary in a constant way) among individuals with similar own income. Since the simulation uses pre-event incomes, the assumption fails if individuals bunch in anticipation, adjusting their earnings to thresholds that are not yet effective. In addition to (i) density tests around the thresholds, it can be inspected through (ii) earnings growth of the two groups before the event, which should not differ, and (iii) earnings growth in the region of partner income where the event changes no rate, which should not vary with partner income. All three lend themselves to transparent, graphical inspection.

The variation has two advantages over that of a tax reform. Under a reform the same key assumption must hold between affected and unaffected groups that differ in income, and two biases then threaten it: (i) unusually high incomes tend to fall back towards the average (mean reversion), and (ii) growth rates differ along the distribution as inequality changes. Controlling for income is the remedy a large literature has refined in different ways (Blundell et al., 1998; Burns & Ziliak, 2017; Graber et al., 2026; Gruber & Saez, 2002; Jakobsen & Søgaard, 2022; H. J. Kleven & Schultz, 2014; Kumar & Liang, 2020; Weber, 2014),[6] but it is not straightforward given that the variation itself –which taxpayers a reform affects– is a function of income and risks being absorbed. By contrast, our variation is *within-income*, in the sense that exposure to the tax change depends on partner income. Own income can then be controlled for without absorbing the identifying variation, and both biases are alleviated. The second advantage is that the tax changes come from *shifts* of the thresholds of an otherwise

[6]Gruber and Saez (2002) and H. J. Kleven and Schultz (2014) use richer income controls, Weber (2014) longer lags of the income that defines exposure, and Jakobsen and Søgaard (2022) and Graber et al. (2026) differencing against placebo periods with graphical validation; grouping estimators use group-level rather than individual variation (Blundell et al., 1998; Burns & Ziliak, 2017), and Kumar and Liang (2020) allow responses to vary across individuals.

unreformed schedule. The individual's tax rate is then more likely to move in the direction the event predicts (monotonicity) than under a reform,[7] and the rate change comes without the other provisions of a reform, which may confound the estimate (Kopczuk, 2005; Kumar & Liang, 2020; Saez et al., 2012).[8]

Our estimates describe parents at the time their first-born leaves the tax unit, but the strategy can be used to study other "events" and heterogeneity within that group. Parents are 51 to 53 on average when their first-born leaves, and the estimates partly reflect the higher responsiveness of people approaching retirement. Marriage, which we use as a second design (Appendix G), and the birth of a child are such events. This paper focuses on the child's exit because it generally does not coincide with the child moving out, which makes the assumptions easier to believe, and because parents' ages at the event vary widely (85 percent of the women and 82 percent of the men are 45 to 59), which allows (i) age fixed effects in every estimation and (ii) sample splits by age as a further heterogeneity analysis. The split shows that elasticities are smaller below the minimum retirement age.[9]

To the literature on heterogeneity in tax responsiveness by income, the paper contributes a profile of tax elasticities along the income distribution, estimated within a single setting, design and income measure. Figure 1 collects the quasi-experimental estimates of earnings responses published in top field or general-interest journals (Bastani & Selin, 2014; Blomquist & Selin, 2010; Chetty et al., 2011; Chetty et al., 2013; Gelber, 2014; He et al., 2021; Jakobsen & Søgaard, 2022; H. Kleven et al., 2025; H. J. Kleven & Schultz, 2014; H. J. Kleven & Waseem, 2013; Lehmann et al., 2013; Miao et al., 2025; Saez, 2010; Sigaard, 2023). The estimates vary widely (Neisser, 2021); each is either local to a narrow part of the income distribution or one average over a wide range, and most concern the top three deciles. Because the studies differ

[7] Monotonicity requires that no individual's realised marginal rate moves against the change the schedule predicts. Under a reform that raises the rates of one income group, for example, an individual who stops working in response ends up at a zero rate where a rise was predicted. At an event that changes the composition of the tax unit, a violation needs a much larger fall in income than under a reform. This is because such events move only the bracket thresholds, and because partner income limits how far the household's rate can fall when own income falls (Appendix H gives the formal argument and a simulation).

[8] A reform is salient, and responses to salient and to unnoticed tax changes differ (Saez et al., 2012). It rarely moves one rate alone and usually changes the tax base with it (Kopczuk, 2005). It also moves rates elsewhere in the schedule, over which individuals optimise too, and an estimate based on the rate at their own income omits them (Kumar & Liang, 2020).

[9] Restricting the sample to parents below 58 in the year before the event lowers women's elasticities but keeps the shape of their profile, and cuts men's earnings response from 0.17 to 0.06 (own incomes from €10,000 up).

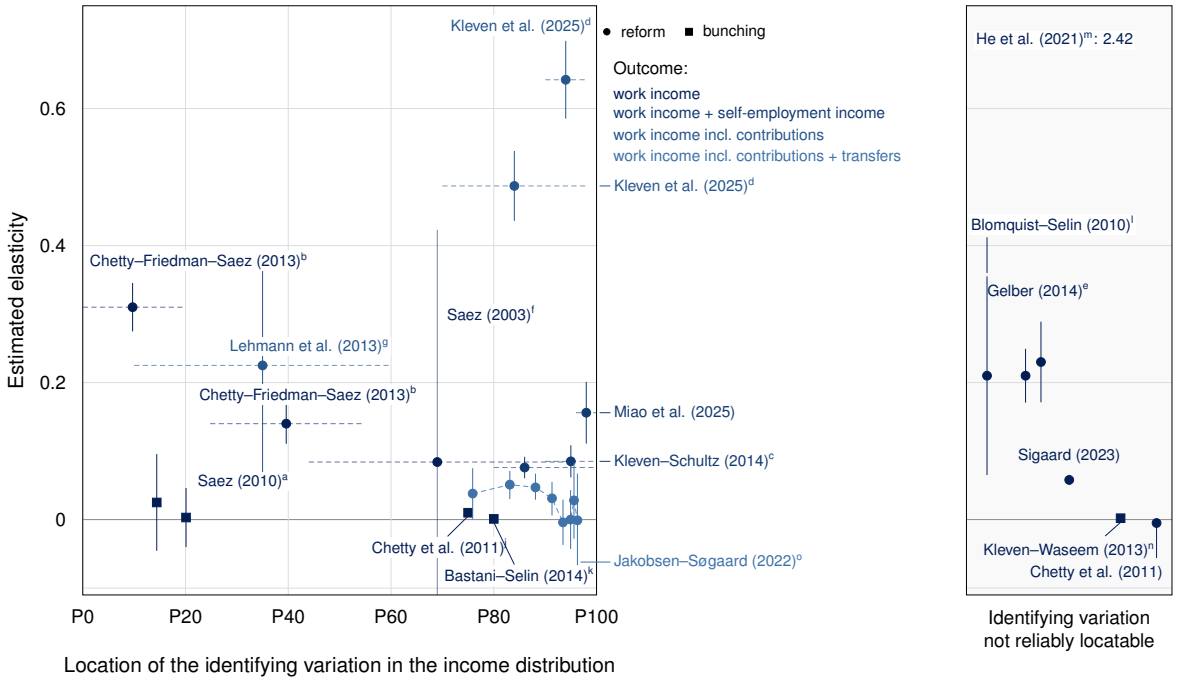


Figure 1: Published estimates of work-income responses

Notes: The figure shows quasi-experimental estimates of work-income tax elasticities from studies published in top field or general-interest journals. Colours denote the outcome (in-figure key); circles denote reform designs and squares bunching designs. Work income incl. contributions is measured gross of employer contributions (payroll contributions in Lehmann et al., employer-administered pension contributions in the Danish studies). Point markers sit at kinks or local-regression centres; horizontal dashed segments give the income range from which the identifying variation comes. Estimates without a locatable range appear in the right panel, where horizontal position is uninformative. Vertical lines are 95% confidence intervals where an elasticity standard error is reported. Percentile ranks refer to each study's ranking universe. Papers estimating taxable-income elasticities for populations broader than wage earners[h] or Frisch elasticities[i] are not included. Exact populations, horizons, estimands, treated ranges and mapping grades are reported in Appendix Table I.1, whose notes also explain the letters at the markers.

in design and income measure, it is difficult to establish how much of the variation reflects heterogeneity in responsiveness.

To the literature on the taxation of couples, the paper contributes evidence along partner income, the second dimension of our variation. Normative work gives opposing prescriptions on optimal rates depending on the assumed difference between primary- and secondary-earner elasticities (Boskin & Sheshinski, 1983; Golosov & Krasikov, 2025).[10] Positive work on switches between joint and individual taxation, or on comparable policy variation, finds that joint taxation lowers the employment and earnings of secondary earners but not of primary

[10]With a common elasticity, optimal marginal rates on secondary earners are on average *higher* than on primary earners (Golosov & Krasikov, 2025) while the inverse-elasticity rule (rates inversely proportional to elasticities) prescribes the opposite (Boskin & Sheshinski, 1983) (with the secondary earner assumed more responsive).

earners (who are typically men) (Herold et al., 2025; Herold & Wallossek, 2024; Isaac, 2023; Kalíšková, 2014; LaLumia, 2008; Selin, 2014) – consistent with the heterogeneity we map by own and partner income.

The paper also contributes evidence on heterogeneity by income measure, gender and time horizon. Broad earnings elasticities, which add self-employment income to work income, are higher at every level of income.[11] Women respond more than men in our estimates, but by less than in the literature (Blomquist & Selin, 2010; Eissa, 1996; Eissa & Liebman, 1996)[12] (attributable to the life-cycle stage we cover). Recent work argues that the time horizon is central to the gap between micro and macro estimates (H. Kleven et al., 2025). We provide estimates by year since the event, up to four years.

The rest of the paper is structured as follows. Section 2 describes the French tax system and the construction of the individual-level panel. Section 3 develops the empirical framework and Section 4 presents the results. Section 5 examines how accounting for heterogeneity affects optimal tax schedules and the evaluation of reforms towards more individual taxation. Section 6 concludes.

2 Background

2.1 Institutional setting

The variation explored in this paper stems from two features of the French tax system: (i) that the marginal tax schedule has a step-wise shape and (ii) that the thresholds of those steps depend on the composition of the tax unit.

What makes the French tax system one of "family taxation" is that the tax burden (equation (1)) is a function of the *family quotient*, which increases with the number of family members: the quotient is 1 for singles, 2 for childless married (or civil-union) couples, 2.5 with

[11] Our variation does not identify responses of taxable or capital income, the object of a large literature (e.g., Feldstein, 1995; Gruber & Saez, 2002; Neisser, 2021; Saez et al., 2012).

[12] The literature finds elasticities for married women and single mothers several times those of men.

one dependent child and 3 with two; each further child raises it, and the tax advantage a child confers is capped.[13,14]

The tax liability is defined (i) under individual filing and (ii) under joint filing as

$$(1) \qquad \text{(i)}\ \ T_t\big(z_{it}\big), \qquad\qquad \text{(ii)}\ \ T_t^H(z_{it}+z_{pt};\, Q_{it}) \;=\; Q_{it}\, T_t\!\left(\frac{z_{it}+z_{pt}}{Q_{it}}\right),$$

where T_t denotes the year-t French income tax formula for a single individual, z_{it} and z_{pt} are own and partner taxable income, and Q_{it} is the family quotient.

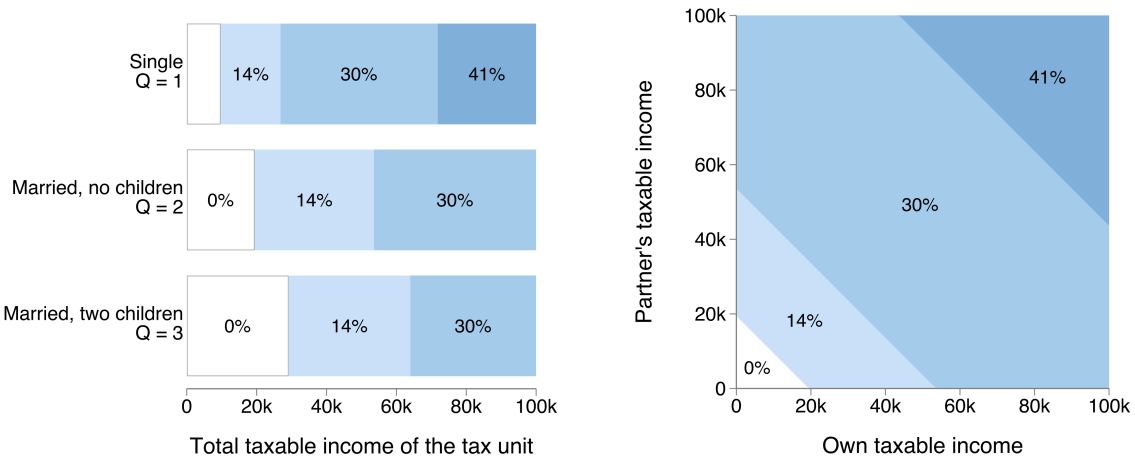
Figure 2 shows the 2015 schedule. Panel (a) gives the marginal tax rate by household composition. The brackets of the single schedule scale with the quotient. A childless couple reaches each rate at twice the income of a single, and the child-related shifts are truncated by the cap. Panel (b) shows the implication for a married couple. Because only the sum of the two incomes enters the tax formula for a given quotient, the marginal rate is constant along diagonals. At given own income, it rises with partner income. The resulting tax liabilities are shown in Appendix A.

The effective schedule differs from the statutory one through the *décote*, a tax reduction for households with a small liability, which is phased out as the liability rises. Over the phase-out range the effective marginal rate exceeds the statutory rate (in 2015, 24.5 instead of 14 percent). Above this range, the statutory rate applies. At the child's fiscal exit, the instrument and the realised rates of Section 3 are computed on the effective schedule, including the *décote* and the cap on the child advantage.

Events that change the composition of the tax unit shift the tax function that applies to an individual's income. These shifts are the variation we study. The fiscal exit of a dependent child lowers the family quotient and leaves the tax base unchanged. By merging two tax units into one, marriage pools the spouses' incomes and doubles the quotient. Table 1 summarises what each event changes.

[13] The cap on the child advantage limits the benefit from an additional child to a fixed amount.

[14] Other factors also influence the family quotient. For example, it increases for single parents and parents of disabled children. We do not consider these cases in our analysis.



(a) By household composition

(b) Married couple, by own and partner income

Figure 2: Marginal tax rates under family taxation

Notes: Statutory schedule of 2015, before the *décote*, with the cap on child-related advantages applied; Q is the family quotient. Panel (a): by household composition and total taxable income of the tax unit. Panel (b): married couple without children (two fiscal parts), by own and partner taxable income; the 14, 30 and 41 percent brackets begin at total taxable income of €19,400, €53,582 and €143,652, twice the single thresholds. The corresponding tax liabilities are shown in Appendix Figure A.1. Source: Own calculations based on OpenFisca France.

Table 1: Events and the tax liability function

Event	Tax base	Family quotient Q
Child's fiscal exit	unchanged: the couple's pooled income	$2.5 \to 2$ with one dependent child; $3 \to 2.5$ with two
Marriage	own income $\to$ the spouses' pooled income	1 for each spouse $\to$ 2 for the joint unit

Notes: What each event changes in the tax liability function of equation (1). Households with more dependent children see the analogous quotient change, subject to the cap on the child advantage; the marriage row refers to two previously single, childless individuals.

2.2 Data

We use the French administrative tax panel known as POTE.[15] The dataset covers all income tax returns filed in France between 2003 and 2023. POTE includes all variables submitted in the tax return (Form No. 2042), intermediate calculations for determining the final tax liability, and all information shown to the household on its official tax notice. It also includes detailed tax-relevant characteristics, such as the number of dependent children, marital status and filing status, gender, and the municipality.

[15]POTE—Panel On The Taxation of Households, INSEE–DGFiP. Data accessed via CASD (https://www.casd.eu/).

Tax returns are filed at the tax-unit level (the *foyer fiscal*), but the POTE database includes individual-level identifiers. It permits the (to our knowledge) first construction of a longitudinal individual-level panel, essential to follow incomes before and after changes to the composition of the tax unit (such as marriage).

The result is a panel of approximately 50 million individuals per year. In 2023, 62 percent of men and 57 percent of women declare positive work income, and 50 and 44 percent, respectively, are married or in a civil union. From this panel we draw a sample of individuals experiencing the fiscal exit of their first child and a sample of individuals marrying. Appendix F.1 describes the construction and characteristics of the two event samples.

3 Empirical framework

This section sets out the model and the identification problem (Section 3.1) and introduces the reform-based and event-based designs (Section 3.2). Section 3.3 defines the cells within which comparisons are made and states the identifying assumptions under which the estimands have a causal interpretation. Sections 3.4 and 3.5 describe aggregation, estimation and inference.

3.1 The model and the identification problem

We model a jointly taxed couple, extending the standard one-person model (Jakobsen & Søgaard, 2022; Saez et al., 2012) to a unitary household maximising a quasi-linear and iso-elastic utility function, with the costs of earning income additively separable across spouses.

The household solves

$$\max_{z_{it},\, z_{pt}} \quad c_t \;-\; \sum_{s\in\{i,p\}} \frac{n_{st}}{1+\frac{1}{\epsilon_s}} \left(\frac{z_{st}}{n_{st}}\right)^{1+\frac{1}{\epsilon_s}} \qquad \text{s.t.} \qquad c_t \le z_{it} + z_{pt} - T_t^H(z_{it}+z_{pt}),$$

where z denotes income, with subscripts i for the individual and p for the individual's partner, c household consumption, and T_t^H the household tax function of Section 2.1.[16] The cost of earning income is governed by two individual parameters, the elasticity ϵ_i and potential income

[16]The model is written for a couple filing jointly, the tax unit of the child's fiscal exit design, where the event lowers the quotient Q_{it} of equation (1). Marriage instead merges two individually filing units into one joint unit. Before marriage, the household pays $T_t(z_{it}) + T_t(z_{pt})$, so the rate in i's first-order condition, τ_{it}^H, is the own rate $1 - T_t'(z_{it})$.

n_{it}, the income i would earn at a zero marginal tax rate (Saez et al., 2012). Any change in income that is not a response to the rate runs through n_{it}. Because quasi-linearity rules out income effects, compensated and uncompensated elasticities coincide. Appendix D adds income effects.

Because the couple is taxed jointly on the sum of spousal incomes, the marginal net-of-tax rate is common to the two spouses, $\tau_{it}^H \equiv 1 - T_t^{H\prime}(z_{it} + z_{pt}) = \tau_{pt}^H$. With separable costs, the spouses' incomes are linked only through the household rate. The household problem then separates into two individual problems, and each spouse's first-order condition yields the income-supply function $z_{it} = n_{it}\left(\tau_{it}^H\right)^{\epsilon_i}$ (Appendix B relaxes separability).[17] Taking logs and first differences gives the estimating equation

$$\Delta \ln z_{it} = \epsilon_i \, \Delta \ln \tau_{it}^H + \Delta \ln n_{it}. \tag{2}$$

The parameter of interest is ϵ_i, the earnings elasticity: the elasticity of own income with respect to the marginal net-of-tax rate the individual faces.[18] The error term, $\Delta \ln n_{it}$, varies across individuals and years—transitory shocks, life-cycle and secular growth—and, at an event that changes the composition of the tax unit, may respond to the event itself; the design does not assume it exogenous, as Section 3.3 will make clear. The objects of interest are averages of the individual elasticities ϵ_i over subpopulations defined by own and partner income (characterised in Section 3.4). The iso-elastic form makes equation (2) exact and ϵ_i a constant. For any convex cost of earning income, log-linearising the first-order condition around the pre-event optimum gives the same equation, with ϵ_i the local elasticity at the pre-event income and rate (Appendix B). Our estimates are averages of local elasticities for the rate changes the events induce. The design does not distinguish whether they differ across the income distribution because preferences differ or because a common cost function is evaluated at different incomes and rates.

[17] This is the formal counterpart of the standard practice of pairing the one-person model with the marginal rates a household tax system produces (Bick & Fuchs-Schündeln, 2018; Gruber & Saez, 2002).

[18] With separable costs, ϵ_i in equation (2) is the elasticity of own income with respect to the own rate, ϵ_{ii}. Without separability, equation (2) still holds, but ϵ_i must be read as the elasticity of own income with respect to the household rate, $\epsilon_i^H \equiv \epsilon_{ii} + \epsilon_{ip}$ (the sum of its responses to the own and to the partner's rate when the household re-optimises jointly; Appendix B). The child's exit changes the rate common to both spouses, and the design recovers ϵ_i^H. Because spouses are taxed individually before marriage, the event can change the own rate alone, and the marriage design isolates ϵ_{ii} among couples for whom it does.

A naive estimation of equation (2) would not produce causal estimates of ϵ_i. Because the tax schedule depends on income, the marginal net-of-tax rate is a deterministic function of the dependent variable, which means that any change in potential income changes realised income, and with it $T_t^{H\prime}$—the regressor of equation (2) is thus correlated with its error term by construction.

The literature handles this endogeneity in two ways. Bunching designs impose enough structure to infer the elasticity from excess mass at kinks of the schedule (H. J. Kleven, 2016; Saez, 2010). IV designs draw arguably exogenous variation from tax reforms. We build on this second approach and compare our design with it in the next subsection.

3.2 Our variation relative to reform variation

The reform-based instrument of Gruber and Saez (2002) applies the post- and pre-reform schedules to the same pre-reform income, predicting each individual's rate change from the legislated change alone,[19]

$$\Delta \ln \tau_{it}^{\text{reform}} \equiv \ln\big(1 - T_t'^{\,\text{post}}(z_{i,t-k})\big) - \ln\big(1 - T_t'^{\,\text{pre}}(z_{i,t-k})\big).$$

Validity requires

$$\operatorname{cov}\big(\Delta \ln \tau_{it}^{\text{reform}},\, \Delta \ln n_{it} \,\big|\, z_{i,t-k}\big) = 0:$$

conditional on lagged income, potential-income growth must be unrelated to the predicted rate change. The assumption is demanding because exposure to reforms is usually *between*-income. A reform typically changes rates in one bracket implying that the instrument sorts treated and untreated individuals by lagged income. Because the level of income predicts income growth for reasons unrelated to taxes, the instrument is correlated with potential-income growth. A high base-year income is in part transitory and reverts. Incomes at different points of the distribution have also grown at different rates over time. The literature addresses these concerns with richer income controls (Gruber & Saez, 2002; H. J. Kleven & Schultz, 2014), longer instrument lags (Weber, 2014), and differencing against placebo periods or cohorts with graphical validation (Graber et al., 2026; Jakobsen & Søgaard, 2022).

[19] See Saez et al. (2012) for an overview of the literature.

A further concern is that the IV estimand weights individual responses by the realised change in their marginal rate, and reform variation can move realised rates in opposite directions (Appendix H). When a reform raises the rate, an individual may respond by leaving employment or by cutting hours enough to fall below the reformed bracket. In both cases she faces a lower rate where a rise was predicted. Monotonicity (Assumption 3 below) is then violated, and the estimand no longer aggregates behavioural responses with convex weights. Because the events we use shift the thresholds of the schedule and leave its gradient unchanged, a violation requires income to fall below the lower threshold of the pre-event bracket. Income that falls back below the threshold the event pushed it across only returns to that bracket. Partner income also places a floor under the rate an individual can reach by reducing own income.

We exploit the fact that the tax function that applies to an individual's income changes when the composition of the tax unit changes (Table 1). Our main design uses the fiscal exit of a dependent child, which lowers the family quotient and leaves the tax unit otherwise unchanged (Section 2.1). Our second design uses marriage, which merges two tax units. Unlike a reform, the event leaves the bracket formula unchanged and changes the income to which it is applied. We index the two schedules by $g \in \{0, 1\}$: the individual is taxed under the pre-event schedule ($g = 0$) or the post-event one ($g = 1$). From here on, "schedule" refers to the household tax function T^H under g, and "formula" to the calendar-dated bracket formula T_t of equation (1). Dates are expressed in event time, $r \equiv t - m_i$, where m_i is the calendar year of the event. For the fiscal exit, m_i is the year in which fiscal attachment of the first-born ends. Attachment lasts until the child turns 18 and can be extended to 21, or to 25 for a child in education. The schedule indicator in event time is $g_{ir} \equiv \mathbb{1}[r \geq 0]$. Every individual in our designs is taxed under the pre-event schedule for $r < 0$ and under the post-event schedule from $r = 0$ onwards.[20] Comparisons are made within cells of individuals who all experience the event. Section 3.3 states what identification requires of the event.

The bracket formula applies to income per part, $b_{ir}(g)$, which is total tax-unit income under schedule g divided by the quotient. For an individual in a couple filing jointly, $b_{ir}(g) = (z_{ir} + z_{pr})/Q_g$. For an unmarried individual, $b_{ir}(0) = z_{ir}$. Like the reform instrument, the event instrument is a simulation. It applies the pre-event year's formula twice to pre-event

[20] The design compares the pre-event year $r = -1$ with one post-event year, and the event-time profiles of Section 4 make this comparison at each horizon.

$(r = -1)$ income per part, under $g = 0$ and under $g = 1$, and takes the difference in log net-of-tax rates. The two applications differ only in g, which sets the quotient (and at marriage also the tax unit):

$$Z_{ir} \equiv g_{ir} \times \underbrace{\Big[\ln\big(1 - T'_{-1}\big(b_{i,-1}(1)\big)\big) - \ln\big(1 - T'_{-1}\big(b_{i,-1}(0)\big)\big)\Big]}_{\equiv Z_i,\ \text{the } \textit{dose}}, \tag{3}$$

where T_{-1} is the formula of the individual's pre-event calendar year. The dose Z_i is fixed at that year, and the indicator g_{ir} switches it on once the post-event schedule applies. Within a cell the dose takes few values, the log-rate differences between brackets, and each dose δ corresponds to a bracket transition. It is zero for individuals whose income per part is predicted to remain in its bracket.[21] At the child's fiscal exit the tax unit is unchanged and the quotient falls from 2.5 to 2 with one dependent child and from 3 to 2.5 with two. With a lower quotient, income per part weakly rises at every level of household income, and for almost every individual the event predicts a weakly higher marginal rate, $Z_i \leq 0$.[22]

This instrument definition leaves *within-income* variation: conditional on own income and the pre-event bracket, whether or not an individual is pushed over a bracket threshold is a function of partner income (Figure 2, Panel (b)).[23] Own income can therefore be held fixed without absorbing the identifying variation.

3.3 Identification and an illustration

Given the two schedules, we let $Y_{ir}(g)$ denote potential log income at event time r under schedule $g \in \{0, 1\}$, and $\text{NTR}_{ir}(g)$ the corresponding potential log net-of-tax rate; the post-event schedule applies from $r = 0$ onwards (g_{ir}, Section 3.2). $Y_{ir}(0)$ is not a no-event counterfactual, because the event occurs under both potential outcomes and only the schedule is varied. The

[21] The dose and the realised net-of-tax rate τ_{ir}^{H} of Section 3.5 come from the same effective schedule: the hypothetical bracket under the post-event quotient and the realised bracket of every year are assigned from the same set of kinks and effective marginal rates, by year and number of parts (Section 2.1), so that these features enter instrument and outcome identically.

[22] For 5 percent of the women and 6 percent of the men, the effective schedule of Section 2.1 predicts a lower rate at the child's exit; these rate cuts enter the estimation as doses of their own. At marriage, two tax units with quotient 1 merge into a single unit with quotient 2; predicted increases and decreases coexist, and we restrict the analysis to observations with predicted increases.

[23] Jakobsen and Søgaard (2022) develop the associated identifying assumption of constant trend differentials and its graphical validation.

schedule change shifts the log net-of-tax rate by $\Phi_{ir} \equiv \mathrm{NTR}_{ir}(1) - \mathrm{NTR}_{ir}(0)$. The construction is stated for earnings, with log work income as the outcome. The participation indicator, $P_{ir} \equiv \mathbb{1}\{z_{ir} > 0\}$, is treated in the same way, with its model counterpart in Appendix D and its estimation in Section 3.5. The potential outcomes map to observed outcomes through

$$
\begin{aligned}
Y_{ir} &= Y_{ir}(0) + \mathbb{1}[Z_{ir} \neq 0]\,\big[Y_{ir}(1) - Y_{ir}(0)\big], \\
\mathrm{NTR}_{ir} &= \mathrm{NTR}_{ir}(0) + \mathbb{1}[Z_{ir} \neq 0]\,\big[\mathrm{NTR}_{ir}(1) - \mathrm{NTR}_{ir}(0)\big].
\end{aligned} \tag{4}
$$

For zero-dose individuals the two schedules prescribe the same marginal rate at their pre-event income—the schedule change is a treatment only where $Z_i \neq 0$—so the mapping assigns the predicted-control group its untreated outcomes at every horizon. The realised rate changes of this group form its first stage and provide a direct check (Appendix E).

In line with the model of Section 3.1, we assume potential log income is linear in the potential rate, with the slope invariant across the two schedules (as in Graber et al., 2026; Saez et al., 2012):

$$
Y_{ir}(\mathrm{NTR}, g) = \epsilon_i \times \mathrm{NTR} + \ln n_{ir}(g), \tag{5}
$$

where ϵ_i is the individual elasticity of Section 3.1 and $n_{ir}(g)$ the zero-rate income of Section 3.1 under schedule g, whose schedule argument collects every channel from the schedule change to income other than the marginal rate—the argument Assumption 2 below restricts.[24]

For both the identification argument and the estimation that follows, we compare at a disaggregated level, because this makes the identifying assumptions more credible and because the elasticities are heterogeneous. Comparisons are made within cells of individuals alike in the determinants of income growth, and each dose δ in the support of Z is compared separately to the zero-dose group of its cell. The identifying assumptions then need to hold only among near-identical individuals, where they are easiest to defend and to test. Comparing each dose separately also avoids the negative weights that pooling doses can place on indi-

[24] With non-separable costs, the household model of Appendix B delivers the same form locally, with ϵ_i read as the elasticity of own income with respect to the household rate, $\epsilon_i^H = \epsilon_{ii} + \epsilon_{ip}$. The proof of Proposition 1 does not need linearity either. Equation (5) enters it only through $Y_{ir}(1) - Y_{ir}(0) = \epsilon_i\,\Phi_{ir}$. For any model in which the schedule affects income only through the rate and through $n_{ir}(g)$—so that, under Assumption 2, an unchanged rate means unchanged income—the proposition holds with ϵ_i read as the ratio of the individual's income change to her rate change, $[Y_{ir}(1) - Y_{ir}(0)]/\Phi_{ir}$. That ratio may then differ across horizons and with the size of the change, which the linear form rules out.

vidual responses when elasticities are heterogeneous (Callaway et al., 2026; Graber et al., 2026).[25] Formally, cells fix own income within narrow bins, the event year and the event type exactly, and partner income within an interval. Let $X_i \equiv (z_{i,-1},\, z_{p,-1},\, m_i,\, e_i)$ collect these predetermined variables, with e_i the event type (the number of dependent children and single or joint filing before the event); let the cells $\mathcal{I}_c$ be the elements of the resulting partition of the support of X, and write $c(X_i)$ for the cell of individual i.[26] Throughout, expectations subscripted by c, $\mathbb{E}_c[\,\cdot\,]$, are taken over the individuals of cell $\mathcal{I}_c$.

Within cells, we impose three assumptions:

Assumption 1 (Conditional common trends). *For every cell, horizon r, and dose δ in the within-cell support of Z—writing $\Delta_r V \equiv V_r - V_{-1}$ for growth relative to the pre-event year—*

$$\mathbb{E}_c\big[\Delta_r Y(0) \mid Z_i = \delta\big] = \mathbb{E}_c\big[\Delta_r Y(0) \mid Z_i = 0\big],$$
$$\mathbb{E}_c\big[\Delta_r \mathrm{NTR}(0) \mid Z_i = \delta\big] = \mathbb{E}_c\big[\Delta_r \mathrm{NTR}(0) \mid Z_i = 0\big].$$

Assumption 1 holds whenever untreated growth does not vary within cells at all, $\mathbb{E}[\Delta_r Y(0) \mid X_i = x] = f_y^r(c(x))$ for all x, and likewise for $\Delta_r \mathrm{NTR}(0)$: since own income, the event year, and the event type are fixed by the cell, this requires that the exact position of partner income within the cell's partner-income tercile does not predict untreated growth. By equation (5), $\Delta_r Y(0) = \epsilon_i\, \Delta_r \mathrm{NTR}(0) + \Delta_r \ln n(0)$, so the restriction concerns the untreated evolution of net-of-tax rates and of zero-rate income. Zero-rate income includes the direct effects of the event, which occur under both schedules and difference out between the two groups. Although stated conditional on the constructed dose Z_i, the assumption restricts only predetermined variation: within a cell, Z_i is a deterministic function of the position of the pre-event incomes within the cell's own-income bin and partner-income tercile (predominantly of partner income, because own income moves only within its narrow bin). Conditioning on $Z_i = \delta$ partitions the cell by predetermined income positions, and Assumption 1 is a restriction on untreated growth across these predetermined groups. Sorting of the event's timing on the tax consequence (say,

[25] Conversely, if elasticities were homogeneous within cells, the dose separation and the monotonicity condition below would be dispensable—the pooled within-cell contrast identifies the common ϵ. Even then the cells would remain necessary, because treated and controls differ unconditionally in own income, event year and event type (the dimensions along which untreated growth varies), and Assumption 1 holds only within them.

[26] Given X_i, both the pre-event tax bracket and the instrument of equation (3) are determined.

delaying a child's fiscal exit by extending attachment when the predicted bracket change is costly) would violate this restriction within cells, which fix the event year, and so would anticipatory bunching below thresholds that bind only after the event. The density tests of Appendix E check for both.

Table 2: The example cell: characteristics by predicted treatment status

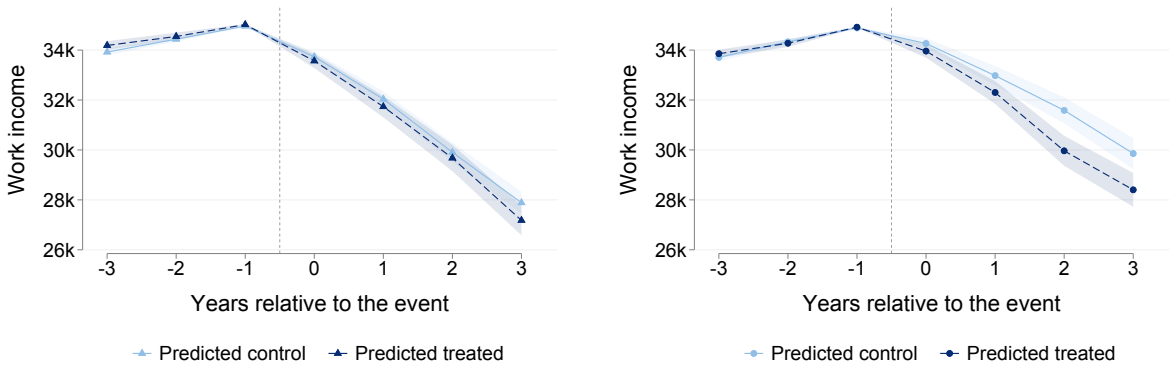
	Men		Women	
	Control	Treated	Control	Treated
Work income (€)	34,960	35,022	34,885	34,919
Partner's work income (€)	22,098	28,979	22,592	30,524
Age	54.0	54.7	53.3	53.8
Partner's age	52.1	53.1	54.3	54.8
Marginal rate at $r = 0$				
Predicted	14%	30%	14%	30%
Realised (mean)	16%	24%	16%	24%
N	4,295	2,586	1,687	1,735

Notes: The cell comprises individuals with pre-event own income in [€ 34,000, € 36,000], a pre-event marginal rate of 14% and 2.5 fiscal parts, whose first-born exits the tax unit at $r = 0$ in 2006–2020. The two groups are balanced on own income and age, and differ in partner income, which determines whether the fall in the family quotient pushes income per part across the threshold of the post-event schedule. Incomes are measured in the pre-event year and ages at the event. The rate rows refer to $r = 0$, the first year under the post-event schedule.

Table 2 and Figures 3 and 4 explain with an example how Assumption 1 can be checked and what it means economically. The example comprises individuals with pre-event own income between € 34,000 and € 36,000, at a pre-event marginal rate of 14%, in 2.5-part tax units whose first-born exits at $r = 0$ (event years 2006–2020). Individuals whose partner earns enough are predicted into the 30% bracket once the quotient falls to 2 ($Z_i = \delta = \ln(0.70/0.86) \approx -0.21$), and the rest are not ($Z_i = 0$). Figure 3 plots the income paths of the two groups, and their alignment before the event is the first check of the assumption.[27] Figure 4 shows the split of the sample by partner income into a *treatment region*, where the exit is predicted to raise the marginal rate, and a *validation region*, where the exit predicts no rate change. Economically, the assumption requires partner income to have no channel to income growth of its own within the cell's partner-income tercile. Such channels are conceivable. A higher-earning partner may allow the individual to reduce work once the child has left, or the individual may adjust own work to changes in the partner's.[28] Either would appear as a gradient of income growth in

[27] Every exhibit of the example applies the civil-status restriction of the estimation sample (Appendix F.1): individuals whose civil status changes at the event are excluded.

[28] Appendices B and D take up cross-spouse responses and income effects.



(a) Men

(b) Women

Figure 3: Work income around the first-born's fiscal exit, by predicted treatment status

Notes: Mean work income in euro by year relative to the event; bands are 95% confidence intervals; the dashed vertical line marks the event. Raw means for the example, without matching or covariate adjustment, so the figure shows the comparison the design uses. The series distinguish the predicted treated ($Z_i = \delta < 0$) from the predicted controls ($Z_i = 0$). Before the event the two paths lie within €150 of each other for women and €270 for men. Two years after it, treated women earn €1,600 (5 percent of the pre-event level) less than control women and treated men €230 less than control men, and the shares in work are 0.89 against 0.94 for women and 0.87 against 0.89 for men. Table 2 describes the sample.

partner income. The predicted rate change is a step function of partner income, while such channels would be smooth in it. The validation region tests for the gradient directly, because partner income varies within it with no accompanying rate change. Absent such a channel, income growth (the vertical distance between the two series of Figure 4) is flat in partner income across the region. In a single cell such as the example, this flatness is a second check of the assumption. Section 4 reports the same check aggregated over all cells of the design, in the pooled analogue of Figure 4 (Figure 5).[29]

Assumption 2 (Exclusion restriction). $n_{ir}(0) = n_{ir}(1)$ *with probability one.*

Zero-rate income—including any direct effect of the event, through time use, home production, or transfers—is unaffected by the schedule change. These direct effects are left unrestricted, and the schedule change affects income only through the marginal net-of-tax rate. This is the potential-outcomes form of the no-income-effects property of the quasi-linear model of Sec-

[29]Partner-income groups finer than terciles weaken the requirement, and the matching implementation of Section 3.5 additionally aligns pre-event income paths within cells.

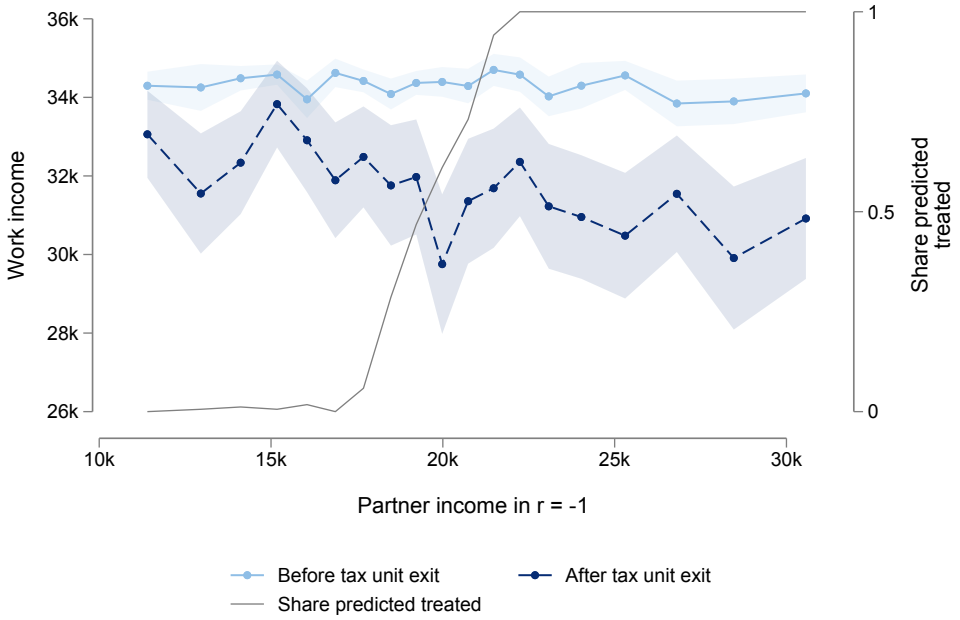


Figure 4: Identifying variation at the first-born's fiscal exit

Notes: Sample: women of the example (pre-event own income in [€34,000, €36,000], pre-event marginal rate 14%, 2.5 fiscal parts, first-born exiting the tax unit at $r = 0$ in 2006–2020). Left axis: mean work income at $r = -2$ (before) and, for each individual, the average of $r = +1$ and $r = +2$ (after), by 20 equal-count bins of partner income (about 170 women per bin; the lowest bin, with mean partner income below €10,000, lies outside the plotted range); bands are 95% confidence intervals. Right axis: the share of the bin predicted into the 30% bracket. Partner income is the household's taxable income at $r = -1$ net of the individual's own earnings; taxable wages carry the 10% deduction for professional expenses, so the measure is about nine tenths of the partner's wage less a tenth of own earnings, which is why the amounts look small. The validation region comprises the bins with mean partner income above €11,000 in which the predicted-treated share is below 5% (partner income up to about €16,900), and the treatment region those in which it exceeds 95% (from about €22,200). The figure follows the graphical validation approach of Jakobsen and Søgaard (2022). In the validation region the event predicts no rate change. A gradient of income growth (the vertical distance between the two series) in partner income there would indicate a direct partner-income channel and a violation of Assumption 1. Appendix E shows the corresponding figure for men (Figure E.1) and the participation counterpart (Figure E.2).

tion 3.1.[30] Under quasi-linearity the assumption holds by construction.[31] With income effects, the change in tax *liability* is a second channel from the schedule change to income: the fall in the quotient raises every household's liability, and households predicted across a threshold face a larger increase. Assumption 2 then fails and the estimands below recover a total response to the schedule change; because the liability change works against the rate change when leisure

[30] Formally, decompose $\ln n_{ir}(g) = \mu_{ir} + \xi_{ir}(g)$ with $\xi_{ir}(0) \equiv 0$: μ_{ir} carries life-cycle and calendar-time growth and the event's direct effects (changed time use, home production, transfers), which occur under both schedules and remain unrestricted. Assumption 2 sets $\xi_{ir}(1) = 0$. Reform settings impose the analogous restriction when they assume potential income unaffected by the reform.

[31] Quasi-linearity also implies that ϵ_i is the compensated elasticity, which coincides with the uncompensated one (Graber et al., 2026, Propositions 3–4).

is normal, the estimates are lower bounds on the compensated elasticity. Appendix D shows how to relax the assumption, and Figure D.1 there illustrates the approach.[32]

Together with the linear income function (5), Assumption 2 also rules out responses to the position of the new threshold. An individual predicted into the higher bracket who stays just below its threshold shows a fall in income with no change in the realised rate ($\Phi_{ir} = 0$). Such responses enter only the reduced form of the estimands below and raise the cell–dose elasticity. The policy elasticity defined below (footnote 33) is unaffected because it divides by the predicted change.

Assumption 3 (Monotonicity). $Z_i \left[\mathrm{NTR}_{ir}(1) - \mathrm{NTR}_{ir}(0)\right] \geq 0$ *with probability one, for all* $r \geq 0$.

The schedule change may not move any individual's rate in the direction opposite to the predicted change ($Z_i < 0$ for a predicted rate increase; Section 3.2). Rates remain free to move either way for reasons unrelated to the event. This is the standard no-defiers condition ensuring that Wald estimands reflect positively weighted averages of individual parameters (Angrist & Imbens, 1995; Imbens & Angrist, 1994).

Section 3.2 explained why reform variation easily violates the condition and why our events make a violation less likely. Appendix H formalises the argument and illustrates it with a simulation.

For each cell, dose δ, and horizon, the reduced-form and first-stage contrasts compare realised income growth $\Delta y_{ir} \equiv \ln z_{ir} - \ln z_{i,-1}$ (the observed counterpart of $\Delta_r Y$) and net-of-tax rate growth $\Delta\mathrm{NTR}_{ir} \equiv \ln \tau^H_{ir} - \ln \tau^H_{i,-1}$ across dose groups:

$$RF^r_c(\delta) \equiv \mathbb{E}_c\big[\Delta y_{ir} \mid Z_i = \delta\big] - \mathbb{E}_c\big[\Delta y_{ir} \mid Z_i = 0\big],$$
$$FS^r_c(\delta) \equiv \mathbb{E}_c\big[\Delta\mathrm{NTR}_{ir} \mid Z_i = \delta\big] - \mathbb{E}_c\big[\Delta\mathrm{NTR}_{ir} \mid Z_i = 0\big],$$

[32] The event-induced liability change scales with the amount of income shifted across the threshold and can be predicted from pre-event incomes in the same way as the rate change. Appendix D reports an attempt to add this simulated liability change to the cell regressions as a second regressor: within a cell it is largely collinear with the dose, so the design cannot separate the two channels.

and our headline estimand is the cell–dose elasticity[33]

$$\beta_c^r(\delta) \;\equiv\; \frac{RF_c^r(\delta)}{FS_c^r(\delta)}.$$

Proposition 1 (Causal interpretation). *Under the mapping* (4)*, the income function* (5)*, and Assumptions 1–3, for every cell, dose* $\delta \neq 0$*, and horizon* $r \geq 0$ *with* $FS_c^r(\delta) \neq 0$*:*

$$\beta_c^r(\delta) \;=\; \mathbb{E}_c\big[\,\omega_{ir}\,\epsilon_i \mid Z_i = \delta\big], \qquad \omega_{ir} \equiv \frac{\Phi_{ir}}{\mathbb{E}_c[\Phi_{ir} \mid Z_i = \delta]},$$

where the weights $\omega_{ir} \geq 0$ *average to one: the cell–dose elasticity is a positively weighted average of individual elasticity parameters among the predicted treated.*[34] *(Proof: Appendix C.)*

3.4 Aggregation

We aggregate in two steps.[35] Each estimate $\beta_c^r(\delta)$ rests on its own regression sample (Section 3.5): the dose's predicted treated and the cell's zero-dose group, observed at horizon r. We write $N_c^r(\delta)$ for its size and $\mathcal{D}_c$ for the within-cell support of non-zero doses. Within a cell, the dose-specific elasticities are combined with weights proportional to their regression samples,

$$\beta_c^r \;\equiv\; \sum_{\delta \in \mathcal{D}_c} w_c^r(\delta)\,\beta_c^r(\delta), \qquad w_c^r(\delta) \;\equiv\; \frac{N_c^r(\delta)}{\sum_{\delta' \in \mathcal{D}_c} N_c^r(\delta')}.$$

The zero-dose group belongs to the regression sample of every dose it is compared with, so it is counted once per dose. In the second step, we aggregate all cells contained in a group h of interest (own-income intervals for the income profiles and own-by-partner-income rectangles for the maps of Section 4):

$$\epsilon_h^r \;\equiv\; \sum_{c \,\in\, h} v_h^r(c)\,\beta_c^r, \qquad v_h^r(c) \;\equiv\; \frac{\sum_{\delta \in \mathcal{D}_c} N_c^r(\delta)}{\sum_{c' \in h} \sum_{\delta \in \mathcal{D}_{c'}} N_{c'}^r(\delta)},$$

[33] Dividing the reduced form by the predicted change instead, $RF_c^r(\delta)/\delta$, scales the response by the predicted rather than the realised rate change (the policy elasticity of Hendren (2016)). It needs only Assumption 1 and is unaffected where frictions or threshold-crossing responses attenuate the realised rate change.

[34] For event-based instruments, Proposition 1 is the analogue of the characterisation in Graber et al. (2026) of when reform-based ETI estimands are causal. Their two conditions are a binary instrument contrast and income controls rich enough to reproduce the conditional mean of the instrument. In our setting, the first is a feature of the design (each estimand contrasts $\mathbb{1}\{Z_i = \delta\}$ against the zero-dose group), and the second is Assumption 1.

[35] The aggregation is stated for earnings and applies to participation in the same way (Section 3.5).

so that $v_h^r(c)$ is cell c's share of the group's estimation sample. Together, the two steps weight every estimate $\beta_c^r(\delta)$ of the group by its regression sample; we call these the frequency weights. Only estimates that pass the screens of Appendix F.2 enter the sums. We report ϵ_h^r as an event-time profile in Appendix F.5. The headline displays of Section 4 pool the post-event years $r = 0$ to 3 in the same way: every estimate enters with its own regression sample, so each year counts in proportion to the group's estimation sample in that year.

Because both steps use non-negative weights, the summary parameters and their pooled averages inherit the causal interpretation of Proposition 1. Substituting, ϵ_h^r is a weighted average of the individual parameters ϵ_i over the predicted treated of the group. An individual with dose δ in cell c has a weight proportional to $\omega_{ir}\, N_c^r(\delta)/n_c^r(\delta)$, where $n_c^r(\delta)$ is the number of the dose's predicted treated in its regression sample. The factor $N_c^r(\delta)/n_c^r(\delta)$ is one plus the ratio of the zero-dose group to the dose's predicted treated, so doses and cells with large comparison groups relative to their predicted treated count more than their share of the treated. Weighting by the predicted treated alone lowers the French-taxpayer aggregates of Section 4.2.1 to 0.18 (earnings) and 0.19 (participation),[36] and the difference is concentrated in the cells of households predicted to cross from the 14 to the 30 percent bracket (Appendix F.2). The aggregation weights $v_h^r(c)$ and $w_c^r(\delta)$ are known sample frequencies—in contrast to pooled regressions across treatment intensities, whose implicit weights can turn negative under heterogeneity (Callaway et al., 2026; Graber et al., 2026)—and alternative policy-relevant weightings can be substituted without affecting the identification argument.[37]

3.5 Estimation and inference

Estimation is by two-stage least squares within cells, separately for women and men, over an event window spanning $r = -3, -2$ and $r = 0, \ldots, 3$, with $r = -1$ the base year. A period is a year r relative to the event. A regression is run for each cell, dose and period, and its coefficient $\beta_c^r(\delta)$ is a cell–dose–period estimate. The regression sample of dose δ in

[36] Weighted by the predicted treated alone, the French-taxpayer aggregates are 0.13 (men), 0.23 (women) and 0.18 (total) for earnings (0.17, 0.29 and 0.23 under the frequency weights; Table 4) and 0.15, 0.23 and 0.19 for participation (0.21, 0.28 and 0.25; Appendix Table F.4). From €10,000 of own work income up, they are 0.13, 0.09 and 0.11 for earnings (0.17, 0.23 and 0.20) and 0.10, 0.08 and 0.09 for participation (0.15, 0.18 and 0.16).

[37] Inverse-variance weights are also non-negative and preserve the causal interpretation, but they weight estimates by their precision. The headline displays of Section 4 use the frequency weights throughout; Section 4.2.1 reports an inverse-variance-weighted aggregate for comparison.

period r pools the dose's predicted treated with the cell's zero-dose group, and we estimate the regression

$$\Delta y_{ir} \;=\; a_c^r(\delta) \;+\; \beta_c^r(\delta)\,\Delta\mathrm{NTR}_{ir} \;+\; W_i'\gamma \;+\; u_{ir}, \tag{6}$$

instrumenting the realised rate change $\Delta\mathrm{NTR}_{ir}$ with Z_{ir}; the controls W_i are fixed effects for own income at $r = -1$, -2 and -3 (in bins), age, marital status, the number of children at $r = -1$, and the event year.

For earnings, which we measure as work income without self-employment income, the outcome is log work-income growth. The coefficient is the earnings elasticity: the elasticity of own earnings with respect to the household net-of-tax rate (one minus the household's marginal tax rate at the individual's income), conditional on positive work income in both years ($r = -1$ and r).[38] The total response of Section 4.2.1 combines earnings and participation (Appendix Table F.5). Its reduced form is estimated by Poisson pseudo-maximum likelihood (PPML) in levels including zeros and divided by the cell's first stage. For participation, $\Delta_r P_i \equiv P_{ir} - P_{i,-1}$ replaces Δy_{ir} in (6). This regression is estimated on the full event sample, which includes individuals without work income at $r = -1$. The coefficient is the participation semi-elasticity, the change in the participation probability per unit change in the log household net-of-marginal-tax rate. We divide it by the cell's participation two years before the event to obtain the participation elasticity with respect to the household net-of-marginal-tax rate.

The marginal rate at household income is not the price used in the participation literature. That price is the participation tax rate, the share of own earnings that the household pays in additional tax when the spouse works, given partner income (Appendix D.2). The child's fiscal exit moves the two rates differently. The marginal rate at household income changes only where the household crosses a threshold, and the cells isolate that step. The participation tax rate changes wherever the spouse's earnings carry the household across a threshold that the exit moves (for zero-dose households as well), and by an amount that varies continuously with partner income and own earnings. Measured by this price, the zero-dose group is treated as well, and the contrast between the predicted treated and the zero-dose group is small. It is

[38] This is the earnings elasticity of optimal-tax models, taken with respect to one minus the marginal rate and defined on the earnings of those who work rather than on hours (H. J. Kleven et al., 2009; Saez, 2001). The reform model of Section 5.2 uses it in the same sense (Bierbrauer et al., 2024).

about one eighth of the marginal-rate contrast over the cells and about one half in the bottom own-income bin, where the first euro of someone without earnings is taxed at the household's rate at partner income. Our participation estimates are responses to what differs between the two groups' schedule changes (mainly the change of bracket) per unit of the marginal-rate contrast. Appendix D.2 re-expresses them per unit of the participation-tax contrast, which raises the French-taxpayer aggregate by a factor of about 3.4 (Appendix F.7). Appendix D.2 also states what that number assumes and describes a comparison within our cells that would estimate the response to the participation tax alone.

The estimation cells fix the own-income bin, the partner-income tercile within the bin and the pre-event bracket. The event year and the number of children (which the cells of Section 3.3 fix exactly) are pooled in estimation and enter through the fixed effects; filing status is fixed by the sample.[39]

A dose is estimated if at least 30 predicted-treated individuals carry it in the cell and if its regression sample holds at least 300 outcomes observed at $r = -3$. In the lowest own-income bin, €0–10,000, the cells are in addition formed separately for individuals with at least €400 of work income at $r = -1$ (the earner sub-sample) and for those below (the entrant sub-sample), so that a cell compares individuals of the same pre-event earner status. Each group has its own partner-income terciles. The bin's estimates pool the cells of both groups with the frequency weights and screens of every other bin (Appendix F.2). Estimating the reduced form and first stage on Z_i at pre-event horizons $r < 0$—where $Z_{ir} = 0$, so (6) itself has no first stage—produces the placebo contrasts implied by Assumption 1, which we report alongside the post-event estimates.

The matching implementation replaces the regression comparison within each cell with nearest neighbours (with replacement) on the Mahalanobis distance of pre-event work income, measured at $r = -3$, $r = -2$ and $r = -1$, and age, with exact matching on the event year and a caliper of 0.8. The reduced form is then the matched average difference in income growth over all predicted-treated and zero-dose individuals of the cell whom the caliper keeps (in Proposition 1, the average is over the predicted treated alone). The first stage is the difference

[39]The reduced form $RF_c^r(\delta)$ and first stage $FS_c^r(\delta)$—Δy_{ir} and ΔNTR_{ir} regressed on Z_i with the same controls—are estimated alongside; the two-stage least squares coefficient equals their ratio, and without the controls the coefficients equal the group-mean contrasts of Section 3.3 divided by the dose δ (a factor common to both that cancels in the ratio).

in realised rate changes between the two groups on the same sample, without matching, and the elasticity is the ratio of the reduced form to the first stage (footnote 29).

Standard errors are clustered by person. Within a comparison group—a cell's zero-dose group and the doses compared with it—the estimates for different doses and periods rest on the same individuals, and their person-clustered covariance propagates to the aggregates β_c^r and ϵ_h^r, whose weights are sample frequencies with negligible estimation error. Different comparison groups contain different individuals and are treated as independent. At the median bin of the income profiles the clustered standard error is 1.39 times the conventional one (which treats every cell–dose–period estimate as independent), and at the median tile of the maps 1.45 times, mostly because an aggregate pools four post-event years of the same people.[40] The pre-event placebo contrasts are pooled with the frequency weights of the estimate they accompany and tested with the same covariance. The estimates on the balanced sample (individuals observed in every year of the event window, $r = -3$ to 3), the matching estimates and the marriage estimates treat cells as independent in their standard errors, as do the placebo tests of the marriage exhibits. The notes to these exhibits say so. Standard errors of totals over both genders and of gender differences treat the women's and the men's estimates as independent, although the two samples contain the two spouses of the same tax units.

4 Results

This section proceeds from validation to estimates. We first describe the comparison cells and assess the identifying assumptions across all cells. The main results then summarise the estimates as population aggregates (Section 4.2.1), along the own-income distribution with their event-time profiles (Section 4.2.2), and over the joint distribution of own and partner income (Section 4.2.3). Section 4.3 examines robustness to specification choices and Section 4.4 reports heterogeneity across household characteristics.

[40] Dose-specific comparisons within a cell share the zero-dose control group; the clustered covariance includes the induced correlation across dose-specific estimates, which is small. Across the bins of the income profiles, the covariance across doses accounts for a median 0.2 percent of the variance of the earnings estimate (0.3 percent for participation), the covariance across the periods of the same dose for 36 percent (52), and the estimates' own variances for 64 percent (47).

4.1 Comparison cells and pooled validation

The cells cover parents in couples,[41] aged 51 to 53 on average at the first-born's fiscal exit, with own work income below €70,000. Among these parents, the estimates are local to those whom the fall in the quotient moves across a threshold. Appendix F.1 describes the sample and Appendix G the marriage design. A cell fixes an own-income bin (€10,000 wide, of which the example of Section 3.3 is a €2,000 slice), the partner-income tercile within the bin and the pre-event bracket; within it, each dose is compared with the cell's zero-dose group (Section 3.5). In the headline specification (work income, all event years), 297 cell–dose comparisons for men and 248 for women reach the estimation floors of Section 3.5, over ten pre-event brackets and 25 distinct doses for each gender.[42] Of these, 93 for men and 102 for women are in the €0–10,000 bin, where they are formed separately for the earner and entrant sub-samples (Appendix F.2). The comparisons differ widely in size because the density of taxpayers varies along the income distribution and because several bracket crossings exist only in the subperiods in which the schedule created them. Table 3 reports the size distribution at the level of the 21 grid cells of own-income bin × partner tercile (the finest level disclosed). Every estimate below is an average over cell–dose comparisons, each of which is weighted by its regression sample (Section 3.4) and screened as in Appendix F.2.

The tests of Assumption 1 for the example cell can be pooled across cells, because every cell splits into a predicted-treated and a predicted-control group. The two groups differ where assignment requires it. The predicted treated have partners who earn more (by 0.24 of a standard deviation for women and 0.30 for men) and a household taxable income higher by about one standard deviation, which is what assigns them. The two groups also differ in family size, which sets the quotient. Own work income and age differ by 0.2 to 0.3 of a standard deviation, and pre-event income growth and participation by less than 0.05 (Appendix Table F.2). Figure 5 pools the validation-region check of Figure 4 across cells and measures household income from each household's hypothetical post-event threshold. Before the event, income growth shows no systematic gradient in household income across the validation region or on

[41] Couples whose civil status changes at the event are excluded, so that the fall in the quotient is the only change to the household's schedule (Appendix F.1).

[42] The brackets are the distinct marginal rates of the effective schedule (Section 2.1) at which the comparisons start, across the schedules in force from 2003 to 2023. The rates and thresholds of these schedules changed over the period. A dose need not lead to the next rate up: the fall in the quotient can carry a household past more than one kink of the schedule.

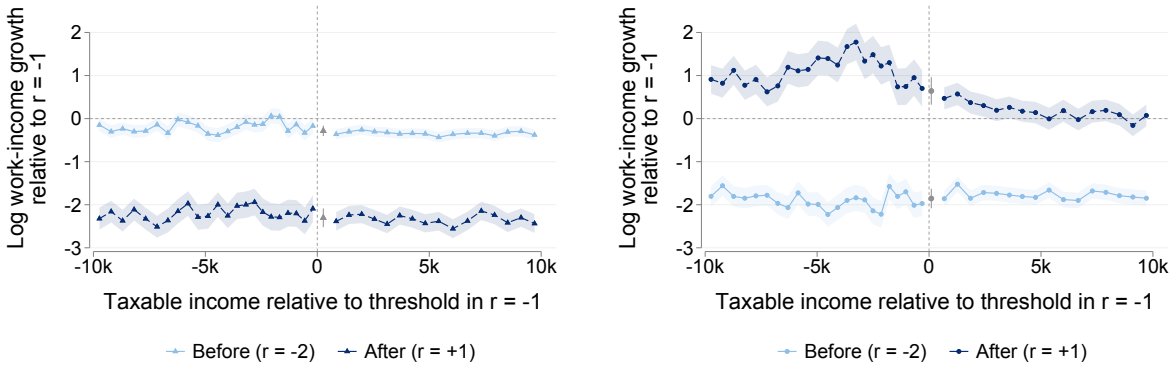
Table 3: The comparison grid

	Women	Men
Event sample below the income cap		
Individuals	1,884,448	1,758,056
Predicted treated (share)	0.30	0.32
Comparisons of the headline estimation		
Cell–dose comparisons	248	297
of which rate cuts	64	73
Pre-event brackets	10	10
Distinct doses	25	25
Grid cells with a comparison (of 21)	18	21
Distinct doses per grid cell (median)	13	13
Individuals in the comparisons	1,401,229	1,388,055
predicted treated (share)	0.39	0.38
Individuals per grid cell: median	78,295	60,858
smallest	391	821
largest	186,175	158,205
Behind the displayed estimates		
Cell–dose comparisons	128	152
Grid cells	18	21
Individuals	1,089,165	1,213,681
predicted treated (share)	0.37	0.35

Notes: The upper panel counts the child's fiscal exit sample below the income cap (Appendix F.1) and its predicted treated. The lower panels count the cell–dose comparisons of the headline estimation and the individuals in them at $r = -1$: the predicted treated of every comparison and the zero-dose comparison group of every cell, each counted once. A grid cell is an own-income bin (€10,000 wide) × within-bin partner tercile; in the €0–10,000 bin the comparisons are formed separately for the earner and entrant sub-samples (Appendix F.2), and a grid cell pools the two sub-samples' terciles of the same rank. The comparison cells of Section 3.5 additionally fix the pre-event bracket; that finer split is not disclosed at cell level. Behind the displayed estimates: comparisons with at least one estimate that passes the screens of Appendix F.2 for earnings or participation.

the other side of the threshold, for either gender (in every €2,000 band of household income, women's work income grows by 1.7 to 2.0 log points from $r = -2$ to $r = -1$ and men's by 0.1 to 0.4). A partner- or household-income channel of the kind Assumption 1 excludes would already show such a gradient. After the event the two genders move in opposite directions. Men (about two years older than women on average) see their work income fall by 2.1 to 2.4 log points from $r = -1$ to $r = +1$ in every band, a little less in the validation region than above the threshold. Women's work income still grows, and more in the validation region (0.9 to 1.5 log points by band) than above the threshold (0.5 or less).[43] Single bands are noisy, since the figure pools different comparison cells along its axis. Because winsorising the

[43] Women's growth is highest €2,000–6,000 below the threshold (1.3 to 1.5 log points), and about 0.9 both further below and just below it. A possible reading is that predicted controls within a few thousand euros of the post-event threshold are partly treated, and that the hump reflects bunching in year $r = 1$. The figure's axis measures income in $r = -1$, and in year $r = 1$ the threshold lies to the left of where the figure draws it.



(a) Men

(b) Women

Figure 5: The validation-region check, pooled across cells

Notes: Log work-income growth relative to $r = -1$ (100 times the change in log work income, in log points) at $r = -2$ (before the event) and $r = +1$ (after), against household taxable income at $r = -1$ minus the household's statutory threshold at post-event fiscal parts, pooled across the comparison cells of the design. An individual enters once per comparison it belongs to (zero-dose controls once per dose comparison of their cell) and once per statutory threshold within €10,000 of its household taxable income, on whichever side its income lies. Points are means of 40 quantile bins of these rows, cut over the whole window and joined within each side of the threshold; the first bin right of the threshold (grey) can therefore also hold rows just below it. Growth is computed for individuals with work income above €1,000 in both years and winsorised at the 5th and 95th percentiles of the gender sample at each horizon. Bands are 95% confidence intervals clustered by person. The figure is the pooled analogue of the validation-region check of Figure 4: a pre-event profile that is flat across the threshold indicates that household income carries no income-growth gradient of its own before the event (Jakobsen & Søgaard, 2022). Sample: the child's fiscal exit sample of Appendix F.1, in the comparison cells of the design (Section 3.5); 874,000 (women) and 934,000 (men) rows at $r = -2$ and 864,000 and 908,000 at $r = +1$, 19,900 to 24,500 per bin.

outcome caps the large falls that carry the response, the figure only checks the validation region. Section 4.2 estimates the response within cells. Sorting of the event's timing and bunching below the post-event threshold would show as a break in the density of household taxable income there; pooled across cells, the estimated log-density discontinuities are −0.002 (s.e. 0.010) for women and −0.001 (s.e. 0.009) for men (Appendix Figure F.1). The pre-event placebo contrasts of Section 3.5 accompany every estimate below, and an estimate is flagged where they reject (Appendix F.2).

4.2 Main results

The main results proceed from population-level averages to the profile along the own-income distribution and then to the map over the joint distribution of own and partner income.

4.2.1 Aggregated elasticities

Table 4 reports the aggregate elasticity that our estimates imply under four of the aggregation schemes most common in the literature, by gender and combined. The participation counterpart is Table F.4 in Appendix F.6.

Table 4: Aggregated earnings elasticities

	Men	Women	Total
Population-weighted: French taxpayers	0.169 (0.035)	0.289 (0.051)	0.229 (0.031)
Population-weighted: event population	0.174 (0.033)	0.322 (0.048)	0.246 (0.029)
Inverse-variance weighted	0.125 (0.023)	0.102 (0.022)	0.114 (0.016)
Income-weighted	0.137 (0.031)	0.217 (0.037)	0.171 (0.024)

Notes: Aggregation of tile-level estimates: cell–dose elasticities pooled within own-by-partner-income rectangles under the profile-level display conventions of Appendix F.2. One bin-level screen applies: a bin contributes only if the cells passing the cell screens cover at least 10 percent of the bin's estimation sample and at least 1,000 individuals underlie them; this removes the men's €0–10,000 bin for earnings (one cell, 5.7 percent of the bin's estimation sample) and, by the floor, the women's €60,000–70,000 bin for earnings and participation (865 and 885 individuals). The €0–10,000 bin pools the earner and entrant sub-samples (Appendix F.2). *French taxpayers*: each own-income bin weighted by its share of prime-age taxpayers, from the INSEE distribution of annual salary income by gender (2021), with the non-employed in the bottom bin; the support of the estimates, own income up to €70,000 in 2015 prices, covers at least 95.6 percent of the men and 97.0 percent of the women filing a return in 2023. The averaged estimates remain those of parents aged around 50 at the event, older than the taxpayers who supply the weights. *Event population*: each own-income bin × partner-tercile group is weighted by its regression samples in the event year ($r = 0$), summed over its doses, with the zero-dose group counted once per dose as in Section 3.4. *Income-weighted*: each bin weighted by its share of aggregate earnings from the same INSEE distribution. In the French-taxpayer and income-weighted rows, a bin's weight is split equally over the partner terciles whose cells pass the screens. The Total column combines genders with equal population shares (French-taxpayer and inverse-variance rows), the genders' shares of the event-population counts (event row), and gender aggregate-income shares (income row). Standard errors in parentheses are clustered by person, and the inverse-variance weights use the same clustered variances; the standard errors of the Total column treat the two genders as independent.

Weighted to the population of French taxpayers, the average earnings elasticity is 0.229 (0.289 for women, 0.169 for men).[44] Weighted to the event population itself (parents at the fiscal exit of their first-born, each bin and partner tercile weighted by its regression samples in the event year), the average is 0.246 (0.322 for women, 0.174 for men). Because the two totals differ by less than 0.02, the summary is not an artefact of the event population's income composition. The difference of 0.12 between women and men under the French-taxpayer weights is 1.9 person-clustered standard errors (95% confidence interval from about zero to 0.24, treating the two genders as independent).

The literature has also aggregated estimates with inverse-variance (efficiency) weights. These produce the smallest average (0.114) because the cells with high elasticities are the less pre-

[44]The French-taxpayer weights are constructed from the INSEE distribution of annual salary income by gender (2021), interpolated to a piecewise-linear CDF with a Pareto tail, with the non-employed entering the bottom income bin; see the table notes.

cisely estimated ones, and precision weighting shifts mass towards mid-income cells where responses are modest.

The convention in the taxable-income literature, however, is to weight each individual by their contribution to the tax base. This yields an earnings elasticity of 0.171 (0.024) overall—0.137 for men and 0.217 for women—below the population-weighted averages. The shortfall arises because weighting by income moves mass away from low incomes, where the responses of both genders are concentrated (Section 4.2.2).

We treat the French-taxpayer weighting as the headline aggregate; across the four schemes, the average earnings elasticity lies between 0.11 and 0.25. The participation elasticities are of comparable size (Appendix Table F.4). Weighted to French taxpayers, their average is 0.247 (0.283 for women, 0.212 for men). For both earnings and participation, the women's aggregate lies above the men's under every scheme of Table 4 except inverse-variance weighting (with treated weights, from €10,000 of own income up, the ordering reverses; footnote 36). Under the French-taxpayer weights, part of the gender differences comes from more women than men having low incomes, where elasticities are largest (45 percent of women and 34 percent of men are in the bottom bin below €10,000, which includes the non-employed), and applied to the same pooled profiles of Figures 6 and 7 the women's weights give averages 0.04 (earnings) and 0.02 (participation) higher than the men's. For earnings, the men's average also leaves out their bottom bin (notes to Table 4) and rests only on the bins from €10,000 up, where elasticities are lower.

Earnings and participation combine in the response of mean work income, which we report as the total (level) response. We estimate the reduced form of each cell in levels, including zeros, by Poisson pseudo-maximum likelihood (PPML) and divide it by the cell's first stage to put it on the scale of the elasticities above. The earnings elasticity conditions on positive work income in the year before the event and in the outcome year, while the total response does not. Weighted to French taxpayers from €10,000 of own income up, it is 0.21 (0.03) for men and 0.20 (0.03) for women, a third (men) to a half (women) smaller than the sum of the earnings and participation elasticities (0.32 and 0.41).[45] For women the total response is at or below

[45]The two elasticities overlap, since a fall to a few hundred euros counts both as an exit and as a large fall in earnings (Appendix Table F.13, *Floor 400*). An exit from small earnings also lowers mean work income only a little.

the earnings elasticity alone in every bin from €10,000 up that rests on more than one cell, so in euros their participation response adds little to the change in mean income. In the bottom bin the two measures describe different objects (for women, 0.08 against a sum of 0.78). The earnings elasticity averages log changes of small incomes, and these changes amount to only a few euros. Appendix Table F.5 reports the earnings and participation elasticities, their sum and the total response by own-income bin.

Our income-weighted earnings elasticity of 0.171 with respect to the household net-of-marginal-tax rate is within the range of quasi-experimental estimates of the elasticity of taxable income surveyed by Neisser (2021).

4.2.2 Profiles along own income

The population averages are made up of elasticities that vary systematically, most of all along own income.

Figure 6 reports elasticities aggregated within own-income bins, pooled over the post-event years $r = 0$ to 3. With men and women pooled, the elasticity is 0.45 in the lowest bin, 0.23 and 0.22 in the €10,000–20,000 and €20,000–30,000 bins and 0.04–0.09 in the three bins from €30,000 to €60,000. The top bin, at 0.18, is the least precise (standard error 0.13). The participation profile (Figure 7) also declines, from 0.32 in the lowest bin to 0.12–0.17 in the three bins from €10,000 to €40,000 and 0.07–0.08 in the next two, with the top bin at 0.13 (standard error 0.07). No bin of either pooled profile fails its pre-event placebo.

Appendix Figure F.4 shows the earnings profile by gender. For women (Panel b), elasticities are largest at the bottom of the distribution and lower everywhere above it. They are 0.45 in the lowest bin, 0.18 and 0.02 in the €10,000–20,000 and €30,000–40,000 bins and 0.19–0.31 in the other bins up to €60,000. The top bin rests on a single cell–dose–period estimate (-0.28) and is drawn in grey. The €10,000–20,000 bin fails its pre-event placebo and is displayed hollow. Its dip relative to the neighbouring bins partly reflects the flagged differential trend. For men (Panel a), the profile over the bins resting on more than one estimate is highest at 0.32 in the €10,000–20,000 bin and lower in every bin above it. It is 0.17 and 0.13 through the middle of the distribution and 0.02–0.19 in the three top bins. The lowest bin rests on a single cell–dose–period estimate and is drawn in grey (it enters no aggregate, and its placebo

and later-year values are in the figure note). On participation the two genders are closer (Appendix Figure F.5), with 0.33 (women) against 0.29 (men) in the lowest bin, 0.14 against 0.17 in the €10,000–20,000 bin and 0.20 against 0.15 in the €20,000–30,000 bin (the three bins that carry most of the taxpayer weight).

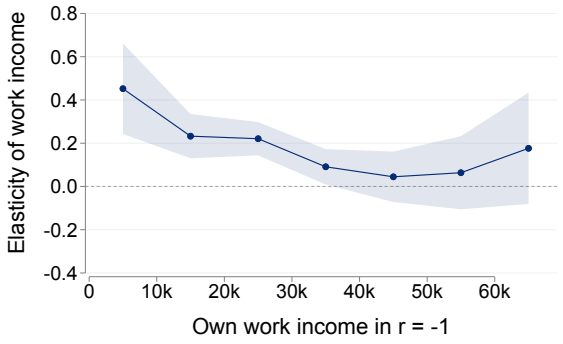


Figure 6: Earnings elasticities along the own-income distribution

Notes: Elasticities are largest at low own incomes. In the raw data, falls in work income of more than one log unit after the event are 0.2 to 1.7 percentage points more frequent among the predicted treated than in their comparison groups in every men's and women's bin from €20,000 up with more than one estimate (Appendix Figure F.4), so part of the response there may reflect a small share of large falls. Sample: the child's fiscal exit sample of Appendix F.1, all event years. Outcome: work-income growth relative to $r = -1$, pooled over $r = 0$ to 3. Construction, screens and weights follow Appendix F.2, with own-income bins as the aggregation groups; the figure runs the cell–dose–period estimates of both genders through the same aggregation, each with its frequency weight. No bin fails its pooled pre-event placebo ($|t| \geq 1.96$). The €0–10,000 bin pools the earner and entrant sub-samples (Appendix F.2): 18 cell–dose–period estimates on 159,108 individuals for women and a single estimate on 4,421 individuals for men, all from the earner sub-sample. Points are at bin midpoints of own work income at $r = -1$; shaded areas are 95% confidence intervals from person-clustered standard errors. The vertical axis is set by the bins with at least two cell–dose–period estimates and is shared with the gender breakdown in Appendix Figure F.4, whose two bins resting on a single estimate (drawn in grey there) enter their bins here with weights of 1 and 2 percent. The participation counterpart is Figure 7.

Event-time profiles. The estimates by year relative to the event serve as a validity check, since the same construction yields placebo estimates for the pre-event years; they are collected in Appendix F.5. Earnings and participation differ in their dynamics. The earnings elasticity is largest in the event year (0.37 for men, 0.36 for women), falls to 0.14 and 0.30 at $r = 1$ and is statistically indistinguishable from zero from $r = 2$. The participation elasticity is more persistent. It appears immediately and remains between 0.15 and 0.23 throughout the window for women, while for men it declines from 0.20 in the event year to 0.08 at $r = 3$. The headline earnings estimates pool $r = 0$ to 3 and lie between the event-year response and the smaller later-year responses. For participation, pooling matters less, and its pre-event placebo estimates are indistinguishable from zero for both genders. For earnings, the women's placebo at $r = -3$ is negative (-0.10, $t = -2.65$) –the same differential drift flagged in the

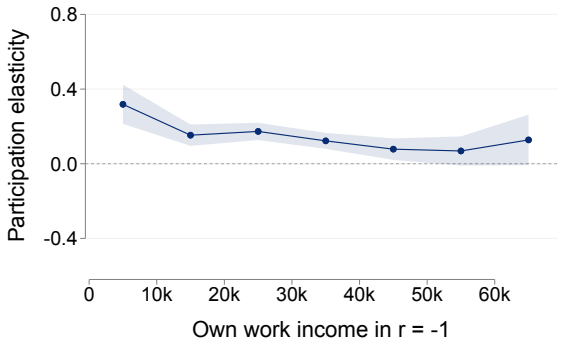


Figure 7: Participation elasticities along the own-income distribution

Notes: As Figure 6, for participation: the outcome is an indicator for work income above €1,000 (2015 prices), and the estimates are participation elasticities with respect to the household net-of-marginal-tax rate, divided by participation two years before the event (Section 3.5). No bin fails its pre-event placebo and no bin rests on a single estimate. The €0–10,000 bin pools the earner and entrant sub-samples of both genders (Appendix F.7 reports their composition by gender). The vertical axis is shared with the gender panels of Appendix Figure F.5, which show the breakdown.

women's €10,000–20,000 bin of Appendix Figure F.4– and so is the men's at $r = -2$ (-0.05, $t = -2.00$).[46] Pooled over all pre-event cells with the weights of the estimates, the placebo reduced form does not reject at the 5 percent level for men's earnings and participation or for women's participation. For women's earnings the test is borderline ($t = -1.9588$, $p = 0.050$; $t = -2.30$ if the pre-event cells are weighted by inverse variance instead). The event-year estimate combines the immediate behavioural response with any income realisation coinciding with the timing of the fiscal exit itself. We are cautious about reading the event-year peak of the earnings response as a steady-state elasticity.

The fiscal exit makes a lasting change to the household's marginal rate, at a date set by the child's age or, if attachment is extended, by the household (Section 3.2). Such a change predicts a lasting response, whether or not the household allowed for it in advance. If it did, the response at the event is an intertemporal (Frisch) one, and otherwise it includes the income effect (Section 3.3). Adjustment frictions delay a response, and for lasting tax changes H. Kleven et al. (2025) find that earnings responses build up over years. A response that is large in the event year and gone two years later points instead to adjustments concentrated around the event, e.g. the timing of the exit from work (Section 4.3). An exit in the course of a

[46] Pre-event points are scaled up relative to post-event ones because they divide the placebo reduced form by the smaller pre-event contrast in realised rates (0.10 to 0.13, against 0.43 to 0.45 in the event year). On the event-year scale the two placebos are -0.03 for women and -0.02 for men, and the women's value is 8 percent of their event-year response.

year leaves a partial year of earnings, a large fall that the earnings outcome (defined on those with work income in the outcome year) records at that horizon only, while the participation outcome records the exit from the following year on. The pooled earnings elasticity of $r = 0$ to 3 averages over these years, and we do not read it as a Frisch or as a steady-state elasticity.

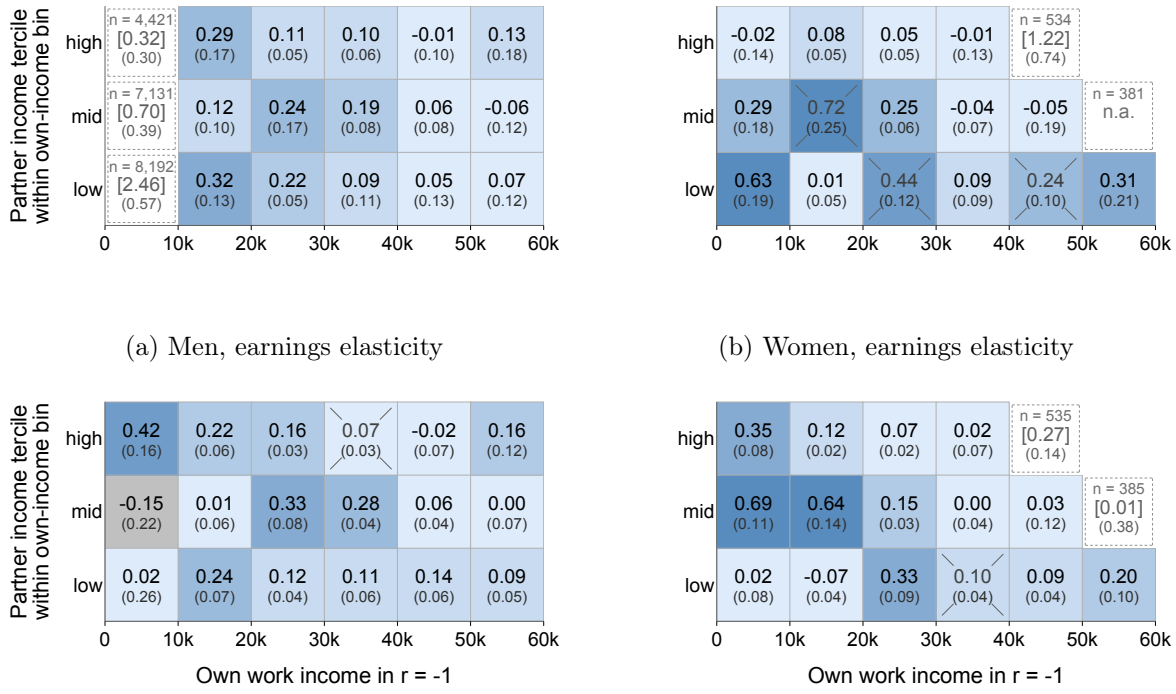
Relation to the literature. The larger responses at low own incomes mirror the concentration of participation and hours responses at low earnings documented in the labour-supply literature.

4.2.3 Maps over own and partner income

The own-income profile uses one of the design's two dimensions. Within an own-income bin, the identifying variation comes from partner income (Section 3.2). Because the comparisons do not need wide partner-income intervals, we can divide the partner-income distribution and study heterogeneity along partner income as well. This lets us ask whether the high elasticities documented at low own incomes are driven by households insured by a high-earning partner or by households in which both partners earn little.[47] Figure 8 disaggregates the own-income profile accordingly and reports one estimate per own-income bin and within-bin partner-income tercile. The vertical axis orders partner income *within* each own-income bin, so the comparison across terciles holds the own-income composition fixed.

For women, the answer differs between earnings and participation. For earnings (Panel b), the large bottom-bin response concentrates among households in which the partner also earns little. The low-partner tercile of the bottom bin (the double-low corner of the map) has an elasticity of 0.63, against 0.29 in the middle and -0.02 in the top tercile, and no other tile of the women's map whose placebo passes is as large (the €10,000–20,000 middle-tercile tile, 0.72, fails its placebo). For participation (Panel d) the pattern reverses. The bottom-bin elasticity is close to zero where the partner also earns little (0.02) and large where the partner earns more (0.69 in the middle and 0.35 in the top tercile). The participation response of women with a higher-earning partner is not confined to the event year. In the lowest bin, the middle- and top-tercile tiles are 0.80 and 0.44 over the three following years, against 0.48 and 0.11

[47] In the lowest own-income bin, the low partner-income tercile ends at €24,700 of partner income for women and at €13,800 for men in the earner sub-sample; Appendix F.2 gives the boundaries of both sub-samples.



(a) Men, earnings elasticity

(b) Women, earnings elasticity

(c) Men, participation elasticity

(d) Women, participation elasticity

Figure 8: Earnings and participation elasticities by own and partner income

Notes: The double-low corner is the lowest own-income bin with a low-earning partner. On the women's earnings map, its tile (0.63) passes the pre-event placebo and is exceeded only by one coloured tile (0.72), which fails its placebo. Sample: the child's fiscal exit sample of Appendix F.1, all event years. Outcome: work-income growth relative to $r = -1$ in Panels (a) and (b), an indicator for positive work income in Panels (c) and (d), pooled over $r = 0$ to 3; the estimates are elasticities with respect to the household net-of-marginal-tax rate, those for participation divided by participation two years before the event. Each tile pools the cell–dose elasticities of its own-by-partner-income rectangle, with the pooled standard error beneath and, for a dashed cell, the number of underlying individuals above. Standard errors are clustered by person, except those of the bracketed (screened) values, which are classical. Colour classes, the four display states and the flag rule are those of Appendix F.2: a coloured tile passes the screens; a dashed cell with a bracketed value fails the standard-error cap of 0.4 or the floor of 1,000 individuals; n.a. marks a rectangle with no cell passing the other screens; a blank rectangle has no cell; corner ticks and a grey value mark a failed pre-event placebo; the estimate is kept. A grey fill marks an estimate below −0.075. The €0–10,000 column pools the earner and entrant sub-samples (Appendix F.2). In the men's low and middle participation tiles the two sub-samples carry opposite signs (0.47 and 0.31 for earners against −0.37 and −0.26 for entrants), and the pooled 0.02 and −0.15 are within two standard errors of zero; neither tile is read. Both incomes are measured at $r = -1$; partner terciles are formed within each own-income bin, so a row is not at constant partner income. The grid ends at €60,000, the own-income range supported for both genders. Tercile boundaries, estimates, standard errors and cell counts for every tile are reported in Appendix Tables F.7 and F.8. Appendix Figures F.16 and F.17 redraw the maps along percentiles of own income, the second for the earnings maps.

in the event year.[48] Women's and men's bottom-bin participation elasticities are of similar size (0.33 and 0.29; Appendix Figure F.5) but come from different couples. Men respond in couples that paid no income tax before the event (0.95 against 0.10 in couples already taxed;

[48]Each value pools the tile's cell–dose–period estimates at $r = 0$, or at $r = 1$ to 3, with the screens and weights of Figure 8; the person-clustered standard errors are 0.10 and 0.07 in the event year and 0.14 and 0.09 over the three following years.

7 and 20 cell–dose–period estimates) and women in couples already taxed (0.73 against −0.01; 39 and 28). Men are on average two years older at the event (Appendix Table F.1), and the retirement margin matters more for them (Section 4.3). Women with a higher-earning partner adjust participation, while women in households where both earn little adjust earnings and largely remain in work. The double-low corner is, however, where the design has least support.

For men, the earnings map (Panel a) shows no clear partner-income gradient within own-income bins; the largest tile is 0.32, at €10,000–20,000 with a low-earning partner, and the €0–10,000 column is screened.[49] On the participation map (Panel c), responses are sizeable across most of the map. The largest (0.42) is in the top partner tercile of the bottom bin, whose two other tiles pool responses of opposite sign and are not interpreted (see the figure notes). In the bottom bin the two sub-samples measure different responses. The cells of the entrant sub-sample can record only entry into work, and those of the earner sub-sample mostly record exit from it. Over the bin as a whole, predicted-treated entrants enter work less often than their comparison groups, and predicted-treated earner men leave work more often. The entrant cells carry 75 percent of the weight of the men's bottom bin (Appendix F.7).

Appendix F.9 shows the equivalent income profiles for an alternative outcome definition: broad earnings, which add self-employment income to work income. With men and women pooled, the broad earnings elasticities track the earnings elasticities, shifted up by about 0.05 from €30,000 of own work income up (Appendix Figure F.7, Panel a).

Two checks concern the sample period and the event definition. Section 4.3 shows that the map survives dropping the 2020–2023 event years, which the headline specification includes (Appendix F.10). For the jump from the 14 to the 30 percent bracket, the marriage design (Appendix G) gives earnings elasticities similar to those at the child's fiscal exit for men and somewhat lower for women. For men, it gives 0.32 in the €10,000–20,000 bin and 0.16 in the €20,000–30,000 bin, against 0.32 and 0.17 at the child's fiscal exit (Appendix Figure F.4, all bracket changes). For women, it gives 0.14 and 0.18, against 0.18 and 0.30. Above €30,000 no bracket jump of the marriage design gives a number that is stable across weightings.

Four readings are consistent with the maps. (i) Partner income may act as insurance. A lower earner whose partner earns considerably more can afford to leave work. (ii) Hours may

[49] No men's earnings cell in that column is within the maps' standard-error cap of 0.4, so its three tiles show bracketed values. The single cell of the men's lowest bin in Appendix Figure F.4 passes the profiles' cap of 0.5.

be more adjustable at low pay, where part-time work is common, and earnings can then be adjusted without leaving work. (iii) The event arrives late in working life, and much of the response from €10,000 up comes from those approaching retirement ages. Below age 58 in the year before the event, the women's earnings profile shifts down but keeps its shape, and the men's profile lies below 0.10 from €10,000 to €60,000 (Appendix Figure F.9). For those approaching retirement, the timing of the exit from work may be the margin of response. (iv) Part of the gender gap may reflect the partner's adjustment transmitted within the household (Appendix B). Since both spouses' responses are estimated within the same cells, the pair of estimates is informative about such reallocation. Under all four, responsiveness depends on the household's circumstances as well as on preferences, in line with Attanasio et al. (2018), who find that women's labour-supply elasticities vary with age, wealth, hours and wages.

In sum, responsiveness is largest at low own incomes for both genders. The participation responses of the lowest-earning women are driven mostly by those with higher-earning partners, while their earnings responses are largest where both partners earn little. The heterogeneity around the aggregates is large; Section 5 examines its implications for tax design.

4.3 Robustness

The earnings map of Section 4.2.3 is robust to the estimation variants of Table 5; the exception is nearest-neighbour matching, which does not reproduce the men's map. The table re-estimates the map under each variant, on the same sample and grid, and reports the correlation of tile values with the headline map, the mean absolute tile difference and the income gradient. The balanced sample and the sample without the 2020–2023 event years reproduce the map, with correlations above 0.8 and gradients within 0.03 of the headline's. Matching in place of the regression comparison within cells (Section 3.5) reproduces only the women's map. For both genders, the matched gradient still has the headline's negative sign. PPML estimates measure a different object, the response of the level of work income with zeros included, and are reported with the aggregates as the total response (Section 4.2.1).

From €10,000 up, the responses come mostly from parents close to leaving work. Because the fiscal exit of a first-born typically comes as the parents approach retirement, Appendix Table F.13 repeats the estimates in restricted samples. Among individuals below 58 in the

Table 5: Specification agreement of the heterogeneity map

	Men				Women			
	Corr.	Rank corr.	mAD	Gradient	Corr.	Rank corr.	mAD	Gradient
Headline	1.00	1.00	0.00	-0.07 (0.03)	1.00	1.00	0.00	-0.07 (0.03)
Balanced sample	0.84	0.91	0.04	-0.06 (0.02)†	0.98	0.94	0.05	-0.10 (0.03)†
Nearest-neighbour matching	0.27	0.24	0.12	-0.03 (0.02)†	0.70	0.82	0.16	-0.09 (0.03)†
Without 2020–2023 event years	0.82	0.92	0.04	-0.06 (0.03)	0.98	0.97	0.05	-0.09 (0.04)

Notes: Each row re-estimates the earnings map of Figure 8 under one estimation variant, with the map's grid and display conventions (Appendix F.2; tiles flagged by the pre-event placebo included). Corr. and Rank corr. are the Pearson and Spearman correlations of tile values with the headline map, and mAD is the mean absolute tile difference, all on the tiles both maps keep (own-income bin × partner tercile). Gradient is the slope of tile values in own income per €10,000, each tile weighted by its estimation sample, with its standard error in parentheses, clustered by person; standard errors marked † treat cells as independent. From €10,000 up the women's correlations are 0.99 (balanced sample), 0.72 (matching) and 0.99 (without 2020–2023). The women's matching gradient of −0.09 comes from the bottom-bin tiles; from €10,000 up it is −0.02, against −0.04 for the headline. The first-stage screen uses the robust t-ratio, except in the balanced sample (classical t-ratio) and under matching (no first-stage screen).

year before the event, the aggregate earnings elasticity falls from 0.17 to 0.06 for men and from 0.23 to 0.10 for women (Appendix Figure F.9). Excluding instead the few percent who already draw some pension income in that year lowers it to 0.01 for men and 0.06 for women, and reduces the combined participation aggregate.

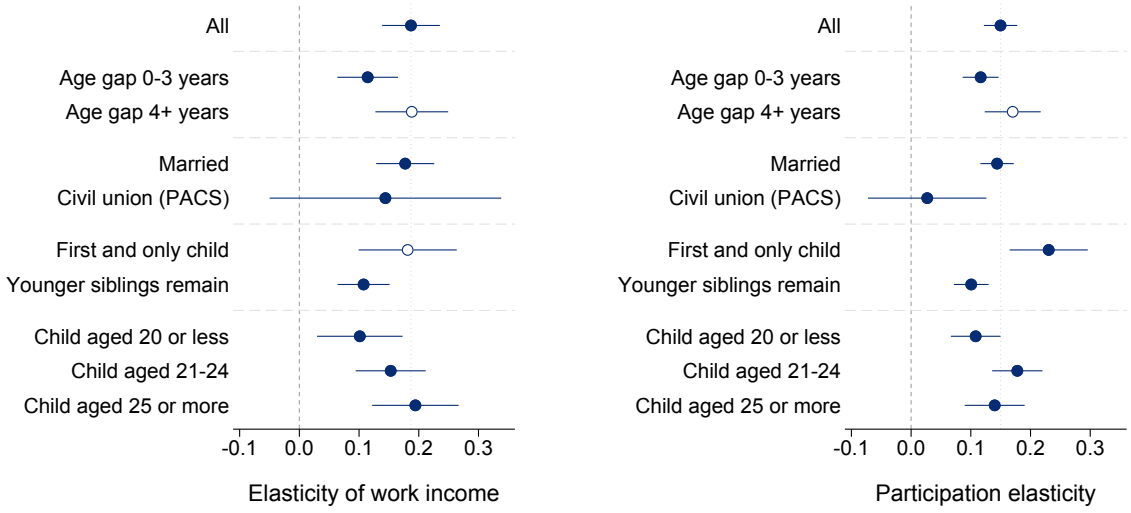
Weak instruments. The displayed estimates and their confidence intervals can be read without a weak-instrument correction. A display screen keeps the weakest cells out, and two checks show that the cells it admits need none. A small predicted-treated group can leave a cell's just-identified first stage weak. The display screen admits a cell to an exhibit only if its heteroskedasticity-robust first-stage F is at least 9 (Appendix F.2). With one instrument, this F is the effective F of Montiel Olea and Pflueger (2013). We use the robust statistic because the first-stage residual variance differs between the predicted treated and the zero-dose group; at the frequency-weighted median displayed cell it is about 0.8 times the classical F. Weak cells carry almost no weight. Those with F below 23.1—the Montiel Olea and Pflueger (2013) critical value for a worst-case bias of 10 percent—hold 0.05 percent of the earnings and 1.4 percent of the participation weight (Appendix Table F.9). Weak-instrument-robust inference agrees with the cells' robust Wald intervals. Robust Anderson–Rubin sets (Anderson & Rubin, 1949) are bounded in every displayed cell with a robust first stage, at the median 0.4 and 1.0 percent longer, and reject zero wherever the Wald intervals do (Appendix Table F.12). The tF correction of Lee et al. (2022) is applied to each cell and carried to the

bin standard errors as an approximation. With it, no income-profile bin loses its significance at 5 percent (Appendix Table F.11).

A stricter screen tests whether the aggregates lean on the weakest displayed cells. The stricter screen requires $F \geq 104.7$, above which the tF critical value is 1.96 and the two-stage least squares t-ratio needs no correction. Under it, every income-profile bin moves by less than one standard error except the two bins of women's participation at €10,000–30,000, which fall by 0.09 and 0.08, about two standard errors (Appendix Table F.10). Weak-instrument bias does not explain those falls. They are driven by seven cells of households at the 14 percent rate predicted to cross into the 30 percent bracket. These cells are not weak by the conventional criteria. Their robust F of 28 to 97 clears 23.1, and their Anderson–Rubin sets exclude zero in five of the seven. With 77 to 403 predicted treated, they fall short only of 104.7. The stricter screen changes the weighting. Under the frequency weights of Section 3.4 the seven cells hold 7 and 11 percent of the two bins' weight but 0.2 percent of their predicted treated, and with predicted-treated weights the same screen moves the bins by -0.01 and 0.00.

4.4 Heterogeneity

The design permits heterogeneity cuts beyond own income, partner income and gender. Figure 9 pools the cells of both genders with own income from €10,000 up and reports elasticities by household subgroup: the age gap between the parents, marriage versus civil union (PACS), whether the exiting first-born is the household's only child, and the age of the exiting child. Responses are larger when the age gap between the parents is four or more years (0.19 against 0.11 for earnings, 0.17 against 0.12 for participation) and when the exiting child is the first and only one (0.18 against 0.11, and 0.23 against 0.10 for participation). The second contrast is in line with the first fiscal exit also being the last for these households. Both contrasts come with a caveat. The four-or-more-years row fails its pooled pre-event placebo for earnings and participation, and the first-and-only-child row fails it for earnings (hollow markers). The age-gap contrast is the men's. Within men, the earnings elasticity is 0.26 with a gap of four or more years against 0.09 below it, and within women it is 0.12 against 0.16. The cut moves only the father's age at the event (53.4 against 51.2 years). The children contrast has the same sign for both genders and is largest for women's participation (0.30 against 0.07). By



(a) Earnings elasticity

(b) Participation elasticity

Figure 9: Elasticities by household subgroup

Notes: Each row pools the cells of one subgroup with own income from €10,000 up, both genders together, weighting each estimate by its regression sample (Section 3.4); the screens are those of Appendix F.2. Estimates are elasticities with respect to the household net-of-marginal-tax rate; those for participation are divided by participation two years before the event. Whiskers are 95 percent confidence intervals, with standard errors clustered by person within gender and the two genders treated as independent. A hollow marker means that the row fails its pre-event placebo at the 5 percent level, tested with the estimate's weights and covariance. A filled row can still hide one gender that fails its own placebo: women in the All and Married rows for earnings and in the Younger siblings remain row for participation (Appendix Figure F.11 tests each gender on its own). The All row (0.19 for earnings, 0.15 for participation) is not weighted to French taxpayers; the French-taxpayer aggregates from €10,000 up are 0.20 and 0.16. Subgroups are defined at $r = -1$, except the child's age, measured at the event; "First and only child" means that no other child is ever observed in the tax unit. The civil-union (PACS) row rests on 20 cells for earnings and 57 for participation (358 and 681 for All), and both its intervals include zero.

the age of the exiting child, households whose child is 20 or younger respond least (0.10 for earnings and 0.11 for participation, against 0.15–0.19 and 0.14–0.18 in the two older groups); above that age the ordering differs by gender. Appendix Figure F.11 reports the by-gender breakdown.

5 On the relevance of accounting for heterogeneity

Section 4 documented substantial heterogeneity in estimated elasticities, both across the income distribution and by the composition of income within the household. In this section, we examine whether this heterogeneity matters for tax design. The two exercises below use the estimated earnings and participation elasticities (both with respect to the household net-

of-marginal-tax rate), pooled over the event year and the three years after it ($r = 0$ to 3).[50] The income distributions they run on are calibrated from aggregate statistics, so the analyses remain illustrative. The elasticities are those of parents in couples at the fiscal exit of a child, and the exercises apply them to everyone in these distributions.

We assess this question against two benchmarks. The first is the normative optimum, the tax schedule that maximises a social welfare function subject to resource and incentive constraints (Mirrlees, 1971; Saez, 2001). For couples, finding it is a multidimensional screening problem (Golosov & Krasikov, 2025; H. J. Kleven et al., 2009). The second is the evaluation of reforms with the perturbation approach of Bierbrauer et al. (2024). This approach takes the observed tax system as given and derives sufficient statistics under which a move towards more individualised taxation of couples is Pareto- or welfare-improving. Section 5.1 conducts the first exercise using our estimates of tax elasticities by own income, and Section 5.2 conducts the second using our results on heterogeneity in tax responsiveness by the division of income within couples.

5.1 In optimal tax models

The optimality conditions of the Mirrlees–Saez framework are local, in the sense that the optimal marginal rate at earnings level z is governed by the elasticity *at* z (Mirrlees, 1971; Saez, 2001). In the framework, a government chooses a tax schedule to maximise a social welfare function. Because ability is unobserved, the schedule can condition only on earnings. Saez (2001) expresses the resulting optimality conditions in terms of sufficient statistics. The optimal marginal rate at z balances the mechanical revenue and welfare effects of a small rate change against the behavioural response of taxpayers at z. The formula does not require the elasticity to be common across the distribution, and the constant-elasticity calibrations used in practice are a convenience.[51]

[50] The pooled earnings elasticities of this section average over a response that is largest in the event year and statistically indistinguishable from zero from $r = 2$. The participation response persists for women and declines for men (Section 4.2.2).

[51] Following Brewer et al. (2010), the calibration infers a skill distribution from observed earnings under a flat reference tax rate and holds that distribution fixed while solving for earnings and taxes under each elasticity schedule. Each modelled type is assigned the fitted cross-sectional elasticity at its reference earnings, which is held fixed when its earnings change. The calibration is an illustrative approximation and does not estimate a complete structural labour-supply model. The profile is estimated at the rates of the existing schedule and could itself shift with rates if local responses depend on them.

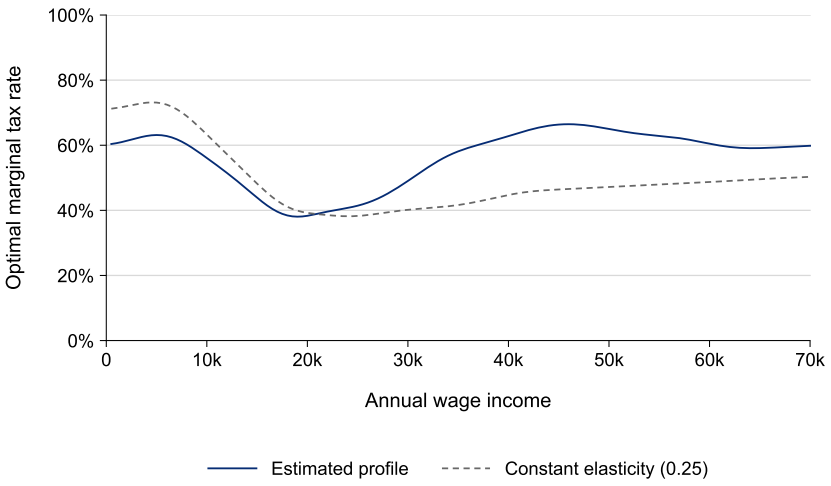


Figure 10: Optimal marginal tax rates under constant and heterogeneous elasticities: 2023 administrative work-income distribution

Notes: The figure plots an illustrative Mirrlees–Saez calculation following the skill-inversion approach of Brewer et al. (2010), with logarithmic social welfare; welfare weights and the revenue-balancing transfer are recomputed for each schedule. The earnings distribution is that of positive earners in the 2023 administrative data, reconstructed from gender-specific counts, participation shares and mean work incomes in nominal €10,000 bins, with men and women weighted equally. Nominal earnings and the revenue target are converted to 2015-price euros. The grey dashed line uses the paper's headline constant elasticity of 0.229. The navy line uses a positive, standard-error-weighted smooth fit to the gender-pooled own-income-bin elasticity estimates. The smoothing penalty is chosen by leave-one-bin-out prediction of the gender-pooled and the men's elasticity estimates. Tax rates are calculated only afterwards. Each modelled type retains the elasticity assigned at its reference earnings as earnings change. The full schedules are solved over the reconstructed distribution; the displayed rates are €500-bin medians smoothed by a symmetric €5,000 Hanning window. The figure is shown through €70,000; above the last estimated midpoint, the fitted elasticity is held fixed. Smoothing of the displayed rates does not enter the calculations or reported numerical results. Appendix Figure F.18 draws the schedules against the percentile of earners; Figure F.19 adds two non-increasing profiles.

We adapt the Mirrlees–Saez calculation of Brewer et al. (2010) to a skill distribution inferred from the 2023 administrative work-income aggregates and compare a constant elasticity with a fitted income profile.[52] Figure 10 provides an illustrative comparison. The grey dashed line assumes the paper's headline French-taxpayer-weighted elasticity of 0.229, and the navy line uses the fitted profile at each modelled type's reference income in 2015-price euros. The inferred skill distribution is held fixed, while earnings, welfare weights, and the revenue-balancing transfer are recomputed for each schedule. The estimated-profile rate is lower by about 10.9 percentage points on average below €10,000 in the unsmoothed solution; the transfer rises from €9,617 to €10,209. The fitted profile remains low around €40,000–€50,000, where the displayed estimated-profile rate peaks at about 66%. Above that range, the rate declines

[52]The fit gives imprecise bin estimates less weight and penalises changes in slope across adjacent income bins; its penalty is selected from elasticity prediction error, without using the resulting tax rates. Pre-trend-flagged estimates remain in the fit. At the two edge bins, the fit uses only the estimate that enters the headline aggregates.

gradually towards 60% near €70,000. Because the administrative export reports nominal-income bins, the within-bin shape and the division of their mass across 2015-price bins are fitted. The revenue requirement applies to the full fitted distribution, including its Pareto tail. The last estimated elasticity is carried forward from its €65,000 midpoint to incomes beyond the estimated €70,000 range. This extrapolation is an assumption of the calculation. The comparison illustrates how the income profile of responses changes the rates prescribed by the calculation. Even with less weight on imprecise bins, the fitted profile implies a higher mid-income rate.

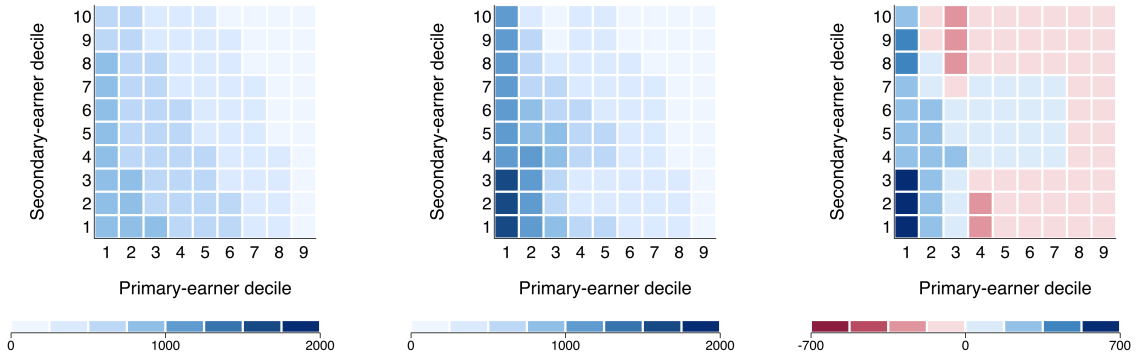
This exercise exploits only the variation of responses with own earnings. By construction, a one-dimensional formula does not include partner income and cannot absorb the variation of elasticities with the composition of household income. Using that dimension requires a framework for couples.

5.2 In the assessment of jointness reforms

For couples, solving for the optimal two-dimensional tax schedule is a multidimensional screening problem, which is analytically demanding even under strong simplifying assumptions (Golosov & Krasikov, 2025; H. J. Kleven et al., 2009). Under assumptions common in the perturbation literature, the optimal schedule cannot be recovered by perturbing the status quo (Bierbrauer et al., 2026). In particular, the analytical characterisation in Golosov and Krasikov (2025) relies on a constant elasticity that is common to all individuals and therefore symmetric across spouses. Its results do not carry over to elasticities that vary with the division of earnings within the couple. We therefore follow Bierbrauer et al. (2024), who evaluate reforms of the *existing* system. Bierbrauer et al. (2026) generalise the approach to multidimensional screening problems and in particular to the taxation of couples. Starting from status-quo couple taxation, they derive sufficient statistics for reforms towards more individualised taxation, which allow the marginal rates of the two spouses to differ and lower them on secondary relative to primary earnings in a revenue-neutral way. The statistics determine whether such a reform is Pareto-improving and otherwise whether it raises welfare under given welfare weights.

Their baseline calibration sets the elasticity of primary earnings to 0.25 and that of secondary earnings to 0.75. To isolate the role of our estimated heterogeneity, our constant-response benchmark instead uses the headline French-taxpayer-weighted earnings elasticity, $e = 0.293$, for each spouse and the corresponding participation elasticity with respect to the household net-of-marginal-tax rate, $\eta = 0.386$, for the household-level participation term. The estimated-response specification replaces these constants with the estimated surfaces over own income and the within-bin partner-income tercile, and pools the men's and women's estimates equally at each observed tile.

Figure 11 displays the welfare effects of local reforms towards a more individualised tax system under equal welfare weights. Panels (a) and (b) use the constant-response calibration and our estimated earnings and participation surfaces, respectively; Panel (c) reports their difference. Appendix Figure F.12 repeats the comparison with near-Rawlsian welfare weights.



(a) Constant responses (b) Estimated responses (c) Difference, (b) minus (a)

Figure 11: Welfare effects of local reforms towards more individualised taxation

Notes: Colours report the calibrated welfare effect of a one-percentage-point jointness-reducing perturbation integrated over a €500-by-€500 local income cell following Bierbrauer et al., 2024; the scales are in model welfare units under this normalization, not euro equivalents. Blue denotes gains; red in Panel (c) denotes that the estimated-response calibration produces a smaller gain than the constant-response calibration. Panel (a) holds the earnings elasticity at $e = 0.293$ for both spouses and the participation elasticity at $\eta = 0.386$; Panel (b) uses the own-income-by-partner-tercile tiles for earnings and participation; Panel (c) is Panel (b) minus Panel (a). Marginal rates apply the statutory 2019 French schedule (Section 2.1) to a two-part couple after the standard 10% wage-income deduction, before tax credits and child-related adjustments. Men and women are pooled equally within each tile; pre-trend-flagged aggregates are retained to avoid selection on pre-period noise. The men's €20,000–€30,000 middle-tercile earnings tile fails a disaggregated pre-trend test and is omitted before interpolation; the women's estimate is used at that position. Primary earners are the higher-earning spouse; secondary-earner deciles are computed conditionally within each primary-earner decile. A decile cell is displayed only when the full simulated income range of both spouses lies within the estimates' support, which omits the tenth primary-earner decile. The household participation term averages the two spouses' predicted participation elasticities. All panels use equal welfare weights.

Heterogeneity changes the size of the effects and where the gains are concentrated. With equal welfare weights, the unweighted mean across the 90 supported local reforms rises from 470 to

510 model welfare units, and the largest gain rises from 965 to 1,654 in the corner where both partners earn little. Under near-Rawlsian weights (Appendix Figure F.12), the mean becomes less negative, from −483 to −442. The large estimated responses where both partners earn little strengthen the local gain in that corner. Elsewhere the effects move in both directions, and with equal weights the welfare effect falls in 50 of the 90 supported cells.

The two exercises show that accounting for heterogeneity changes the shape of the optimal tax schedule and the welfare effects of reforms towards more individual taxation of couples.[53,54]

6 Conclusion

In a panel built from the universe of French tax returns,[55] this paper estimates the first profile of the elasticity of earnings with respect to taxation along the income distribution.[56] It declines with income, most steeply at the bottom, from 0.45 in the bottom decile of the income distribution to 0.22 in the middle deciles and to below 0.1 in the top decile. The decline holds for both genders, with women's earnings elasticity falling from 0.45 in the bottom decile to 0.31 in the top decile and men's from 0.32 in the deciles below the median to 0.19 in the top decile. Averaged over the French taxpayer distribution, the earnings elasticity is 0.23 and the participation elasticity is 0.25. Splitting the estimates by partner income shows who responds. Among the lowest-earning women, the participation elasticity is 0.69 in the middle partner-income tercile and 0.35 in the top, against 0.02 in the bottom. The earnings response arises in the year of the tax change and fades within two years, while the participation response is more persistent.

[53]The direction of the optimal primary–secondary differential is itself unsettled. With a common elasticity for all individuals, optimal marginal rates on secondary earners are on average *higher* than on primary earners, because low secondary earnings identify couples to whom a redistributive planner wants to target transfers (Golosov & Krasikov, 2025). The classic argument of Boskin and Sheshinski (1983) prescribes *lower* rates on the more responsive spouse. Which force dominates depends on the size and location of the elasticity gap between spouses—the object we estimate. Section 4.2.3 discusses possible sources of that gap.

[54]Heterogeneous elasticities may also change whether moves towards individualisation are politically feasible (Bierbrauer et al., 2025). We do not pursue this margin.

[55]The returns of 2003 to 2023, about forty million individuals a year, reshaped from a panel of tax units into an individual-level panel that follows each person across tax units. The code is open source, and the panel can serve other studies of French tax data.

[56]The earnings elasticity is the elasticity of own earnings (work income) with respect to the household net-of-tax rate, one minus the marginal tax rate the household faces at the individual's income.

This heterogeneity matters and existing estimates do not capture it (Figure 1). They are local to one threshold of the tax schedule or to the income bracket a reform affects, but they enter the forecasts of reforms, the design of optimal taxes and the calibration of macroeconomic models. Extrapolating from a local estimate to the rest of the distribution introduces a bias that grows with the degree of heterogeneity. Two exercises in Section 5 exemplify what the estimated profile of elasticities changes. Using it in place of a single elasticity lowers optimal marginal tax rates below €10,000 by about 14 percentage points on average. Using profiles over own and partner income in place of one elasticity per spouse changes the welfare gains of reforms that move couples from joint towards individual taxation.

The estimates use variation that joint taxation supplies throughout the income distribution. Under joint taxation with tax advantages for children, an individual's marginal tax rate may change at an event that changes the composition of the tax unit, such as a child's exit or a marriage. Whether the rate changes depends on partner income (of two women with the same earnings, only the one with a higher-earning partner may be pushed into the bracket above). The strategy formalises such comparisons within narrow cells of own income and partner-income terciles. Pre-event incomes predict who will be pushed into the bracket above (predicted treated) and who will not (predicted controls). The event need not be exogenous. The identifying assumption is instead that the earnings of the two groups would have continued to grow alike without the tax change. The assumption is tested by evaluating whether earnings growth differs between the groups before the event and by evaluating whether, where the event changes no rate, there is variation in earnings growth by partner income (Sections 3 and 4). Since exposure to the tax change depends on partner income, own income can be controlled for without absorbing the identifying variation. And controlling for own income alleviates two biases that threaten comparisons across income groups under a reform. Unusually high incomes tend to fall back towards the average (mean reversion). Growth rates also differ along the distribution as inequality changes.

Further research can use the strategy offered in this paper to study heterogeneity at other events and for other outcomes. A limitation of this application is that it is local to the child's exit. Parents are 51 to 53 on average at this event, and the response is shown to come partly from people approaching retirement. Marriage serves as a second design (Appendix G), and

the birth of a child could serve as a third. The same comparisons can be made for other outcomes the tax returns record and in other countries with joint taxation.

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A Tax liability by household composition

Figure A.1 plots the income tax liability under the 2015 schedule for the household compositions of Figure 2, Panel (a), and for a married couple with one child. The child-related advantage is capped, so beyond the income at which the cap binds the one- and two-child schedules run parallel to the childless couple's, at a constant distance equal to the cap.

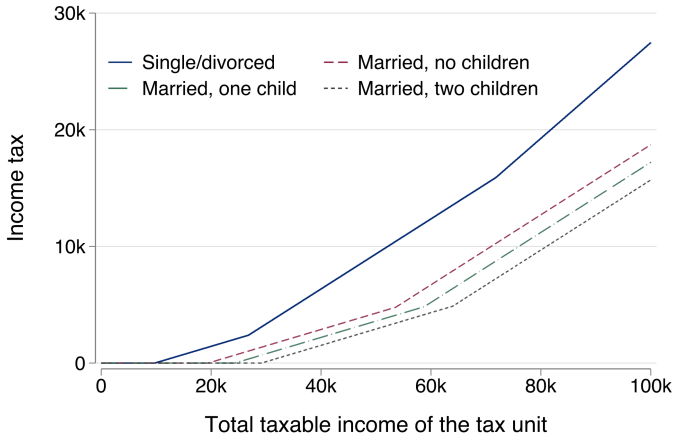


Figure A.1: Income tax liability by household composition

Notes: Statutory income tax liability under the 2015 schedule of Figure 2, by household composition and total taxable income of the tax unit, with the cap on child-related advantages applied. Source: Own calculations based on OpenFisca France.

B The household model and the estimand

This appendix relaxes the separability assumption of Section 3.1. It derives the household estimating equation under a general convex joint cost of earning, shows that joint filing collapses it back to equation (2) with the coefficient read as the elasticity of own income with respect to the household rate, and states which elasticities the two events identify (including for the marriage design, which switches the filing regime).

We model a household choosing both spouses' incomes against a joint budget constraint,

$$\max_{z_{it},\, z_{pt}} \; c_t - \Psi\big(z_{it}, z_{pt};\, n_{it}, n_{pt}\big) \qquad \text{s.t.} \qquad c_t \le z_{it} + z_{pt} - \mathcal{T}_t(z_{it}, z_{pt}),$$

where z_{pt} is the partner's income, $\mathcal{T}_t$ is the couple's tax function, and Ψ is a convex joint cost of earning income. The form of $\mathcal{T}_t$ depends on the filing regime (Section 2.1), and our design uses this dependence. The joint cost may be additively separable across spouses or non-separable. Non-separability can reflect leisure complementarities, home production, or coordinated retirement (Gelber, 2014; Goux et al., 2014).[57]

The potential incomes n_{it} and n_{pt} index each spouse's cost of earning income; we normalise them so that n_{it} is the income i would earn if both spouses faced marginal tax rates of zero, as in the separable case of Section 3.1.

Writing $\tau_{it} \equiv 1 - \partial\mathcal{T}_t/\partial z_{it}$ and $\tau_{pt} \equiv 1 - \partial\mathcal{T}_t/\partial z_{pt}$ for the spouses' marginal net-of-tax rates, the first-order conditions are $\partial\Psi/\partial z_{it} = \tau_{it}$ and $\partial\Psi/\partial z_{pt} = \tau_{pt}$. Totally differentiating the pair around the pre-event optimum and solving the resulting two-equation system yields the elasticities

$$\epsilon_{ii} \;=\; \frac{\tau_{it}}{z_{it}}\,\frac{\Psi_{pp}}{D}, \qquad \epsilon_{ip} \;=\; -\,\frac{\tau_{pt}}{z_{it}}\,\frac{\Psi_{ip}}{D}, \qquad D \;\equiv\; \Psi_{ii}\Psi_{pp} - \Psi_{ip}^2 \;>\; 0, \tag{B.1}$$

where subscripts on Ψ denote second derivatives with respect to the spouses' incomes, evaluated at the optimum, and $D > 0$ by strict convexity. Solving the system also yields the household estimating equation in changes,[58]

$$\Delta \ln z_{it} = \epsilon_{ii}\,\Delta \ln \tau_{it} + \epsilon_{ip}\,\Delta \ln \tau_{pt} + \Delta \ln n_{it}. \tag{B.2}$$

Every coefficient in (B.2) is an elasticity of i's *own* income, and the subscripts index whose rate varies: ϵ_{ii} is the response of i's income to i's own rate, ϵ_{ip} its response to the *partner's*

[57] Quasi-linearity in household consumption shuts down income effects at the household level, in the sense that neither the partner's net income nor the change in the household's tax *liability* affects income choices. It also makes the intra-household allocation of consumption irrelevant for those choices, because with transferable utility any sharing rule supported by lump-sum transfers leaves the first-order conditions unchanged. For the same reason the model's predictions for incomes coincide with those of an efficient collective household à la Chiappori (1992) with quasi-linear private utilities. We need not take a stance on the household decision process. Gelber (2014) rejects the income-pooling restriction of the unitary model. The rejection operates through income effects, which quasi-linearity shuts down, so the income predictions used here do not rest on the restriction. See Appendix D for the extension with income effects.

[58] For additively separable, iso-elastic Ψ the relation is exact and $\epsilon_{ip} = 0$, while in general the elasticities are local to the pre-event optimum. The error term is composite. With non-separable costs, changes in the *partner's* potential income also shift i's income through the same cross-derivative Ψ_{ip}, and $\Delta \ln n_{it}$ stands for the combined contribution of both spouses' potential-income changes to i's income. Assumption 1 requires that this component does not vary with the dose within comparison cells. The model does not deliver this restriction.

rate. As derivatives of the household's optimal income-supply function $z_i^*(\tau_{it}, \tau_{pt})$, both are jointly re-optimised responses in which the partner's income also adjusts. The cross-derivative Ψ_{ip} is the numerator of ϵ_{ip} and enters ϵ_{ii} through D. Even when i's rate alone changes, the partner's re-optimisation raises ϵ_{ii} above the one-person elasticity $\tau_{it}/(z_{it}\Psi_{ii})$ by the factor $1/(1-\rho^2)$, with $\rho \equiv \Psi_{ip}/\sqrt{\Psi_{ii}\Psi_{pp}}$. The split between ϵ_{ii} and ϵ_{ip} tracks whose rate changes, and the partner's re-optimisation enters both. Under quasi-linearity the cross-rate response is transmitted entirely through the partner's behavioural adjustment, and it is zero when effort costs are separable, $\Psi_{ip} = 0$.[59]

Under joint filing, the marginal rate is common to both spouses, $\tau_{it} = \tau_{pt} = \tau_{it}^H$ (Section 2.1), so any change in it is common as well, and equation (B.2) collapses to the estimating equation (2) of the main text,

$$\Delta \ln z_{it} = \underbrace{\left(\epsilon_{ii} + \epsilon_{ip}\right)}_{\equiv\, \epsilon_i^H} \Delta \ln \tau_{it}^H + \Delta \ln n_{it},$$

with the coefficient read as the *elasticity of own income with respect to the household rate*, $\epsilon_i^H \equiv \epsilon_{ii} + \epsilon_{ip}$. This elasticity includes the component transmitted through the partner's adjustment. From (B.1), $\epsilon_i^H = (\tau_{it}^H/z_{it})\,(\Psi_{pp} - \Psi_{ip})/D$. It nests the one-person elasticity at $\Psi_{ip} = 0$ and exceeds it under complementarity ($\Psi_{ip} < 0$) through both the cross term and the feedback.

Under non-separable costs, all three elasticities are couple-level parameters, because both spouses' cost curvatures enter each of them. They are local to the couple—the same individual would exhibit a different elasticity if single—and estimates from couples are not directly comparable to estimates on samples of singles. A gradient of responses in partner income at fixed own income can arise from four sources: sorting of individual elasticities across partner types, income effects (Appendix D), the non-separability studied here, or a common cost function whose local elasticity varies with the household rate, which differs across partner terciles at fixed own income. The design identifies the gradient but cannot separate these sources.

[59] Separability is the assumption underlying most quantitative work on couples taxation (Bick & Fuchs-Schündeln, 2018).

At the child's fiscal exit, the household files jointly before and after the event, so the event-induced rate change is common to both spouses, and the design recovers the elasticity of own income with respect to the household rate, ϵ_i^H.[60]

At marriage, by contrast, spouses are individually taxed before the event, so the two spouses' predicted changes generally differ (often only one spouse is predicted to see any change). Among couples for whom the event predicts a change in i's rate only, the event-induced variation moves $\Delta \ln \tau_{it}$ alone, and the reduced form identifies the own-rate elasticity ϵ_{ii}, which includes the partner's re-optimisation (the factor $1/(1-\rho^2)$ above). Among the same couples, the *partner's* income response to i's rate change identifies the cross-rate elasticity ϵ_{pi} directly—the design of Gelber (2014) reproduced within our setting. The current implementation does not condition on the partner's predicted change. Among treated individuals the partner (typically the higher earner) may simultaneously be predicted to cross a threshold downwards, in which case the reduced form mixes ϵ_{ii} with a term in ϵ_{ip}. The contamination is bounded by the cross-rate elasticity, which is small by the estimates cited below.

The gap between ϵ_i^H and ϵ_{ii} is also likely small. Exploiting the Swedish tax reform of 1990–1991, in which spouses' marginal rates moved independently, Gelber (2014) estimates compensated cross-rate elasticities of 0.05 for husbands and 0.07 for wives, an order of magnitude below his own-rate estimates of 0.41 and 0.47. Because these cross-rate elasticities are positive, ϵ_i^H weakly overstates ϵ_{ii}. Where the distinction matters, the marriage design can in principle provide internal estimates, and external ones can be imputed instead: $\epsilon_{ii} = \epsilon_i^H - \hat{\epsilon}_{ip}$.[61]

[60] The income-weighted average of ϵ_i^H across spouses gives a different object, the response of the couple's *combined* income to the household rate. Revenue and Laffer calculations need this response, whichever way it splits across own- and cross-rate channels.

[61] In more detail, two routes lead from the identified total to the structural own-rate elasticity. **Route 1** (*spousal separability*): with $\Psi_{ip} = 0$, $\epsilon_{ip} = 0$ and $\epsilon_i^H = \epsilon_{ii}$. In the iso-elastic form, the household problem separates into two copies of the one-person model that each face the household rate. The usual practice of pairing the one-person model with household-rate variation rests on this separation. **Route 2** (*external adjustment*): impute $\hat{\epsilon}_{ip}$ from Gelber (2014) and report $\epsilon_{ii} = \epsilon_i^H - \hat{\epsilon}_{ip}$ as an adjustment or as bounds. Because our estimates decline with income, the implied adjustment is modest relative to them at low own incomes and less so at higher ones. No adjustment is applied here. The imputation matches the model's object. Because the model is quasi-linear, ϵ_{ip} is a compensated cross-rate elasticity, as are Gelber's estimates. The uncompensated added-worker channel operates through income effects, which are absent here and treated in Appendix D. This channel is opposite-signed and potentially large after major income shocks (Blundell et al., 2016). For bracket-switch variation, the associated liability changes are small relative to lifetime resources (for the predicted treated, 0.9 percent of household taxable income, about €400–€600 a year), and the channel is of second order. Because Gelber's estimates remain gender-specific and come from a different population and a different tax change, we treat them as orders of magnitude and would apply the adjustment to the earnings elasticities only. The distinction matters most for the heterogeneity results. Gender differences in ϵ_i^H are individual parameters only under Route 1 or after the Route 2 adjustment. Without either, part of a

C Proof of Proposition 1

Proof. By (4), (5) and Assumption 2, $\Delta y_{ir} = \Delta_r Y_i(0) + \mathbb{1}[Z_{ir} \neq 0]\,\epsilon_i \Phi_{ir}$ and $\Delta \mathrm{NTR}_{ir} = \Delta_r \mathrm{NTR}_i(0) + \mathbb{1}[Z_{ir} \neq 0]\,\Phi_{ir}$. By Assumption 1, the conditional means of the untreated components are equal for the $Z_i = \delta$ and $Z_i = 0$ groups of the same cell, so they cancel in the contrasts: $RF_c^r(\delta) = \mathbb{E}_c[\epsilon_i \Phi_{ir} \mid Z_i = \delta]$ and $FS_c^r(\delta) = \mathbb{E}_c[\Phi_{ir} \mid Z_i = \delta]$. Taking the ratio gives the weighted-average representation, with weights averaging to one by construction; under Assumption 3 all Φ_{ir} share the sign of δ, hence the sign of their conditional mean, and the weights are non-negative. □

D Model extensions

D.1 Income effects

Relaxing quasi-linearity, we replace linear consumption with a concave utility $v(c_t)$: the household problem becomes

$$\max_{z_{it},\, z_{pt}} \; v(c_t) - \Psi\big(z_{it}, z_{pt};\, n_{it}, n_{pt}\big) \qquad \text{s.t.} \qquad c_t \leq z_{it} + z_{pt} - T_t^H(z_{it} + z_{pt};\, Q_{it}) - \kappa_{it},$$

where κ_{it} collects lump-sum household expenses, which may change at the event (expenses for the child). The first-order conditions become

$$\frac{\partial \Psi}{\partial z_{jt}} \;=\; v'(c_t)\,\tau_{jt}^H, \qquad j \in \{i, p\}:$$

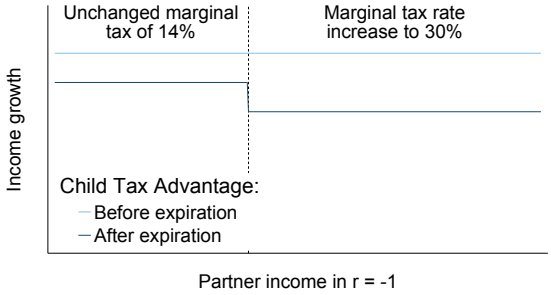
income choices now depend on the marginal utility of household consumption and through it on partner income, the level of the tax *liability* and needs κ_{it}. Log-linearising adds a virtual-income term to equation (2), as in Gruber and Saez (2002). Two remarks follow for our design. First, households predicted to cross a bracket at the event also experience a larger

measured gender gap can mechanically reflect the other spouse's adjustment transmitted within the household. Symmetry of Ψ restricts the cross responses in levels, and in elasticities it implies $\epsilon_{ip}/\epsilon_{pi} = \tau_{pt} z_{pt}/(\tau_{it} z_{it})$, the income ratio under a common rate, so the lower earner's cross-rate elasticity is the larger one (consistent with Gelber's larger estimate for wives). The Route 2 adjustment removes this part of the gap. Because we estimate both spouses' responses to the same household shock within the same cells, the pair of estimates is informative about the transmission within the household, and we return to this when interpreting the heterogeneity results (Section 4.2.3).

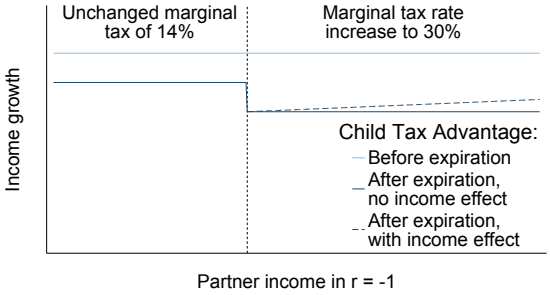
increase in tax liability than their within-cell controls. Absent a virtual-income instrument, our estimates should be read as total responses to the event-induced tax change. The two channels oppose each other—the higher marginal rate reduces income, while the larger liability raises it when leisure is normal—so under income effects our estimates are lower bounds on the compensated elasticity. The end of this subsection reports an attempt to separate the two responses. Second, the needs change is part of the event, not of the schedule change. The child leaves and the expenses change under both schedules (Section 3.3), so the income consequence of the needs change belongs to untreated growth and is restricted by Assumption 1 rather than Assumption 2. The classification matters. With concave v, even a needs change common to every household of a cell moves income through $v'(c_t)$, and consumption varies with partner income within the cell's partner-income tercile, so income effects can convert a common event shock into a partner-income *gradient* in growth. This is the class of violation the validation region detects.[62] Figure D.1 plots income growth against partner income on either side of the threshold, without income effects in Panel (a) and with an income effect that grows with the distance above the threshold in Panels (b) and (c), for a linear and a progressive schedule. Panel (d) replaces the flat growth of Panels (a) to (c) with a constant partner-income gradient in income growth that the event itself does not change.

We tried to separate the two responses within the design by adding the mechanical change in the household's tax liability at the event to the cell regressions as a second regressor. For the predicted treated this change is 0.9 percent of household taxable income on average, about €400–€600 a year. Within a cell it moves largely with the instrument: the predicted rate change is the same for every predicted-treated household of the cell, and the liability change differs from it only through the household's distance above the threshold, which at given own income is mostly partner income. The liability term absorbs about half of the instrument's variation within cells, and in the work-income regressions the unweighted median robust first-stage F over all estimated cells falls from 27 to 6.5 for women and from 23 to 5.0 for men. The coefficient on the liability term is negative: pooled over $r = 0$ to 3, -0.93 (0.20) for men and -0.43 (0.18) for women, for whom it is close to zero at $r = 0$ and negative afterwards (classical standard errors, which treat the event years as independent). Before the event it is close to

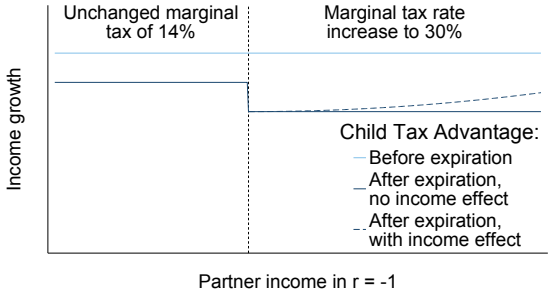
[62] Appendix H discusses income effects as one reason why first-stage monotonicity can fail in reform-based simulated-IV designs and explains why this does not happen in ours.



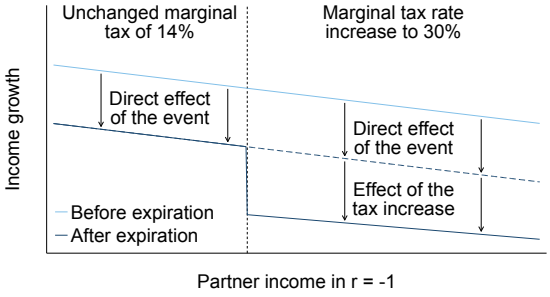
(a) Child's fiscal exit, linear schedule, no income effects



(b) Child's fiscal exit, linear schedule, income effects



(c) Child's fiscal exit, progressive schedule



(d) Child's fiscal exit, constant partner-income gradient

Figure D.1: Income effects

Notes: Stylised income growth against partner income in $r = -1$ for one cell, before and after the event. Above the vertical dashed line the event moves the household into a higher bracket, while below it the marginal rate is unchanged. Panel (a) has no income effects. Panels (b) and (c) add an income effect that grows with the distance above the threshold (dashed curve), under a linear and a progressive schedule. Panel (d) replaces the flat growth of Panels (a) to (c) with income growth that falls linearly with partner income, a constant gradient that the event itself does not change. The dashed curve of Panel (d) is the before-event curve shifted by the event's direct effect alone, and the arrows split the fall above the threshold into that effect and the effect of the tax increase, which includes an income effect.

zero, and on participation it is not distinguishable from zero under the frequency weights of Section 3.4. A negative coefficient is the opposite of an income effect with normal leisure: among the predicted treated, households further above the threshold (whose partners earn more) reduce their income more. We read it as a partner-income gradient of the substitution response. The design does not separate the two channels.

D.2 The participation decision

The model of Section 3.1 cannot generate participation responses. Returning to the quasi-linear case and following H. J. Kleven et al. (2009), consider a spouse who decides whether

to work at all and faces a fixed cost of participation q_{it} (commuting, childcare, forgone home production). Conditional on working, income is chosen optimally, so the value of participation is

$$V_{it}^{w} \;=\; \max_{z} \Big\{ z \;-\; \big[\, T_t^H(z + z_{pt};\, Q_{it}) - T_t^H(z_{pt};\, Q_{it}) \,\big] \;-\; \psi(z;\, n_{it}) \Big\},$$

where the bracketed term is the *participation tax* (Immervoll et al., 2007): the increase in the household's tax liability caused by the spouse's income, given partner income and the household's tax schedule. The spouse works if and only if

$$V_{it}^{w} \;\geq\; q_{it}.$$

The relevant price of participation is the participation tax wedge, which differs from the marginal rate at current income. Partner income and the quotient determine its first-euro marginal rate, $1 - \tau^H$ evaluated at $z = 0$. Because partner income alone places the household above the tax-free threshold, the second earner's first euro is taxed at the household's current bracket. A household pushed into a higher bracket by the child's fiscal exit thus sees the return to participation fall, and with heterogeneous fixed costs q_{it} the participation probability falls with the participation tax—the sign of the participation responses we document. The estimates themselves are not expressed in this price. The implementation instruments the same object for earnings and participation, the event-induced change in the household *marginal* net-of-tax rate evaluated at current own income, whereas the wedge above is governed by the schedule between z_{pt} and $z_{it} + z_{pt}$ and its first-euro rate by partner income and the quotient alone. Within cells both rise for the predicted treated, but the participation tax far less relative to the controls, as the ratio below shows. We read the participation estimates (divided by pre-event participation) as responses to the marginal-rate change (Section 3.5).

To express them in the participation-tax price we compute the participation tax rate implied by the effective schedule at pre-event incomes, $[T_t^H(z_{pt}+z_{it}) - T_t^H(z_{pt})]/z_{it}$, for every household of the estimation sample and under the pre-event and the post-event quotient. We then multiply each cell estimate by the ratio of the cell's change in the log net-of-marginal-tax rate to its change in the log net-of-participation-tax rate. Both changes are contrasts between the treated and the control households. The contrast in the participation tax is small because the control households' participation tax also rises: losing the half part raises the inframarginal tax on

the spouse's earnings even where the marginal rate at household income is unchanged (for women earning €10,000–€20,000, the control households' participation tax rate rises from 11.2 to 13.9 percent when the half part is lost, and the treated households' from 6.8 to 11.3 percent). The ratio averages 8.0 over the estimation cells (6.6 for women, 9.3 for men) and rises with own income, from about 2 in the bottom own-income bin to between 9 and 15 above €30,000. In the French-taxpayer aggregate, the participation elasticities with respect to the net-of-participation-tax rate are about 3.4 times those of the main text. Their pattern across income also differs. The bottom bin no longer stands out, and the rescaled profiles are largest and least precise in the middle and upper bins (Appendix F.7). The rescaling takes the realised pass-through of the mechanical change to be the same for both prices; a participation tax rate evaluated after the event at the pre-event own wage (with the household's realised taxable base and quotient) gives a ratio of 3.9 on the cell-periods for which it can be formed.

Read as a participation elasticity with respect to the net-of-participation-tax rate, the rescaled estimate assumes more than the main design does. The zero-dose group is not untreated in this price, so the cell contrast compares two groups whose participation tax both rise. It recovers the response per unit of the participation-tax change only if that response is on average the same in both groups (which Proposition 1 does not need) and only if the schedule change affects participation through the participation tax alone. Replacing the regressor of equation (6) by the realised change in the log net-of-participation-tax rate (instrumented by the dose as now) would put a realised first stage in place of the equal pass-through but would rest on the same conditions. Its confidence intervals would be wider by about the factor above, because the reduced form is unchanged and the contrast about eight times smaller.

The variation the participation literature uses (a change in the participation tax with no change in the marginal rate) exists within our cells. Take a cell of the 14 percent bracket. A zero-dose household whose partner income alone lies below the first threshold at its new position sees the spouse's participation tax rise by the exit's liability change on the income between that threshold's old and new position, divided by her earnings. A zero-dose household whose partner income alone lies above the first threshold at its old position sees the same change fall on partner income; her participation tax is unchanged. The household's marginal rate is unchanged in both cases. Within cells, participation growth compared across these two groups is a response to the participation tax alone, at the same event date and with the event's other

effects common to both, as in the main design. It rests on Assumption 1 in its continuous form, that untreated participation growth has no gradient in partner income across the zero-dose group, and the two groups' participation paths before the event test it as in Figure 3. As in the participation literature, the participation tax of individuals without earnings is evaluated at a reference level of own earnings, a last observed or a predicted work income.

E Further case-study exhibits

This appendix collects the remaining exhibits of the example cell of Section 3.3: the men's validation-region figure (Figure E.1), the participation counterpart (Figure E.2), the first stage, the second-stage estimates by year relative to the event (Figure E.3) and the density test (Figure E.4).

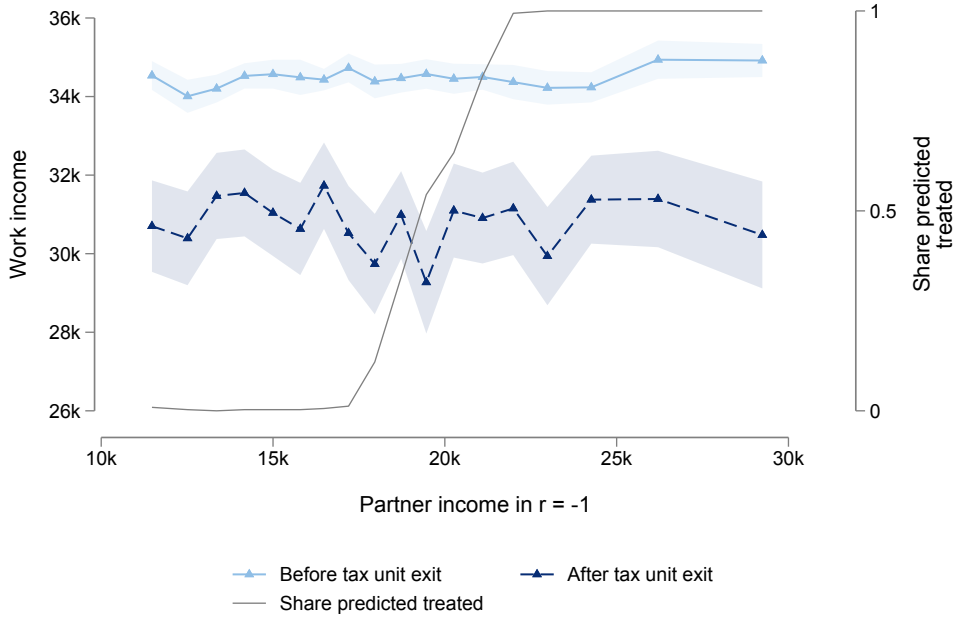
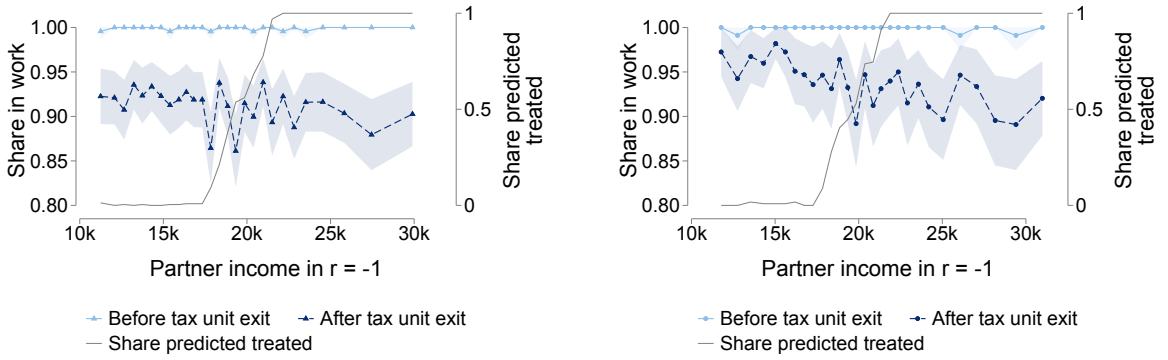


Figure E.1: Identifying variation at the first-born's fiscal exit: men

Notes: Male counterpart of Figure 4: men of the example, same construction and horizons, with partner income in 20 equal-count bins of about 345 men (the two lowest bins have mean partner income below €10,000 and lie outside the plotted range). The validation region reaches up to about €17,200 and the treatment region starts at about €22,000.

The predicted-control group's realised marginal rate is a direct check of the mapping of Section 3.3: between $r = -1$ and $r = 0$ its household marginal rate rises by 1.9 percentage points on average for women and 1.7 for men, with a median change of zero, against 10.5 and



(a) Men

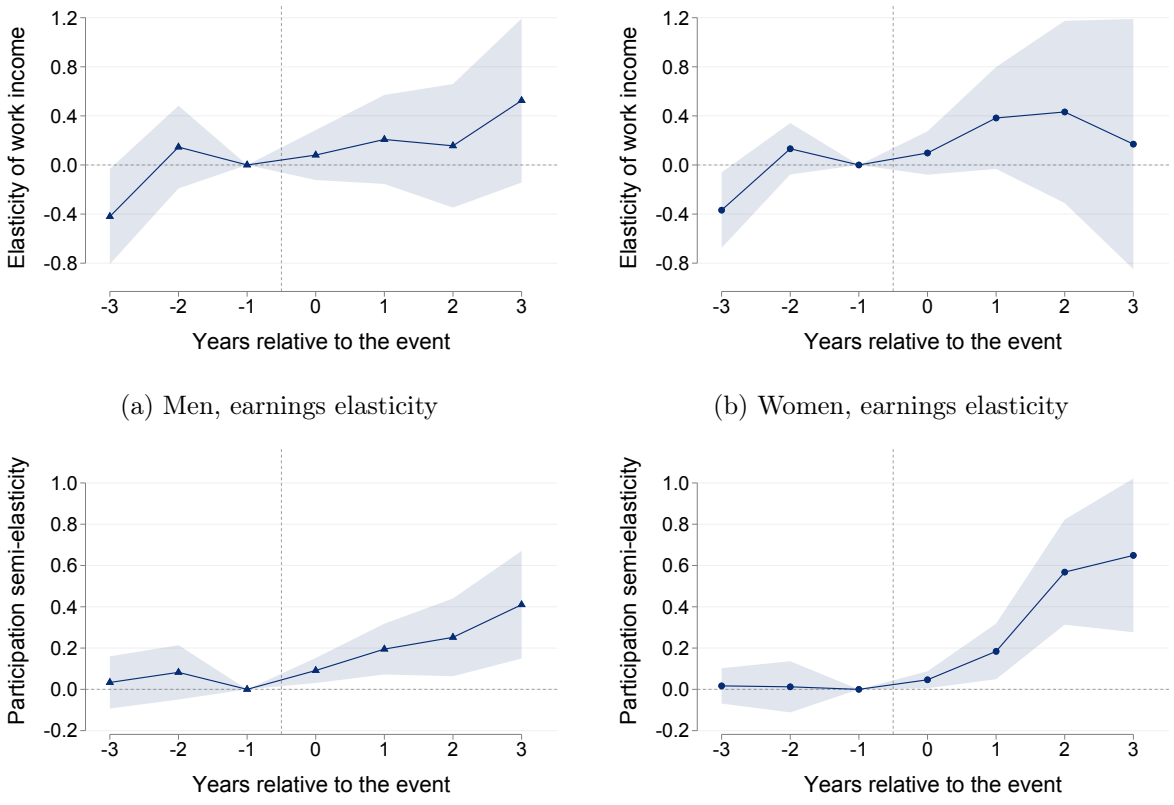
(b) Women

Figure E.2: Identifying variation at the first-born's fiscal exit: participation

Notes: Participation counterpart of Figures 4 and E.1: the outcome is an indicator for positive work income. Left axis: the share of the bin in work at $r = -2$ (before) and the average of $r = +1$ and $r = +2$ (after), in 30 equal-count bins of partner income with about 115 women and 230 men per bin. The share at $r = -2$ is close to one by construction, since the cell requires positive work income at $r = -1$. Bins with mean partner income below €11,000 are not shown. Bands are 95% confidence intervals, capped at one. Right axis and sample as in Figure 4.

10.0 points for the predicted treated, whose predicted change is 16 points. Per unit of the predicted change in the log net-of-tax rate, the realised change is 0.53 (s.e. 0.01) for women and 0.52 (0.01) for men at $r = 0$, and declines to 0.38 and 0.40 by $r = 3$ as incomes move across brackets.

Translating the earnings paths of Figure 3 into elasticities with the estimation approach of Section 3 yields the second-stage estimates shown in Figure E.3.



(a) Men, earnings elasticity

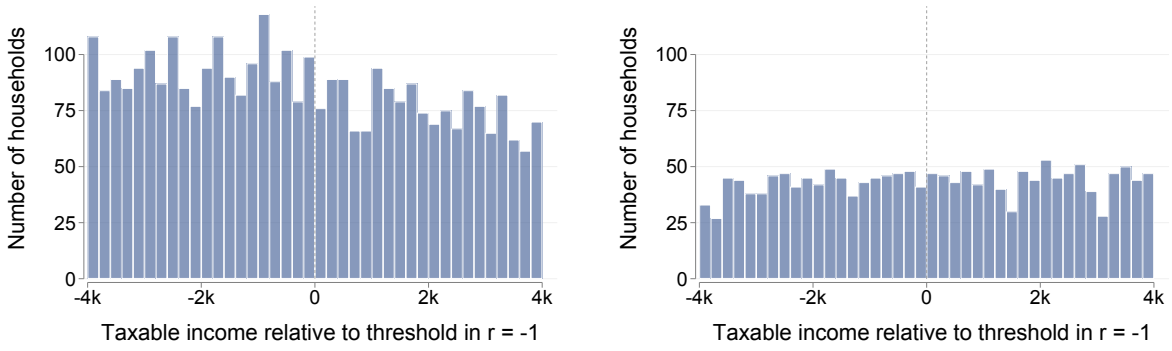
(b) Women, earnings elasticity

(c) Men, participation semi-elasticity

(d) Women, participation semi-elasticity

Figure E.3: Second-stage estimates for the case-study cell

Notes: Two-stage least squares estimates for the example cell of Figure 3, with the estimator of the headline analysis (Section 3), by year r relative to the first-born's fiscal exit ($r = -1$ is the reference); shaded areas are 95% confidence intervals from classical standard errors. The upper panels report the elasticity of own earnings with respect to the household net-of-tax rate and the lower panels the participation semi-elasticity (the change in the share in work per unit change in the log household net-of-marginal-tax rate). At $r = +1$ and $r = +2$ the earnings elasticities are 0.38 (s.e. 0.21) and 0.43 (0.38) for women and 0.21 (0.19) and 0.16 (0.26) for men. The participation semi-elasticities at $r = +1$, $+2$ and $+3$ are 0.18 (0.07), 0.57 (0.13) and 0.65 (0.19) for women and 0.20 (0.06), 0.25 (0.10) and 0.41 (0.13) for men. The pre-event earnings estimates are 0.13 (0.11) for women and 0.15 (0.17) for men at $r = -2$, and -0.37 (0.16) and -0.42 (0.20) at $r = -3$. They rest on a pre-event first stage of 0.08 and 0.06 at $r = -2$ and of 0.12 and 0.09 at $r = -3$, against 0.38 to 0.53 after the event.



(a) Men (b) Women

Figure E.4: McCrary density test at the hypothetical joint-income threshold

Notes: Number of households in €200 bins of the distance of pre-event joint income to the threshold at which the marginal tax rate rises from 14% to 30% once the first-born's exit reduces the family quotient. The panels cover the example cell of Figure 3 within €4,000 of the threshold. Households above the threshold are predicted treated and households below it are predicted controls. If households anticipated the exit by bunching below the future threshold or by timing it, the pre-event density would show excess mass around zero. The McCrary test does not reject continuity: the estimated log-density discontinuity is −0.12 (s.e. 0.13, $p = 0.35$, 3,388 households) for men and 0.04 (s.e. 0.17, $p = 0.79$, 1,735 households) for women.

F Further results

F.1 The event samples

Cleaning of the raw panel drops households in the overseas departments and French residents abroad (1.29 million records in 2023), returns without any individual identifier (107,000) and individuals aged 75 and over (7.4 million), and resolves the individual-year duplicates that multiple declarations create (1.39 million in 2023). Table F.1 describes the two event samples, and the figure notes of Section 4 refer to it. Panel A traces their construction. The child's fiscal exit sample comprises individuals in married or civil-union tax units (i) whose first-born leaves the tax unit between 2004 and 2023 (1,932,140 women and 1,959,441 men); (ii) whose civil status is unchanged at the event, so that the fall in the quotient is the only change to the household's schedule (the restriction removes 0.9 percent of the women and 1.2 percent of the men); and (iii) whose own work income at $r = -1$ lies below €70,000 in 2015 prices (the cap removes a further 1.6 percent of the women and 9.2 percent of the men). In nominal terms, 95.6 percent of men and 97.0 percent of women filing a return in 2023 earn less than €70,000. The second estimation floor below (300 outcomes at $r = -3$) is counted on work income at $r = -3$, which is observed for 87 percent of the sample. The marriage sample comprises first

Table F.1: The two event samples

	Child's fiscal exit		Marriage	
	Women	Men	Women	Men
Panel A: construction (individuals)				
Event observed (2004–2023)	1,932,140	1,959,441	2,295,222	2,297,699
Marriage: predicted rate increase	–	–	575,517	203,268
Civil status unchanged; marriage: design screens[a]	1,914,489	1,936,085	309,067	121,554
Own work income at $r = -1$ below €70,000	1,884,452	1,758,070	308,483	121,154
Positive work income at $r = -1$	1,497,580	1,490,879	269,731	97,744
Panel B: characteristics of the event sample at $r = -1$				
Annual event cohort size (mean)	95,724	96,804	89,249[b]	79,209[b]
Age at $r = 0$ (mean)	51.09	53.28	33.61[c]	35.49[c]
Partner's age at $r = 0$ (mean)	53.39	50.94	35.65[c]	33.52[c]
Married at $r = 0$ (share)	0.98	0.98	0.53[c]	0.55[c]
Dependent children at $r = -1$ (mean)	1.92	1.90	0.27[c]	0.35[c]
Own work income (€, mean)	18,126	27,021	19,160	22,500
Partner's income (€, mean)	42,481	21,632	27,239	22,363
Positive work income (share)	0.79	0.85	0.93	0.93
Predicted into a rate change (share)	0.30	0.32	0.25	0.11
Mean |predicted $\Delta \ln(1 - \text{MTR})$| if treated	0.12	0.12	0.12[c]	0.12[c]

Notes: Panel A counts individuals at each step of the sample construction described in the text. The fourth row is the event sample used throughout and the fifth is the subsample on which the earnings outcome is defined. The marriage columns follow the spouses for whom the wedding predicts a rate increase. The design compares them with the zero-dose spouses of the same cells (941,102 women and 1,004,916 men below the income cap). The marriage estimation sample includes both groups and holds 1,249,585 women and 1,126,070 men. The child's fiscal exit counts of the fourth row exceed the grid totals of Table 3 by the 4 women and 14 men whose own work income at $r = -1$ is exactly €70,000. The grid's bins exclude this value. Panel B reports descriptive statistics for the event samples below the income cap, measured in the pre-event year ($r = -1$) unless otherwise indicated. The marriage columns describe the estimation sample, which includes the predicted-increase and the zero-dose spouses. Partner's income sums work, self-employment, and pension income. Counts respect the data provider's disclosure rules. [a] Both spouses childless and separately taxed in the year before the wedding and filing jointly the year after. [b] Mean over the 2005 to 2017 cohorts; later cohorts are much smaller. [c] All first marriages below the income cap (2,282,283 women and 2,250,589 men), the marriage design's sample before its screens.

marriages of spouses who were childless and separately taxed in the year before the wedding and file jointly the year after. Within cells, the design compares the spouses for whom the wedding predicts a rate increase (Section 3.2) with the zero-dose spouses. Each sample is used for two outcomes. The earnings outcome is log work-income growth and requires positive work income at $r = -1$ and in the outcome year. The participation outcome is an indicator for work income above €1,000 (2015 prices), and its sample also includes individuals without work income at $r = -1$. Two years before the event, 33 percent of the women and 22 percent of the men in the €0–10,000 comparisons are in work, against 97 percent or more in every bin from €10,000 up. In the €0–10,000 bin the comparison cells are formed separately for individuals with at least €400 of work income at $r = -1$ (the earner sub-sample) and for

those below (the entrant sub-sample; Appendix F.2). At $r = -1$, 61 percent of the predicted-treated women and 85 percent of the predicted-treated men in the bin's comparisons have less than €400 of work income, and 36 and 14 percent are in work. Comparison cells enter estimation if (i) at least 30 predicted-treated individuals carry the dose within the cell at $r = -1$ and (ii) the dose's estimation sample (predicted treated and zero-dose comparison group together) holds at least 300 earnings outcomes at $r = -3$. The same floors apply to the participation regressions of the dose, and displayed estimates are further screened under the display conventions of Appendix F.2.

Panel B characterises the samples in the pre-event year. Parents at the first-born's fiscal exit are on average 51 to 53 years old and almost all married; 85 percent of the women and 82 percent of the men are aged 45 to 59 at the event. The exiting first-born is aged 19 to 26 in 86 percent of households (modal age 23), 18 or younger in 3 percent and 27 or older in 10 percent. The age is known for 92 percent of households and is an upper bound where an older child left the tax unit before 2003. Partner incomes differ by gender. Women's partners earn on average €42,481 and men's €21,632, against own work income of €18,126 and €27,021. Between 79 and 85 percent have positive work income at $r = -1$. About 30 percent are predicted into a rate change, with a mean predicted change in the log net-of-tax rate of 0.12 among them. The marriage sample is younger (34 to 36 years old at the event) and has fewer dependent children and higher participation. A quarter of the women and a tenth of the men are predicted a rate increase at the wedding.

F.2 Display conventions for the cell-level exhibits

The headline exhibits of Section 4 share a two-layer display convention. It applies to the two-stage least squares cell–dose estimates of Section 3.5 (equation (6)), which are pooled over the post-event years $r = 0$ to 3 with the frequency weights of Section 3.4. Each cell–dose–period estimate is weighted by its regression sample, which includes both the predicted treated and the zero-dose comparison group. Under these weights a cell whose comparison group is large relative to its predicted treated counts for more than its share of the treated. This over-weighting is largest in the cells of households at the 14 percent rate predicted to cross into the 30 percent bracket, in which a few hundred predicted treated share regression samples of

up to about 51,000 (Section 3.4 and Appendix Table F.10). As in Section 3.5, standard errors and pre-event placebo tests are person-clustered within a comparison group, and comparison groups and genders are treated as independent. The balanced-sample, matching and marriage exhibits and the bracketed map values instead assume independence across cells.

The lowest own-income bin. In the €0–10,000 bin the comparison cells are formed separately for individuals with at least €400 of work income at $r = -1$ (the earner sub-sample) and for those below (the entrant sub-sample), and each sub-sample cuts its own partner-income terciles. In the earner sub-sample the terciles are cut at €24,700 and €38,300 of partner income for women and at €13,800 and €26,200 for men, and in the entrant sub-sample at €27,900 and €51,700 for women and at €11,600 and €26,400 for men. The bin's aggregates pool the cell–dose–period estimates of both sub-samples with the frequency weights and screens of every other bin, and a map tile of the bin pools the two sub-samples' cells of the same tercile rank.

Cell screens. The first layer decides which cell–dose–period estimates enter a displayed aggregate. An estimate enters if (i) its standard error is positive and at most 1 on the outcome scale and at most 0.5 on the elasticity scale in the income profiles, the event-time and subgroup exhibits and the aggregation table, and at most 0.4 in the maps (where a tile shows no confidence interval);[63] and (ii) its heteroskedasticity-robust first-stage F is at least 9, a t-ratio of 3 (Appendix F.11).[64] An aggregate that fails the pre-event placebo is flagged and kept (*Pre-trend annotation* below).

Maps. The second layer decides how an aggregate is shown. Every rectangle of the own-income-by-partner-tercile grid is drawn in one of four states over the income range supported for both genders (own income up to €60,000). A rectangle is filled if at least one of its cells passes the cell screens, the standard error pooled over these cells is at most 0.4 and at least 1,000 individuals underlie them, predicted treated and comparison group together.[65]

[63]The outcome scale is that of the coefficient. For earnings the coefficient is the elasticity, while for participation the elasticity divides the coefficient by pre-event participation (a share of at most one), and the elasticity-scale caps imply the outcome-scale bound of 1. Estimates above 5 in absolute value are dropped as outliers (the elasticity screen).

[64]The classical first-stage t-ratio is used for the balanced-sample and marriage estimates, the level column of Appendix Table F.5 and one headline participation cell and dose (displayed at $r = 0$, and in Appendix Figure F.3 as a pre-event point at $r = -2$); the matching estimates are not screened on the first stage.

[65]Individuals are counted as each dose's largest regression sample over $r = 0$ to 3, summed over doses, so a comparison group shared by several doses is counted once per dose. The coverage of a bin in the aggregation table uses the same count. The cap uses the pooled standard error that treats the rectangle's cells

A rectangle is n.a. when none of its cells passes the elasticity, outcome-scale standard-error and first-stage screens. A rectangle is drawn dashed when its cells pass those screens but none of them is within the elasticity-scale cap, with the bracketed value pooled over these cells. A rectangle with cells within the cap is drawn dashed in the same way when their pooled standard error exceeds 0.4 or fewer than 1,000 individuals underlie them. From top to bottom, a dashed cell prints the number of individuals behind it, the bracketed value and the standard error. An n.a. cell prints the number behind all its cells. A rectangle with no cell is blank.

Profiles, event-time and subgroup exhibits. These exhibits apply the cell screens (the event-time exhibits to their pre-event points as well) and no aggregate-level screen; an imprecise bin appears with a wide confidence interval. In the income profiles the vertical axis of a figure is set by the bins with at least two cell–dose–period estimates, and a bin resting on a single estimate is drawn in grey with its interval cut at the axis; the figure note gives the full interval.

The aggregation table. Table 4 adds one bin-level screen: a bin contributes only if the cells passing the cell screens cover at least 10 percent of the bin's estimation sample and at least 1,000 individuals underlie them. The first condition is one of representativeness and the second one of absolute size; neither is a precision rule, since the cells have passed the standard-error screens. Coverage matters for the table because the table multiplies a bin's estimate by the population share of the bin. The men's €0–10,000 bin carries 0.34 of the men's French-taxpayer weight, but its one earnings cell that passes the screens holds 4,421 of the bin's 77,460 individuals (5.7 percent). Coverage removes that bin for earnings, and the floor removes the women's €60,000–70,000 bin for earnings and participation (865 and 885 individuals, with full coverage). The maps use the floor and the pooled cap in place of the bin-level screen, and a bin shown in the income profiles may enter no aggregate.

Colour. The fill of a tile is a function of its printed estimate alone, with class boundaries at ±0.075, 0.15, 0.225, 0.3, 0.4, 0.55, 0.8, 1.2 and 1.7. Estimates between −0.075 and 0.075 share the lightest tint, and estimates below −0.075 are grey. Precision is shown by the standard error printed beneath every value. Every map uses these classes, including those of the marriage design (Appendix G).

as independent. Among the filled tiles, the person-clustered standard errors printed on the maps are at most 0.26.

Pre-trend annotation. For every displayed aggregate, the pre-event placebo reduced forms of its cells at $r = -3$ and $r = -2$ (Section 3.5) are pooled with the aggregate's own frequency weights—each placebo estimate weighted by its regression sample, as in the aggregate it annotates—and the pooled contrast is tested with the person-clustered covariance of the reduced forms. Aggregates with $|t| \geq 1.96$ are marked with four corner ticks and a grey value (maps), a dagger (tables) or a hollow marker (profiles and subgroup exhibits). The flag only marks the aggregate, and its value and its place on the colour scale stay as they are. The marriage exhibits use the same weights, with cells treated as independent. The event-time profiles carry no flag because their pre-event points are themselves placebo estimates.

F.3 Balance across predicted groups

Table F.2 summarises the balance of predetermined characteristics in the pooled validation of Section 4. It gives means for the predicted-control and predicted-treated groups and the standardised mean difference averaged across comparisons. The two groups are closely comparable in the outcomes before the event (income growth and participation differ by less than 0.05 of a standard deviation). Differences of 0.2 to 0.3 appear in own and partner income levels and in age: within an own-income bin, the predicted treated are those whose partner earns more and whose household income is higher. Own income, partner income and age are correlates of household taxable income, which determines assignment. The largest differences are in household taxable income itself and in family size, which shifts the number of fiscal parts and with it the predicted rate change. In the displayed profile bins, the two arms differ in pension receipt at low incomes. Between €10,000 and €30,000, 7 to 16 percent of the predicted treated in the earnings cells drew a pension at $r = -1$, against 1 to 4 percent of their comparison groups (Table F.3).

Table F.2: Balance across predicted-treated and predicted-control comparisons

	Women			Men		
	Control	Treated	SMD	Control	Treated	SMD
Pre-event outcomes						
Work income, $r = -1$ (€)	16,558	17,167	0.23	27,116	27,710	0.23
Log work-income growth, $r = -3$ rel. $r = -1$ (%)	−2.35	0.68	0.04	11.24	14.28	0.02
Positive work income, $r = -1$ (%)	79.4	81.0	0.03	89.4	89.0	−0.01
Composition						
Partner work income, $r = -1$ (€)	25,783	28,382	0.24	16,085	17,679	0.30
Partner has positive work income, $r = -1$ (%)	84.6	87.2	0.04	77.3	82.3	0.10
Age	49.5	50.6	0.25	51.8	52.8	0.22
Partner age	52.1	52.9	0.17	49.4	50.6	0.28
Number of dependent children	2.24	1.97	−0.47	2.29	1.95	−0.56
Fiscal parts, $r = -1$	3.37	3.15	−0.45	3.41	3.13	−0.55
Assignment and first stage (differ by design)						
Household taxable income, $r = -1$ (€)	39,584	45,112	1.06	39,596	44,416	1.04
Predicted change in log net-of-tax rate (×100)	0.00	−11.42	–	0.00	−11.30	–
Realised net-of-tax-rate change, $r = +1$ (pp)	−1.36	−5.56	–	−1.49	−5.59	–
Sample						
Unique individuals	842,950	534,851		836,555	512,991	
Comparison-year cells	964			1,215		

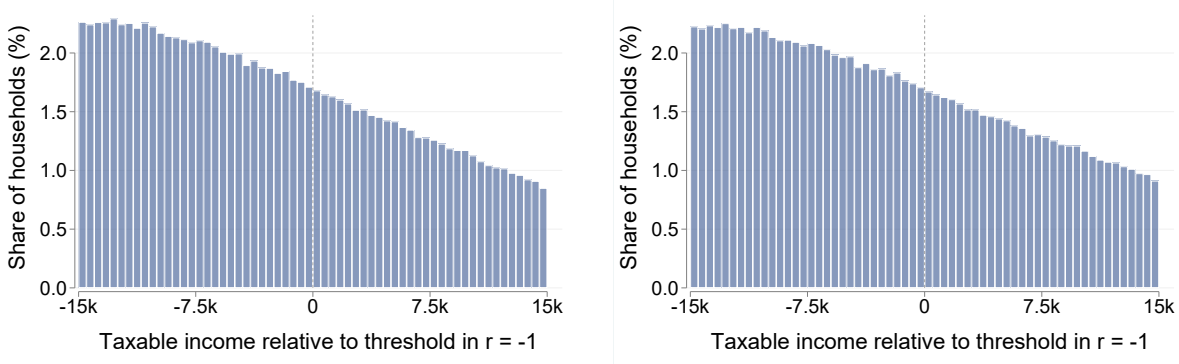
Notes: Each treated tax-change dose is compared with the zero-change controls in the same own-income bin, partner-income tercile, pre-event bracket, and event year. Means are computed within each comparison-year arm and averaged across comparison-year cells with the cell's predicted-treated count as weight; SMD is the mean of the signed within-cell standardised differences with the same weights. Household taxable income determines assignment and differs between arms by construction (standardised difference 1.06 for women, 1.04 for men). The predicted rate change is zero in the control arm by construction, but controls whose income drifts across a threshold realise a change. No standardised difference is reported for either row. The log-income-growth row requires positive work income at $r = -1$ and at $r = -3$. Controls may enter several dose comparisons; the counts report unique individuals. Levels are in nominal euros of the event year. Sample: the child's fiscal exit sample of Appendix F.1, in the comparison cells of the design. The complete table is available on request. It reports all balance variables, the 90th percentile of absolute standardised differences and per-row comparison counts.

Table F.3: Composition of the two arms by own-income bin

	Earnings cells				Participation cells			
	Draws a pension		Age		Draws a pension		In work	
Own income	Treated	Control	Treated	Control	Treated	Control	Treated	Control
Women								
0–10k	0.15	0.06	51.3	49.3	0.23	0.11	0.78	0.76
10–20k	0.08	0.02	50.0	48.7	0.11	0.02	1.00	1.00
20–30k	0.07	0.01	51.3	49.8	0.08	0.01	1.00	1.00
30–40k	0.02	0.01	52.3	50.9	0.02	0.01	1.00	1.00
40–50k	0.02	0.01	52.3	51.5	0.02	0.01	1.00	1.00
50–60k	0.02	0.01	52.0	52.4	0.02	0.01	1.00	1.00
60–70k[a]	0.02	–	52.5	51.9	0.02	–	1.00	1.00
Men								
0–10k[a]	0.62	0.49	58.6	56.7	0.63	0.61	0.22	0.23
10–20k	0.16	0.04	52.7	51.4	0.17	0.05	1.00	1.00
20–30k	0.13	0.02	52.2	50.9	0.13	0.02	1.00	1.00
30–40k	0.12	0.02	53.2	51.5	0.10	0.02	1.00	1.00
40–50k	0.06	0.01	53.3	51.9	0.05	0.01	1.00	1.00
50–60k	0.03	0.01	53.3	52.2	0.04	0.01	1.00	1.00
60–70k	0.02	0.01	53.6	52.3	0.02	0.01	1.00	1.00

Notes: Predicted-treated and comparison arms at $r = -1$ behind the displayed profile bins of Figures F.4 and F.5. Each cell–dose–period estimate is weighted by its regression sample, as in the bins; 'in work' is work income above €1,000, which holds by construction from €10,000 up, and – marks a share resting on fewer than 11 individuals.
[a] The earnings entries rest on a single cell–dose–period estimate (the grey bin of Figure F.4).

F.4 Pooled density tests at the threshold



(a) Men (b) Women

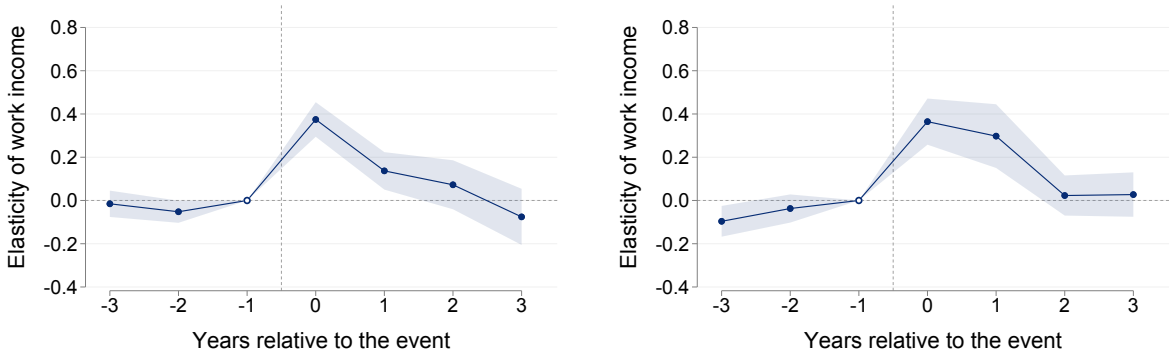
Figure F.1: Density tests at the hypothetical threshold, pooled across cells

Notes: Distribution of household taxable income at $r = -1$, measured relative to the hypothetical threshold at which the household would enter the 30 percent bracket after the child's exit (at its post-exit number of fiscal parts) and pooled across the comparison cells of the design. The vertical axis is the share of households in each €500 bin, in percent. Figure E.4 shows counts of households instead. The estimated log-density discontinuities are −0.002 (s.e. 0.010, $p = 0.80$, $N = 782{,}811$) for women and −0.001 (s.e. 0.009, $p = 0.92$, $N = 776{,}649$) for men. They give no evidence of sorting of the event's timing or of bunching below the threshold.

Figure F.1 shows the density of household taxable income around the hypothetical threshold, pooled across the comparison cells; its note gives the estimated discontinuities.

F.5 Event-time profiles

The event-time profiles of the estimates serve as validity checks, since the same construction yields placebo estimates for the pre-event years (Figures F.2 and F.3). The earnings response is an impulse concentrated in the event year, and for women also the year after, while the participation response is persistent throughout the window for women and declines for men.

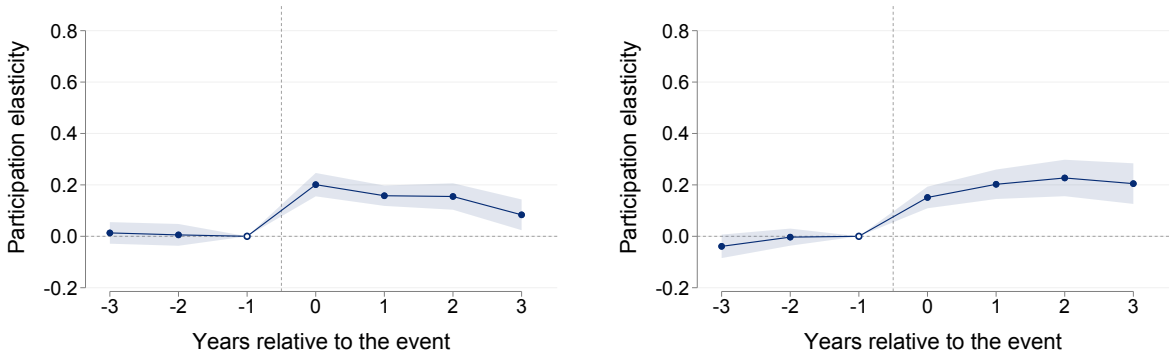


(a) Men

(b) Women

Figure F.2: Event-time earnings elasticities around the child's fiscal exit

Notes: Sample: the child's fiscal exit sample of Appendix F.1 (all event years). The outcome is work-income growth relative to $r = -1$. Each point pools the cell–dose elasticities of one year relative to the event across all own-income bins and weights each by its regression sample. Because the aggregates of Table 4 weight the bins by their population shares, the points do not average to them. The €0–10,000 bin holds 13 percent of the women's weight at $r = 0$ (3 to 4 percent at $r = 2$ and 3). For men, it holds a single cell at $r = 0$. Shaded areas are 95% confidence intervals from person-clustered standard errors. Because each point pools a single year, its standard error involves no covariance across years. The year before the event, $r = -1$, is the reference and is shown at zero (hollow marker: a reference point, not a failed placebo); estimates for $r = -3$ and $r = -2$ are placebo checks. They are scaled up relative to the post-event estimates. Each divides the placebo reduced form by that year's pre-event contrast in realised rates, 0.10 to 0.13 against 0.43 to 0.45 in the event year. On the event-year scale, the women's placebo at $r = -3$ is -0.03 (8 percent of the event-year response), and the men's placebo at $r = -2$ is -0.02. The placebo reduced form is pooled over all pre-event cells with the weights of the estimates and tested with the person-clustered covariance. The test does not reject for men ($t = 0.53$) and is borderline for women ($t = -1.9588$, $p = 0.050$; $t = -2.30$ if the pre-event cells are weighted by inverse variance instead). Only the cell screens of the income profiles apply (Appendix F.2). The participation counterpart is Figure F.3.



(a) Men

(b) Women

Figure F.3: Event-time participation elasticities around the child's fiscal exit

Notes: As Figure F.2, for participation: the outcome is an indicator for positive work income and the displayed estimates are participation elasticities. The cell coefficients are divided by participation two years before the event. The €0–10,000 bin holds 21 to 25 percent of the women's weight and 5 to 7 percent of the men's in each post-event year. The pooled pre-event placebo reduced form does not reject for either gender ($t = -1.08$ for women, 1.32 for men).

F.6 Aggregated elasticities

Weighted to the population of French taxpayers, the average participation elasticity is 0.247 (0.283 for women, 0.212 for men); across the four weighting schemes the average lies between 0.11 and 0.25. As with earnings, women's participation elasticities exceed men's under every scheme of Table F.4 except inverse-variance weighting. From €10,000 of own income up (Panel B of Table F.4), the French-taxpayer aggregate is 0.162 (0.178 for women, 0.147 for men). In this range it is an exit elasticity of established workers, whose participation two years before the event is 0.99.

F.7 Profiles along own income by gender

This subsection reports the gender breakdowns of the pooled earnings and participation profiles of Section 4.2.2 (Figures 6 and 7).

For participation, the outcome is an indicator for work income above €1,000 (2015 prices); bin estimates are participation elasticities (the change in the participation probability per unit change in the log household net-of-marginal-tax rate, divided by participation two years before the event). They are constructed as described in Section 3 with the samples, pooling window ($r = 0$ to 3) and screens of the earnings profiles. Because the earnings outcome requires only

Table F.4: Aggregated participation elasticities

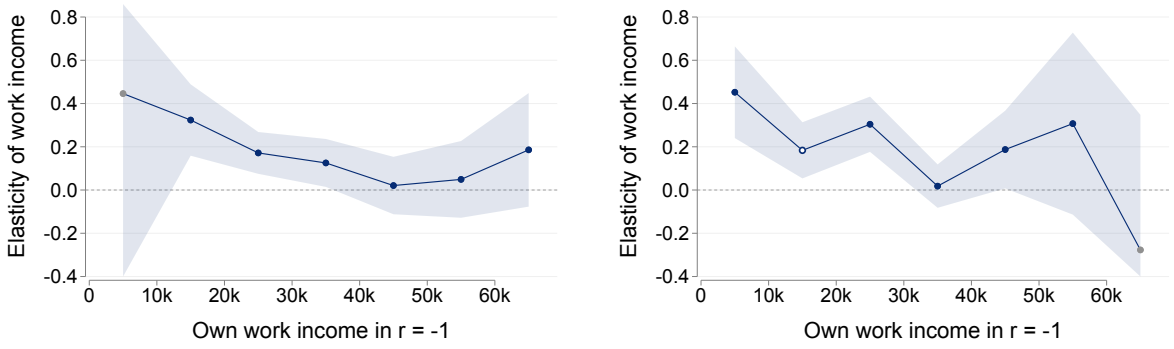
A. All own-income bins

	Men	Women	Total
Population-weighted: French taxpayers	0.212 (0.039)	0.283 (0.029)	0.247 (0.024)
Population-weighted: event population	0.205 (0.033)	0.230 (0.033)	0.218 (0.023)
Inverse-variance weighted	0.127 (0.012)	0.096 (0.011)	0.111 (0.008)
Income-weighted	0.145 (0.016)	0.177 (0.022)	0.158 (0.013)

B. Own work income at $r = -1$ of €10,000 or more

	Men	Women	Total
Population-weighted: French taxpayers	0.147 (0.017)	0.178 (0.024)	0.162 (0.015)
Population-weighted: event population	0.147 (0.018)	0.174 (0.027)	0.158 (0.015)
Inverse-variance weighted	0.124 (0.012)	0.086 (0.011)	0.105 (0.008)
Income-weighted	0.135 (0.015)	0.159 (0.023)	0.145 (0.013)

Notes: As Table 4, for participation: the outcome is an indicator for positive work income, and the entries are participation elasticities with respect to the household net-of-marginal-tax rate. The cell coefficients are divided by participation two years before the event (Section 3.5). With respect to the net-of-participation-tax rate (Appendix D.2) the French-taxpayer aggregate is about 3.4 times larger (0.84 against 0.25; Table F.6). Panel B repeats the aggregation over the own-income bins from €10,000 up.



(a) Men (b) Women

Figure F.4: Earnings elasticities along the own-income distribution, by gender

Notes: As Figure 6, with the cell–dose–period estimates aggregated within each gender. A hollow marker flags a bin that fails its pooled pre-event placebo ($|t| \geq 1.96$; here only the women's €10,000–20,000 bin); flagged bins are kept (Appendix F.2). The vertical axis is set by the bins with at least two cell–dose–period estimates and is shared with Figure 6. Bins resting on a single estimate are drawn in grey with their interval cut at the axis: the men's €0–10,000 bin, 0.45 with interval −0.50 to 1.39, and the women's €60,000–70,000 bin, −0.28 with interval −0.90 to 0.35. Both pass the cell screens and are shown, but neither enters an aggregate; in Figure 6 they enter their bins with weights of 1 and 2 percent. The men's cell covers 5.7 percent of its bin's estimation sample, below the 10 percent the aggregation requires, and the women's rests on 865 individuals, below the aggregation's floor of 1,000 (Appendix F.2). The men's estimate has a robust first-stage F of 615 and a classical standard error of 0.49, the value the cap of 0.5 is applied to; its placebo estimates at $r = -3$ and -2 are 0.00 and −0.03, and the same comparison gives 0.22, 0.27 and 0.33 at $r = 1$ to 3. Figures F.13 and F.15 redraw the panels along percentiles of own income.

positive work income, a fall in work income to between €1 and €1,000 counts as an exit

Table F.5: Earnings, participation and total responses by own income

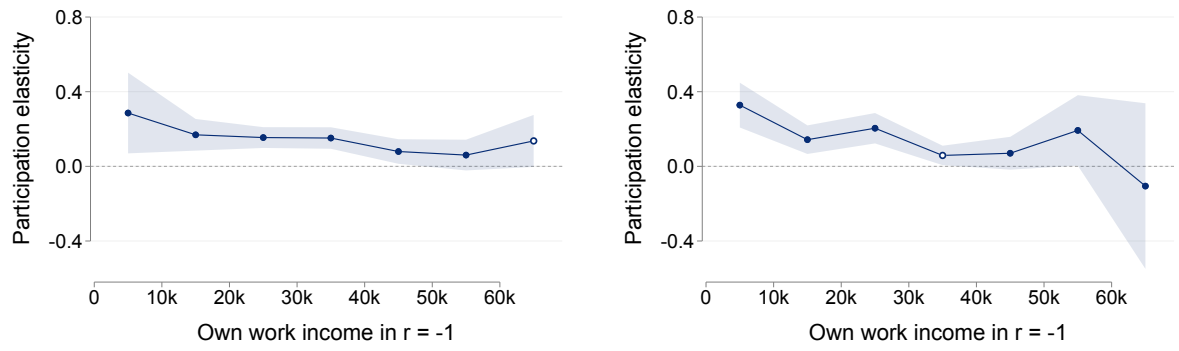
Own work income at $r = -1$	Earnings	Participation	Sum	Level (PPML)
Women				
0–10k	0.45 (0.11)	0.33 (0.06)	0.78 (0.12)	0.08 (0.10)
10–20k	0.18† (0.07)	0.14 (0.04)	0.33 (0.08)	0.13 (0.04)
20–30k	0.30 (0.07)	0.20 (0.04)	0.51 (0.08)	0.29† (0.05)
30–40k	0.02 (0.05)	0.06† (0.03)	0.08 (0.06)	0.02 (0.04)
40–50k	0.19 (0.09)	0.07 (0.04)	0.26 (0.10)	0.16 (0.07)
50–60k	0.31 (0.21)	0.19 (0.10)	0.50 (0.24)	0.27 (0.14)
60–70k[a]	-0.28 (0.32)	-0.11 (0.23)	-0.38 (0.39)	0.08 (0.24)
Population-weighted, 10k and up	0.23 (0.05)	0.18 (0.02)	0.41 (0.05)	0.20 (0.03)
Men				
0–10k[a]	0.45 (0.48)	0.29 (0.11)	0.73 (0.50)	2.40† (0.39)
10–20k	0.32 (0.08)	0.17 (0.04)	0.49 (0.09)	0.31 (0.07)
20–30k	0.17 (0.05)	0.15 (0.03)	0.33 (0.06)	0.25 (0.04)
30–40k	0.13 (0.06)	0.15 (0.03)	0.28 (0.06)	0.18 (0.04)
40–50k	0.02 (0.07)	0.08 (0.03)	0.10 (0.08)	0.10 (0.05)
50–60k	0.05 (0.09)	0.06 (0.04)	0.11 (0.10)	0.07 (0.06)
60–70k	0.19 (0.13)	0.14† (0.07)	0.32 (0.15)	0.18 (0.11)
Population-weighted, 10k and up	0.17 (0.03)	0.15 (0.02)	0.32 (0.04)	0.21 (0.03)

Notes: By gender and own-income bin at $r = -1$, pooled over $r = 0$ to 3: the earnings elasticity of Appendix Figure F.4, conditional on positive work income in the year before the event and in the outcome year; the participation elasticity of Appendix Figure F.5; their sum; and the total (level) response of Section 4.2.1, the reduced form of each cell estimated in levels, including zeros, by Poisson pseudo-maximum likelihood (PPML) and divided by the cell's first stage. The population-weighted rows apply the French-taxpayer weights of Table 4 to the bins from €10,000 up. † The entry's pooled pre-event placebo fails ($|t| \geq 1.96$). [a] The earnings entry rests on a single cell–dose–period estimate (for the men's €0–10,000 bin also the Level entry); neither bin enters the population-weighted rows. Standard errors in parentheses are clustered by person; in the Level column each cell's first-stage sampling variance is added as if the cells' first stages were independent, and the standard error of the Sum treats the earnings and participation estimates as independent, although they are estimated on overlapping samples of the same people. Cells enter the Level column if their classical first-stage t-statistic is at least 3, and the other columns if their robust one is. Three of the 278 women's Level cells have a negative first stage, all at doses that predict a rate cut (Section 3.2); dropping them changes the women's Level response from €10,000 up from 0.203 to 0.204 (standard error 0.026 in both cases).

in the participation outcome and as a large negative log change in earnings. Figure F.6 and Table F.6 show the same bins rescaled by the participation tax rate (Appendix D.2).

One difference from the earnings profiles concerns the men's €0–10,000 bin. A single cell in this bin passes the screens for earnings, but the bin carries the men's largest participation elasticity (0.29, against 0.33 for women). The bin pools the earner and entrant sub-samples (Appendix F.2); the cells of the entrant sub-sample can record only entry into work, and those of the earner sub-sample record mostly exits. In the raw data, predicted-treated entrants enter work at 2.4 to 4.7 percent a year (men) and 3.0 to 4.0 percent (women) from $r = 0$ to 3, against 3.2 to 4.6 and 4.4 to 7.6 percent in their comparison groups. In the same years, 41 to 61 percent of predicted-treated earner men leave work, against 33 to 58 percent (gaps

of 3 to 12 percentage points). The women's bottom-bin value of 0.33 (67 cell–dose–period estimates, 360,546 individuals) pools 60 estimates of the earner sub-sample (0.17) and seven of the entrant sub-sample (1.11), all seven at one dose (the move from the 14 to the 30 percent bracket); without them the bin is 0.17, and leaving out one cell–dose at a time gives 0.20 to 0.40. The men's value of 0.29 (27 estimates, 130,117 individuals) pools 16 estimates of the earner sub-sample (0.38) and 11 of the entrant sub-sample (0.26), which carry 75 percent of the bin's weight; leaving out one cell–dose at a time gives 0.19 to 0.50. Within the earner sub-sample, the split by the couple's tax status before the event (Section 4.2.3) is the same as in the bin as a whole. Men respond in couples that paid no income tax (0.74 against 0.04 in couples already taxed; 6 and 10 estimates), and women respond in couples already taxed (0.50 against −0.01).

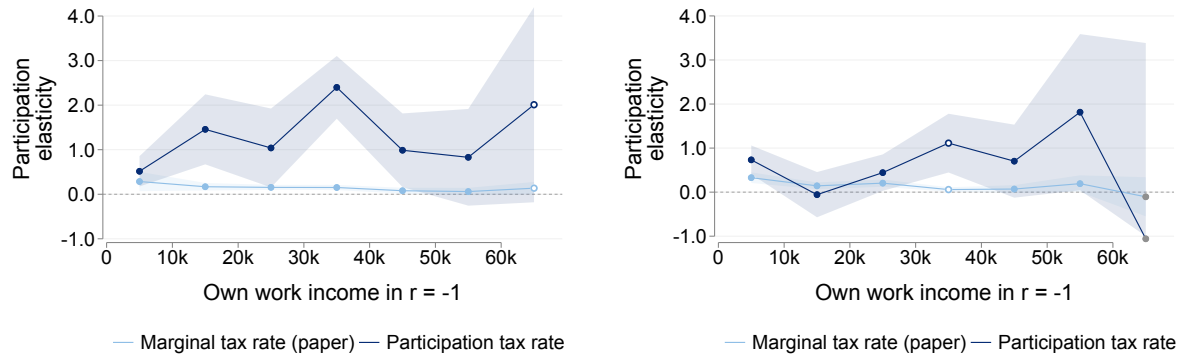


(a) Men

(b) Women

Figure F.5: Participation elasticities along the own-income distribution, by gender

Notes: As Figure 7, with the estimates aggregated within each gender and displayed as in Figure F.4. The women's bottom bin includes seven estimates of the entrant sub-sample at one dose (1.11); without them it is 0.17. Hollow markers: the women's €30,000–40,000 bin and the men's €60,000–70,000 bin ($t = 1.96$) fail their pre-event placebo. The women's €60,000–70,000 bin rests on three cell–dose–period estimates and 885 individuals and enters no aggregate. No bin rests on a single estimate; the vertical axis is shared by the two genders. Figure F.14 redraws the panels along percentiles of own income.



(a) Men

(b) Women

Figure F.6: Participation elasticities with respect to the marginal and the participation tax rate

Notes: The bins of Figure F.5 (light blue: per unit change in the log household net-of-marginal-tax rate, the paper's series) and the same bins rescaled to the log net-of-participation-tax rate (navy). Each cell–dose–period estimate is multiplied by the cell's ratio of the change in the log net-of-marginal-tax rate to the change in the log net-of-participation-tax rate (Appendix D.2); cells whose participation-tax contrast is below 0.005 log points or of the wrong sign are dropped from the rescaled series (cell counts in Table F.6), and the bins are then aggregated with the frequency weights of Section 3.4. Hollow markers identify bins whose pooled pre-event placebo fails; shaded bands are 95% confidence intervals from person-clustered standard errors. The rescaling widens these intervals roughly in proportion to the factor. The women's €60,000–70,000 bin rests on 885 individuals and enters no aggregate and is drawn in grey; its rescaled interval (−5.50 to 3.39) is cut at the axis. Sample, pooling window ($r = 0$ to 3) and screens as in Figure F.5.

Table F.6: Participation elasticities rescaled by the participation tax rate

Own income at $r = -1$	Marginal rate	Factor	Participation tax	Cells	
	elasticity		elasticity	paper	rescaled
Women					
0–10k	0.33 (0.06)	2.4	0.73 (0.17)	67	67
10–20k	0.14 (0.04)	6.7	-0.06 (0.26)	102	82
20–30k	0.20 (0.04)	8.6	0.44 (0.21)	102	84
30–40k†	0.06 (0.03)	12.1	1.11 (0.34)	60	51
40–50k	0.07 (0.04)	9.9	0.70 (0.42)	17	17
50–60k	0.19 (0.10)	9.3	1.81 (0.90)	5	5
60–70k	-0.11 (0.23)	10.0	-1.06 (2.27)	3	3
Men					
0–10k	0.29 (0.11)	1.6	0.52 (0.17)	27	27
10–20k	0.17 (0.04)	7.8	1.46 (0.40)	87	78
20–30k	0.15 (0.03)	8.9	1.04 (0.45)	122	93
30–40k	0.15 (0.03)	11.8	2.40 (0.36)	91	73
40–50k	0.08 (0.03)	11.9	0.99 (0.42)	49	48
50–60k	0.06 (0.04)	12.9	0.83 (0.55)	25	24
60–70k†	0.14 (0.07)	14.7	2.01 (1.12)	18	18

Notes: By gender and own-income bin at $r = -1$: the participation elasticity with respect to the household net-of-marginal-tax rate (the series of Figure F.5), the bin's rescaling factor and the elasticity with respect to the net-of-participation-tax rate, each with its person-clustered standard error in parentheses; the last two columns count the cell–dose–period estimates behind the paper's bin and behind the rescaled bin. The rescaled bin drops the cells whose participation-tax contrast is below 0.005 log points or of the wrong sign. The factor of a bin is the frequency-weighted mean over its cells of the ratio of the treated-minus-control change in the log net-of-marginal-tax rate to that in the log net-of-participation-tax rate. Because factor and estimate vary together within a bin, the rescaled bin is not the paper's bin times the factor. † Bin whose pooled pre-event placebo fails. Weighted to the population of French taxpayers, the rescaled participation elasticity is 0.84, about 3.4 times the 0.25 of Table F.4. In the €0–10,000 bin each cell takes the factor of the matching cell of the whole bin.

F.8 Maps over own and partner income

Tables F.7 and F.8 report the estimate, standard error and number of underlying cell–dose–period estimates for every rectangle of the earnings and participation maps in its display state. Five rectangles (four of them women's) fail their pre-event placebo; their pooled pre-event $|t|$ lies between 2.09 and 2.63.

Table F.7: Tile-level estimates: women

			Earnings		Participation	
Own income	Partner tercile	Partner income	Estimate	Cells	Estimate	Cells
0–10k	low	≤24.7 / ≤27.9	0.63 (0.19)	4	0.02 (0.08)	24
	mid	24.7–38.3 / 27.9–51.7	0.29 (0.18)	4	0.69 (0.11)	21
	high	>38.3 / >51.7	-0.02 (0.14)	4	0.35 (0.08)	9
10–20k	low	≤23.1	0.01 (0.05)	23	-0.07 (0.04)	41
	mid	23.1–34.1	0.72 (0.25)†	13	0.64 (0.14)	30
	high	>34.1	0.08 (0.05)	10	0.12 (0.02)	22
20–30k	low	≤25.3	0.44 (0.12)†	26	0.33 (0.09)	41
	mid	25.3–36.9	0.25 (0.06)	11	0.15 (0.03)	31
	high	>36.9	0.05 (0.05)	11	0.07 (0.02)	22
30–40k	low	≤29.0	0.09 (0.09)	11	0.10 (0.04)†	28
	mid	29.0–42.3	-0.04 (0.07)	8	0.00 (0.04)	13
	high	>42.3	-0.01 (0.13)	5	0.02 (0.07)	7
40–50k	low	≤32.8	0.24 (0.10)†	6	0.09 (0.04)	8
	mid	32.8–50.8	-0.05 (0.19)	3	0.03 (0.12)	6
	high	>50.8	[1.22] (0.74)	1	[0.27] (0.14)	1
50–60k	low	≤36.3	0.31 (0.21)	3	0.20 (0.10)	4
	mid	36.3–60.3	n.a.	–	[0.01] (0.38)	1

Notes: One row per own-income bin (thousand euro) × within-bin partner-income tercile of Figure 8, for earnings and participation. Partner income gives the tercile's range of partner income at $r = -1$ in thousand euro; the lowest tercile includes negative values and the top tercile is open-ended. In the €0–10,000 bin the terciles are formed separately for the earner and entrant sub-samples (Appendix F.2) and pooled by rank; the first range is the earner sub-sample's and the second the entrant sub-sample's. Estimates follow the map-level display conventions of Appendix F.2 and match the printed tiles; standard errors in parentheses are clustered by person, and for bracketed entries they treat the cell estimates as independent. Daggers mark the corner-ticked tiles of the maps, whose pooled pre-event placebo fails ($|t| \geq 1.96$); bracketed entries are estimates screened out of that map (dashed cells), n.a. marks rectangles with no usable cell, and rectangles with no cell in either map are omitted. Cells columns give the number of underlying cell–dose–period estimates; with the pooling window $r = 0$ to 3, one cell–dose can contribute up to four.

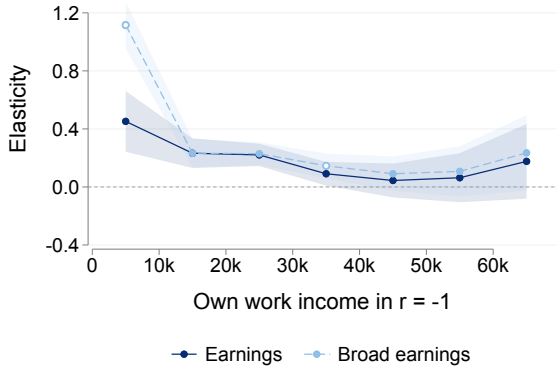
Table F.8: Tile-level estimates: men

			Earnings		Participation	
Own income	Partner tercile	Partner income	Estimate	Cells	Estimate	Cells
0–10k	low	≤13.8 / ≤11.6	[2.46] (0.57)	2	0.02 (0.26)	2
	mid	13.8–26.2 / 11.6–26.4	[0.70] (0.39)	5	-0.15 (0.22)	9
	high	>26.2 / >26.4	[0.32] (0.30)	4	0.42 (0.16)	8
10–20k	low	≤13.6	0.32 (0.13)	8	0.24 (0.07)	28
	mid	13.6–22.0	0.12 (0.10)	12	0.01 (0.06)	26
	high	>22.0	0.29 (0.17)	5	0.22 (0.06)	19
20–30k	low	≤15.4	0.22 (0.05)	29	0.12 (0.04)	47
	mid	15.4–23.6	0.24 (0.17)	22	0.33 (0.08)	35
	high	>23.6	0.11 (0.05)	12	0.16 (0.03)	29
30–40k	low	≤16.5	0.09 (0.11)	20	0.11 (0.06)	37
	mid	16.5–27.1	0.19 (0.08)	9	0.28 (0.04)	22
	high	>27.1	0.10 (0.06)	8	0.07 (0.03)†	16
40–50k	low	≤16.6	0.05 (0.13)	9	0.14 (0.06)	19
	mid	16.6–28.9	0.06 (0.08)	5	0.06 (0.04)	13
	high	>28.9	-0.01 (0.10)	4	-0.02 (0.07)	9
50–60k	low	≤15.8	0.07 (0.12)	5	0.09 (0.05)	12
	mid	15.8–29.9	-0.06 (0.12)	4	0.00 (0.07)	6
	high	>29.9	0.13 (0.18)	2	0.16 (0.12)	5

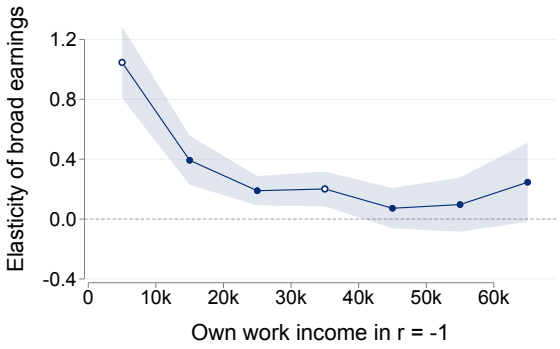
Notes: As Table F.7, for men. No men's earnings cell in the €0–10,000 bin is within the standard-error cap; the three bracketed entries pool the cells that pass the other screens. The €20,000–30,000 mid-tercile earnings tile (0.24) is dominated by a single comparison whose dose group fails a disaggregated pre-event placebo test. Without this comparison the tile is about zero. It has no dagger because the rectangle-level placebo does not flag it.

F.9 Alternative outcomes

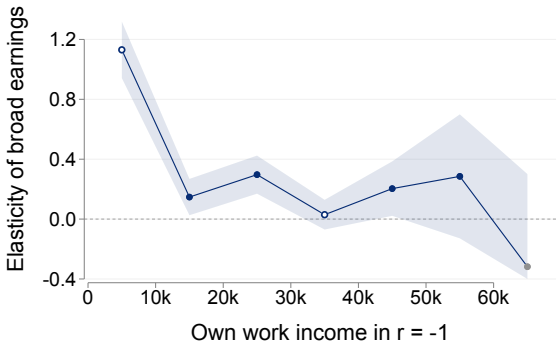
Claimed professional-expense deductions (*frais réels*) were estimated as an outcome and are not reported. Because the predicted bracket is computed from taxable income net of the deductions themselves, households predicted to cross a threshold are selected on low deductions in the base year within their cell, and the pre-event placebo contrasts reject in the income bins that hold over 90 percent of the cells (four of six bins for men and three of five for women; reduced forms of -0.03 to -0.05 at $r = -3$ and -2 for both genders). Estimates for self-employment income on its own are not reported either. Few individuals in the event sample have it (7 percent of men and 3 percent of women before the event) and its log change is about four to six times as dispersed as that of work income (standard deviation 0.64 against 0.11 to 0.18), implying that two to three percent of the cell–dose–period estimates pass the display screens against 21 percent for work income. The pre-event estimates also reject zero in 30 to 57 percent of the participation cells. Broad earnings, which add self-employment income to work income, are reported in Figure F.7.



(a) Men and women pooled



(b) Men



(c) Women

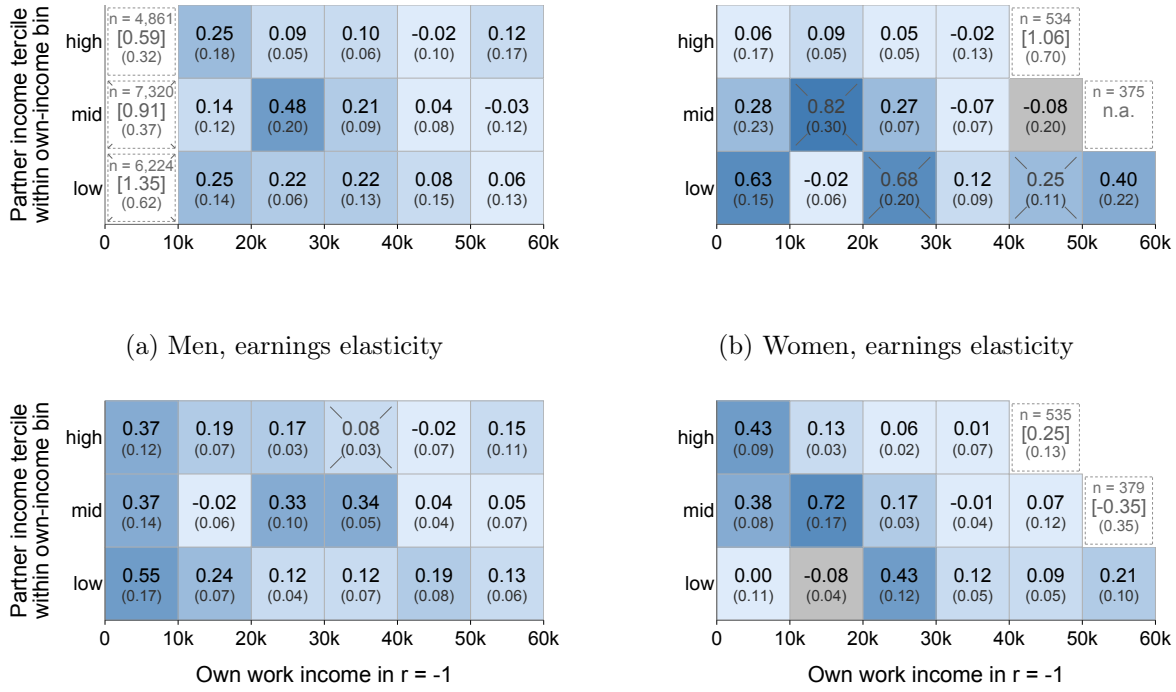
Figure F.7: Broad earnings elasticities along the own-income distribution

Notes: As Figure 6, with broad earnings (work plus self-employment income) as the outcome, on cells defined by own work income as in the headline. Self-employment income sums the declared industrial, commercial, non-commercial and agricultural profit boxes (turnover before the abatement under the micro regimes, net profit under the real regimes); turnover declared under the micro-entrepreneur flat-rate scheme (*versement libératoire*) is not included. Panel (a) pools men and women as in Figure 6 and overlays that figure's earnings profile (dark navy, solid) with the broad earnings profile (light blue, dashed). The broad earnings profile is within 0.01 of the earnings profile between €10,000 and €30,000 and about 0.05 above it from €30,000 up. From €10,000 of own work income up, the broad earnings profile lies within 0.05 of the earnings profile for women (Panel c) and 0.02 to 0.08 above it for men (Panel b), the gender profiles of Appendix Figure F.4. In all three panels the bottom bin and the €30,000–40,000 bin of the broad earnings profile fail their pre-event placebo (hollow markers); the bottom-bin estimates of 1.12 (pooled), 1.13 (women) and 1.05 (men) are not interpreted. The women's €60,000–70,000 bin rests on a single cell–dose–period estimate and is drawn in grey with its interval cut at the axis: −0.32, interval −0.94 to 0.30. The broad earnings outcome has no participation counterpart.

F.10 Excluding the 2020–2023 event years

The headline specification includes all event years. Figure F.8 re-estimates the maps without the 2020–2023 event years. Dropping these years removes about a sixth of the predicted-treated individuals (8 to 22 percent in the bins below €40,000). The maps change little. From €10,000 of own income up, the tiles correlate with those of Figure 8 at 0.99 (women) and

0.82 (men) for earnings and at 0.99 and 0.96 for participation, with mean absolute differences of 0.02 to 0.05 (on all earnings tiles 0.98 and 0.82; Table 5). Every tile shown in the headline maps is shown here, and the women's three earnings tiles whose pre-event placebo fails remain flagged and are larger (0.82, 0.68 and 0.25 against 0.72, 0.44 and 0.24). For participation, the only tile whose flag differs is the women's €30,000–40,000 tile in the bottom partner tercile, which fails its placebo in the headline map only.



(a) Men, earnings elasticity

(b) Women, earnings elasticity

(c) Men, participation elasticity

(d) Women, participation elasticity

Figure F.8: Earnings and participation elasticities by own and partner income: excluding event years 2020–2023

Notes: As Figure 8, excluding the 2020–2023 event years, with the same screens and display conventions (Appendix F.2).

F.11 First-stage strength

Appendix Table F.9 reports the robust first-stage F for all post-event cell estimates and for the cells behind each exhibit. Appendix Tables F.10, F.11 and F.12 show the income profiles under the stricter requirement $F \geq 104.7$, the tF-adjusted bin standard errors and the Anderson–Rubin sets. Section 4.3 summarises what they show.

Table F.9: First-stage strength of the cell estimates behind each exhibit

	Women						Men					
Cells	Number	Median F	Ratio	$F < 16.4$	$F < 23.1$	$F < 104.7$	Number	Median F	Ratio	$F < 16.4$	$F < 23.1$	$F < 104.7$
Panel A: earnings												
All estimates	835	275	0.89	14.7	16.2	33.8	948	353	0.82	15.2	17.9	29.3
Passing the elasticity and s.e. screens	174	1,402	0.87	0.0	0.1	3.3	203	1,016	0.79	0.0	0.0	1.0
Displayed: income and event-time profiles	174	1,402	0.87	0.0	0.1	3.3	203	1,016	0.79	0.0	0.0	1.0
of which in the maps	142	1,825	0.88	0.0	0.1	0.8	154	1,197	0.79	0.0	0.0	0.4
of which in the aggregation table	173	1,402	0.87	0.0	0.1	3.3	202	1,016	0.79	0.0	0.0	1.0
Panel B: participation												
All estimates	939	362	0.86	12.0	13.5	28.8	1,124	311	0.83	13.8	15.9	29.6
Passing the elasticity and s.e. screens	362	694	0.86	0.7	1.4	13.9	422	706	0.79	0.3	1.8	10.9
Displayed: income and event-time profiles	356	697	0.86	0.3	1.0	13.6	419	706	0.79	0.3	1.7	10.9
of which in the maps	307	728	0.86	0.1	0.5	11.7	342	759	0.78	0.2	1.3	8.1
of which in the aggregation table	353	697	0.86	0.3	1.0	13.6	419	706	0.79	0.3	1.7	10.9

Notes: First-stage strength of the cell–dose–period two-stage least squares estimates behind the exhibits of Section 4, $r = 0$ to 3 (in the €0–10,000 bin, the cells of the earner and entrant sub-samples). Each regression has one instrument and one endogenous regressor; F is the heteroskedasticity-robust first-stage F, i.e. the squared robust first-stage t-ratio. With one instrument it equals the effective F of Montiel Olea and Pflueger (2013). For the one displayed cell screened on the classical t-ratio, F is the classical F. Ratio is the median of the robust over the classical F. Median F and Ratio weight each cell by its regression sample; the last three columns give the share of that weight, in percent, in cells below the Stock and Yogo (2005) critical value for a maximal two-stage least squares size of 10 percent (16.38), below the Montiel Olea and Pflueger (2013) critical value for a worst-case bias of 10 percent (23.1), and below 104.7, the value above which the two-stage least squares t-ratio has correct 5 percent size (Lee et al., 2022). Displayed cells pass every cell screen of Appendix F.2, including the robust first-stage screen $F \geq 9$ ($|t| \geq 3$); the two 'of which' rows count the subsets that enter a displayed tile and the aggregation table.

Table F.10: Income profiles under a stricter first-stage requirement

	Displayed ($F \geq 9$)		$F \geq 104.7$			
Own income	Estimate (s.e.)	Cells	Estimate	Cells	Weight kept	Difference
Panel A: earnings						
Women						
0–10k	0.45 (0.11)	18	0.45	18	100.0	0.00
10–20k	0.18 (0.07)	58	0.18	58	100.0	0.00
20–30k	0.30 (0.07)	57	0.26	48	92.4	-0.04
30–40k	0.02 (0.05)	28	0.02	20	94.8	0.00
40–50k	0.19 (0.09)	9	0.19	8	99.1	0.01
50–60k	0.31 (0.21)	3	0.31	3	100.0	0.00
60–70k	-0.28 (0.32)	1	-0.28	1	100.0	0.00
Men						
0–10k	0.45 (0.48)	1	0.45	1	100.0	0.00
10–20k	0.32 (0.08)	41	0.32	41	100.0	0.00
20–30k	0.17 (0.05)	74	0.18	68	99.3	0.00
30–40k	0.13 (0.06)	44	0.13	36	97.2	0.01
40–50k	0.02 (0.07)	22	0.02	20	99.6	0.00
50–60k	0.05 (0.09)	13	0.05	13	100.0	0.00
60–70k	0.19 (0.13)	8	0.20	7	98.1	0.01
Panel B: participation						
Women						
0–10k	0.33 (0.06)	67	0.33	64	98.3	0.00
10–20k	0.14 (0.04)	102	0.05	72	86.7	-0.09
20–30k	0.20 (0.04)	102	0.13	53	77.1	-0.08
30–40k	0.06 (0.03)	60	0.05	28	85.6	0.00
40–50k	0.07 (0.04)	17	0.08	9	83.7	0.01
50–60k	0.19 (0.10)	5	0.20	4	96.5	0.01
60–70k	-0.11 (0.23)	3	-0.02	1	33.8	0.09
Men						
0–10k	0.29 (0.11)	27	0.29	27	100.0	0.00
10–20k	0.17 (0.04)	87	0.20	59	82.3	0.03
20–30k	0.15 (0.03)	122	0.17	83	91.2	0.01
30–40k	0.15 (0.03)	91	0.17	55	88.6	0.02
40–50k	0.08 (0.03)	49	0.08	26	90.0	0.00
50–60k	0.06 (0.04)	25	0.06	16	89.9	0.00
60–70k	0.14 (0.07)	18	0.08	9	77.6	-0.05

Notes: Own-income bins of the income profiles (Figures F.4 and F.5), frequency weights, $r = 0$ to 3: the displayed estimate with its person-clustered standard error, and the same construction with only the first-stage requirement raised from $F \geq 9$ to $F \geq 104.7$ (Lee et al., 2022). Weight kept is the share of the bin's frequency weight retained, in percent; Difference is the change in the bin estimate. The stricter construction is reported as a point estimate only. Largest change: 0.041 (earnings) and 0.090 (participation) over bins with at least two displayed cells. The two bins that move by more than one displayed standard error are women's participation at €10,000–20,000 and €20,000–30,000 (by −0.090 and −0.078). Both move because of cells of households at the 14 percent rate predicted to cross into the 30 percent bracket whose F lies below 104.7. Dropping only those cells moves the bins by −0.079 and −0.097. These cells hold 7.0 and 10.5 percent of the bins' frequency weight but 0.2 percent of their predicted treated; with predicted-treated weights the stricter screen moves the bins by −0.013 and 0.002.

Table F.11: tF-adjusted standard errors of the income-profile bins

Own income	Estimate	s.e.	s.e. tF	Ratio	Max factor	Weight adjusted	$\lvert t\rvert$	$\lvert t\rvert$ tF
Panel A: earnings								
Women								
0–10k	0.45	0.108	0.108	1.000	1.00	0.0	4.2	4.2
10–20k	0.18	0.066	0.066	1.000	1.00	0.0	2.8	2.8
20–30k	0.30	0.065	0.066	1.007	1.11	7.6	4.7	4.6
30–40k	0.02	0.051	0.051	1.004	1.36	5.2	0.4	0.4
40–50k	0.19	0.092	0.092	1.001	1.39	0.9	2.0	2.0
50–60k	0.31	0.215	0.215	1.000	1.00	0.0	1.4	1.4
60–70k	-0.28	0.319	0.319	1.000	1.00	0.0	0.9	0.9
Men								
0–10k	0.45	0.483	0.483	1.000	1.00	0.0	0.9	0.9
10–20k	0.32	0.084	0.084	1.000	1.00	0.0	3.8	3.8
20–30k	0.17	0.049	0.049	1.000	1.20	0.7	3.5	3.5
30–40k	0.13	0.057	0.057	1.001	1.40	2.8	2.2	2.2
40–50k	0.02	0.068	0.068	1.000	1.20	0.4	0.3	0.3
50–60k	0.05	0.091	0.091	1.000	1.00	0.0	0.5	0.5
60–70k	0.19	0.134	0.134	1.000	1.01	1.9	1.4	1.4
Panel B: participation								
Women								
0–10k	0.33	0.061	0.061	1.000	1.04	1.7	5.4	5.4
10–20k	0.14	0.039	0.040	1.032	1.44	13.3	3.7	3.5
20–30k	0.20	0.042	0.043	1.037	1.79	22.9	4.9	4.7
30–40k	0.06	0.027	0.028	1.062	1.52	14.4	2.2	2.1
40–50k	0.07	0.045	0.046	1.031	1.40	16.3	1.6	1.5
50–60k	0.19	0.096	0.098	1.013	1.39	3.5	2.0	2.0
60–70k	-0.11	0.226	0.262	1.158	1.20	66.2	0.5	0.4
Men								
0–10k	0.29	0.110	0.110	1.000	1.00	0.0	2.6	2.6
10–20k	0.17	0.043	0.044	1.015	1.23	17.7	3.9	3.8
20–30k	0.15	0.028	0.033	1.176	1.58	8.8	5.4	4.6
30–40k	0.15	0.029	0.031	1.041	1.76	11.4	5.2	4.9
40–50k	0.08	0.034	0.034	1.018	1.66	10.0	2.4	2.3
50–60k	0.06	0.042	0.043	1.023	1.65	10.1	1.4	1.4
60–70k	0.14	0.071	0.074	1.043	1.25	22.4	1.9	1.8

Notes: The displayed person-clustered bin standard error (s.e.) against a tF-adjusted version (s.e. tF). Each cell's robust standard error is multiplied by the 5 percent adjustment factor $c_{0.05}(F)/1.96$ of Lee et al. (2022, Table 3, Panel A) at its robust first-stage F ($c_{0.05}(F)$ is the tF critical value for $|t|$). The factor is linearly interpolated between the published grid points and equals 1 for $F \geq 104.7$. Ratio is the bin standard error aggregated from the adjusted cell standard errors over the one aggregated from the unadjusted ones (cells treated as independent), and s.e. tF is the displayed standard error times Ratio. Max factor is the largest cell factor in the bin; Weight adjusted is the share of the bin's frequency weight in cells with $F < 104.7$, in percent; $|t|$ is the bin estimate over the respective standard error. The adjustment is a heuristic bound because the tF size guarantee is for a single just-identified t-ratio and a bin estimate is a weighted mean of cells. Largest ratio over bins with at least two cells: 1.007 (earnings), 1.176 (participation).

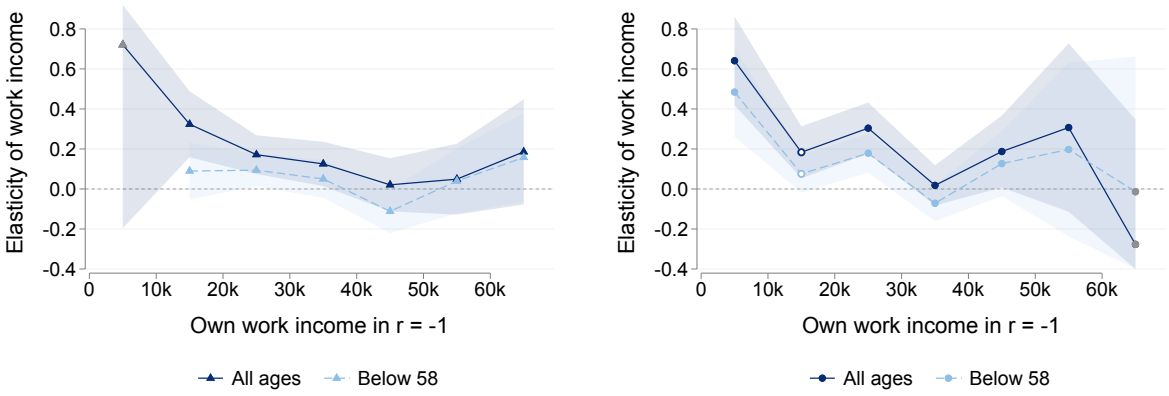
Table F.12: Anderson–Rubin sets and Wald intervals, displayed cells

Own income	Cells	Bounded	Unbounded	Width ratio, median	Width ratio, p90	Both reject	Wald only	AR only
Panel A: displayed cells, earnings								
Women								
0–10k	18	18	0	1.003	1.006	4	0	0
10–20k	58	58	0	1.003	1.013	8	0	0
20–30k	57	57	0	1.004	1.025	8	0	0
30–40k	28	28	0	1.007	1.065	5	0	0
40–50k	9	9	0	1.003	1.134	1	0	0
50–60k	3	3	0	1.005	1.008	0	0	0
60–70k	1	1	0	1.013	1.013	0	0	0
All bins	174	174	0	1.004	1.023	26	0	0
Men								
0–10k	1	1	0	1.003	1.003	0	0	0
10–20k	41	41	0	1.005	1.010	1	0	0
20–30k	74	74	0	1.004	1.018	11	0	1
30–40k	44	44	0	1.004	1.041	4	0	0
40–50k	22	22	0	1.003	1.018	2	0	0
50–60k	13	13	0	1.005	1.010	2	0	0
60–70k	8	8	0	1.009	1.023	0	0	0
All bins	203	203	0	1.004	1.018	20	0	1
Panel B: displayed cells, participation								
Women								
0–10k	67	67	0	1.005	1.015	22	0	0
10–20k	102	102	0	1.008	1.056	29	0	1
20–30k	102	102	0	1.016	1.091	18	0	0
30–40k	60	60	0	1.024	1.148	8	0	2
40–50k	17	17	0	1.017	1.135	1	0	0
50–60k	5	5	0	1.008	1.140	1	0	0
60–70k	3	3	0	1.031	1.075	0	0	0
All bins	356	356	0	1.010	1.080	79	0	3
Men								
0–10k	27	27	0	1.003	1.014	5	0	0
10–20k	87	87	0	1.010	1.035	18	0	0
20–30k	121	121	0	1.010	1.062	31	0	1
30–40k	91	91	0	1.011	1.127	19	0	0
40–50k	49	49	0	1.019	1.134	6	0	0
50–60k	25	25	0	1.011	1.052	1	0	1
60–70k	18	18	0	1.021	1.081	1	0	0
All bins	418	418	0	1.011	1.079	81	0	2

Notes: Heteroskedasticity-robust Anderson–Rubin 95 percent sets (Anderson & Rubin, 1949) per cell, computed from the robust reduced-form and first-stage estimates of each regression on its instrumental-variable sample: with $\hat{\gamma} = \hat{\beta}\hat{\pi}$ the reduced form, V_γ and V_π the robust reduced-form and first-stage variances and C their robust covariance, the set is $\{\beta : (\hat{\gamma} - \beta\hat{\pi})^2 \leq 3.84\,(V_\gamma - 2\beta C + \beta^2 V_\pi)\}$, bounded if and only if $F > 3.84$. The robust two-stage least squares standard error satisfies $\text{s.e.}(\hat{\beta})^2\hat{\pi}^2 = V_\gamma - 2\hat{\beta}C + \hat{\beta}^2 V_\pi$ on every displayed cell up to numerical precision. Cells counts the displayed cells with a robust first stage (all but the one screened on the classical t-ratio). Width ratio is the length of the Anderson–Rubin set over the length of the robust Wald interval $\hat{\beta} \pm 1.96\,\text{s.e.}(\hat{\beta})$; Both reject, Wald only and AR only count cells where both, only the Wald interval, or only the Anderson–Rubin set exclude zero (the Anderson–Rubin test of a zero effect is the robust reduced-form t-test). Over all estimated cells with a robust first stage, 73.4 percent of the sets are bounded for earnings and 77.1 percent for participation.

F.12 Age and retirement restrictions

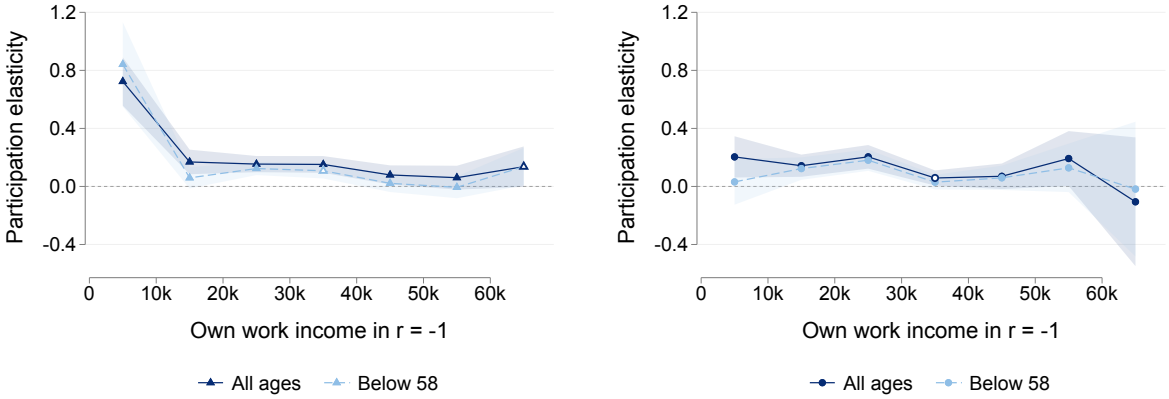
Figures F.9 and F.10 compare the earnings and participation profiles of individuals below 58 in the year before the event with the headline profiles. Table F.13 reports the bins from €10,000 up and their aggregates in three restricted samples (Section 4.3).



(a) Men (b) Women

Figure F.9: Earnings elasticities: all ages versus below 58

Notes: The panels overlay the headline profile (all ages, dark navy, solid) with the profile estimated on individuals below age 58 in the year before the event (light blue, dashed). Below 58, the women's profile shifts down with its shape intact (lowest bin 0.64 to 0.48, s.e. 0.11 in both), and the men's lies below 0.10 in every bin from €10,000 to €60,000 (0.16, s.e. 0.11, in the bin above). No men's bottom-bin estimate passes the screens. The age cut removes 4 percent of the predicted-treated women and 9 percent of the predicted-treated men. Construction, screens and pooling window as in Figure F.4. Shaded areas in the colour of the series are 95% confidence intervals from person-clustered standard errors; hollow markers flag a failed pre-event placebo of the series. Grey markers are bins that rest on a single cell–dose–period estimate and whose interval is cut at the axis: the men's all-ages €0–10,000 bin, 0.72 with interval −0.20 to 1.64, and the women's €60,000–70,000 bin, −0.28 with interval −0.90 to 0.35 (all ages) and −0.01 with interval −0.69 to 0.66 (below 58).



(a) Men (b) Women

Figure F.10: Participation elasticities: all ages versus below 58

Notes: As Figure F.9, for participation; no bin rests on a single cell–dose–period estimate. In the €0–10,000 bin the lowest-earning men below 58 keep a large participation elasticity of 0.84 (s.e. 0.15, on 10 estimates) against 0.72 (s.e. 0.08) at all ages, although the cut removes 57 percent of the predicted-treated men in that bin. Above it, men's elasticities below 58 range from −0.01 to 0.14 (all ages 0.06 to 0.17). Women's are within 0.03 of the all-ages ones from €10,000 to €50,000 and within one standard error in the two top bins; their lowest bin falls from 0.20 to 0.03. The age cut removes 7 percent of the predicted-treated women and 15 percent of the men.

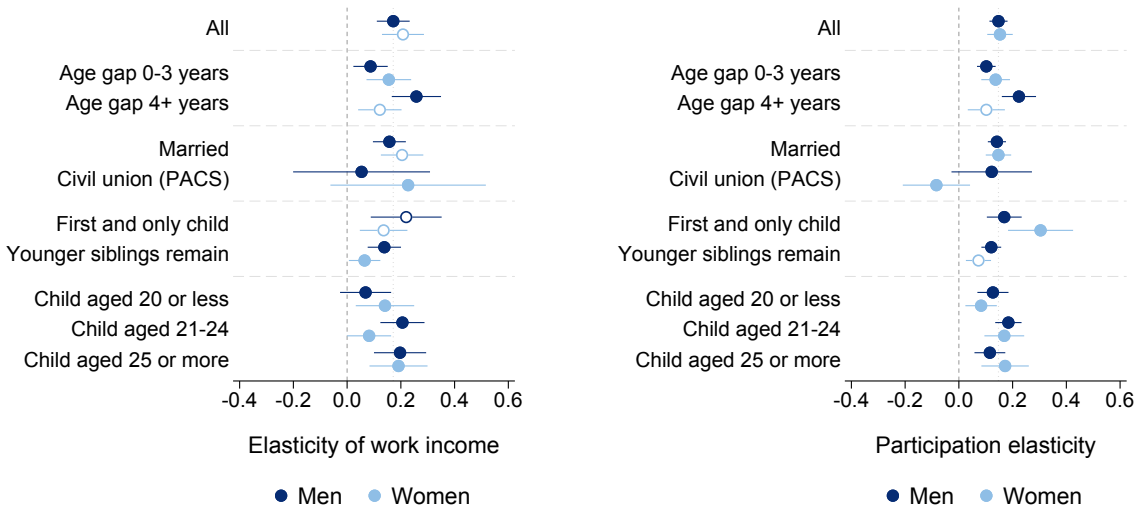
Table F.13: Responses from €10,000 up in three restricted samples

	Women				Men			
Own income	Headline	Below 58	No pension	Floor 400	Headline	Below 58	No pension	Floor 400
A. Earnings elasticity								
10–20k	0.18†	0.08†	0.03	0.16†	0.32	0.09	−0.01	0.09
	(0.07)	(0.05)	(0.05)	(0.06)	(0.08)	(0.07)	(0.08)	(0.06)
20–30k	0.30	0.18	0.08	0.18†	0.17	0.09	0.09	0.14
	(0.07)	(0.05)	(0.04)	(0.06)	(0.05)	(0.04)	(0.04)	(0.04)
30–40k	0.02	−0.07	−0.04	−0.04†	0.13	0.05	0.05	0.02
	(0.05)	(0.05)	(0.05)	(0.04)	(0.06)	(0.05)	(0.04)	(0.04)
40–50k	0.19	0.13	0.15†	0.14†	0.02	−0.11	−0.10	0.03
	(0.09)	(0.08)	(0.09)	(0.07)	(0.07)	(0.06)	(0.06)	(0.05)
50–60k	0.31	0.20	0.32	0.20	0.05	0.04	0.00	0.05
	(0.21)	(0.22)	(0.21)	(0.18)	(0.09)	(0.08)	(0.08)	(0.07)
60–70k	−0.28[a]	−0.01[a]	0.05[a]	−0.33[a]	0.19	0.16	0.18	0.21
	(0.32)	(0.34)	(0.34)	(0.26)	(0.13)	(0.11)	(0.13)	(0.14)
Aggregate[b]	0.23	0.10	0.06	0.15	0.17	0.06	0.01	0.08
	(0.05)	(0.03)	(0.03)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)
Removed (%)[c]		3.5	1.5	0.2		7.7	2.6	0.3
B. Participation elasticity								
10–20k	0.14	0.12	0.08	0.14	0.17	0.06	−0.01	0.17
	(0.04)	(0.04)	(0.05)	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)
20–30k	0.20	0.18	0.11	0.20	0.15	0.12	0.14	0.15
	(0.04)	(0.04)	(0.04)	(0.04)	(0.03)	(0.02)	(0.04)	(0.03)
30–40k	0.06†	0.03	0.03†	0.06†	0.15	0.11†	0.11†	0.15
	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
40–50k	0.07	0.06	0.06	0.07	0.08	0.02	−0.03	0.08
	(0.04)	(0.04)	(0.04)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)
50–60k	0.19	0.13	0.16	0.19	0.06	−0.01	0.01	0.06
	(0.10)	(0.09)	(0.09)	(0.10)	(0.04)	(0.04)	(0.04)	(0.04)
60–70k	−0.11	−0.02	−0.01	−0.11	0.14†	0.14	0.10	0.14
	(0.23)	(0.24)	(0.21)	(0.23)	(0.07)	(0.07)	(0.07)	(0.07)
Aggregate[b]	0.18	0.15	0.11	0.18	0.15	0.11	0.07	0.15
	(0.02)	(0.02)	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Removed (%)[c]		3.7	1.9	0.0		8.3	3.1	0.0

Notes: Each column re-estimates the profiles of Figure F.4 (Panel A) and Figure F.5 (Panel B) on a restricted sample, with the same cells, screens, pooling window ($r = 0$ to 3) and weights; *Headline* repeats those figures. *Below 58*: individuals below 58 in the year before the event. *No pension*: individuals with no pension income of their own and no missing pension record in the year before the event ($r = -1$). *Floor 400*: the earnings outcome is kept only for individuals with more than €400 of work income at $r = -1$ and in the outcome year. Because participation is not restricted, Panel B repeats the headline estimates to two decimals (the t-statistic of the men's €60,000–70,000 placebo falls from 1.96 to 1.94 under the floor). The €0–10,000 bin is omitted. Under *Floor 400*, the men's earnings bin is 0.34 (s.e. 0.21, on six estimates), conditional on outcome-year work income. Standard errors, clustered by person, in parentheses. † The bin's pooled pre-event placebo fails ($|t| \geq 1.96$). [a] A single cell–dose–period estimate. [b] Bins entering Table 4 from €10,000 up, with its weights. [c] Share of the panel's estimation sample from €10,000 up that the restriction removes, within the dose groups estimated with and without it.

F.13 Heterogeneity

Figure F.11 gives the gender breakdown of the subgroup estimates of Section 4.4.



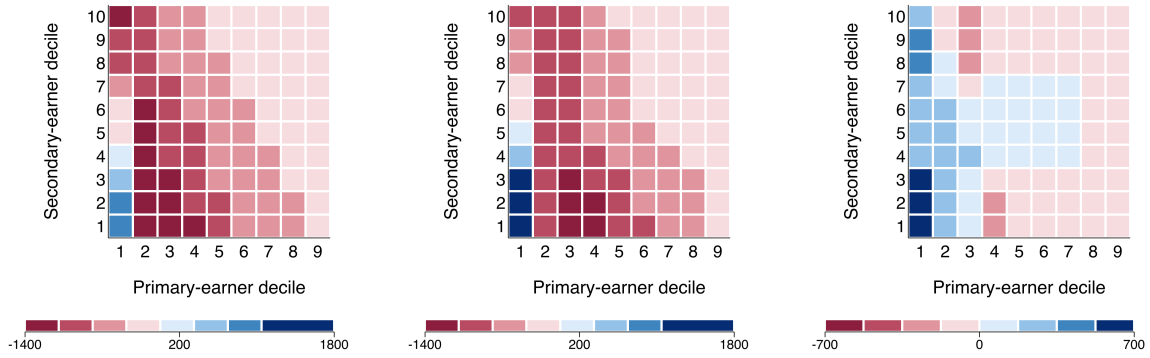
(a) Earnings elasticity (b) Participation elasticity

Figure F.11: Elasticities by household subgroup and gender

Notes: As Figure 9, separately by gender. A hollow marker means that the gender's own pre-event placebo fails at the 5 percent level when tested with the estimate's weights and covariance.

F.14 Welfare effects under near-Rawlsian weights

Figure F.12 repeats the exercise of Figure 11 with near-Rawlsian welfare weights.



(a) Constant responses (b) Estimated responses (c) Difference, (b) minus (a)

Figure F.12: Welfare effects of local reforms under near-Rawlsian weights

Notes: As Figure 11, using near-Rawlsian welfare weights concentrated on the lowest-income households. Panels (a) and (b) show the constant- and estimated-response calibrations; Panel (c) reports estimated minus constant. Colour scales are in model welfare units under the €500-by-€500 local-cell and one-percentage-point reform normalization. These units are not euro equivalents. In Panels (a) and (b), blue denotes gains and red denotes losses. In Panel (c), blue denotes a positive difference and red denotes a negative one.

F.15 The income axis in percentiles

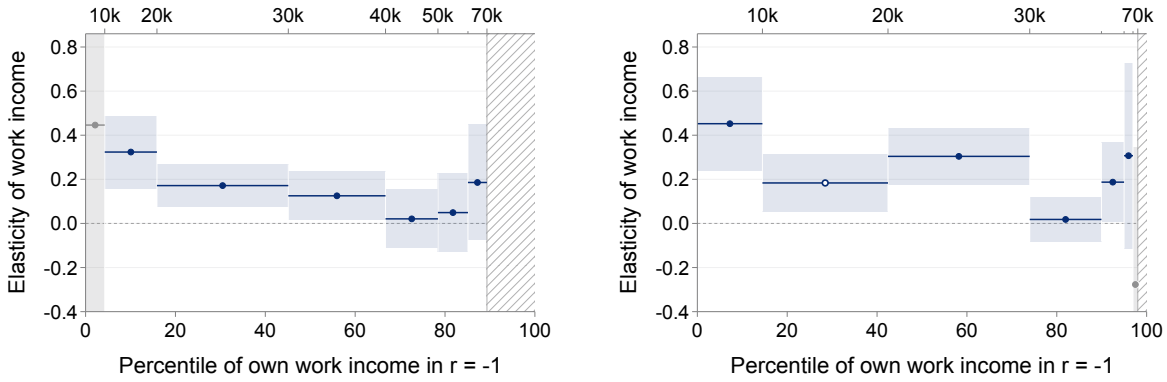
The profiles of Figures F.4 and F.5 and Figures 8 and 10 have income in euros on the horizontal axis. This subsection redraws them with percentiles instead. A bin is as wide as the share of individuals in it. The elasticity estimates are unchanged.

Figures F.13, F.14 and F.16 rank the individuals of the child's fiscal exit sample before its income cap of €70,000 (the third row of Table F.1) by own work income at $r = -1$. The earnings panels rank those with positive work income, on whom the earnings outcome is defined. The participation panels rank all individuals, since the participation estimates include those without work income at $r = -1$. The €10,000 bins then differ in width. The lowest bin holds 14 percent of the women with positive work income and 32 percent of all women, against 4 and 17 percent of the men. With both genders ranked together, the lowest bin holds 9.1 percent of those with positive work income, and incomes below €70,000 account for 93.5 percent. Incomes from €40,000 up account for 10 and 8 percent of the women, against 33 and 29 percent of the men. The width of a bin does not show the precision of its estimate.

Figures F.15 and F.17 place the bins of the earnings estimates in the 2023 work-income distribution of positive earners of the same gender (the two distributions that Figure 10 weights equally). Parents at the child's fiscal exit earn more than the positive earners of 2023. An

income of €30,000 in 2015 prices is the 45th percentile among the men ranked in Figure F.13 and about the 73rd among male earners in 2023, and for women the two percentiles are the 74th and about the 83rd. With men and women ranked together, the upper edges of the seven bins are at about the 18th, 48th, 78th, 90th, 94th, 96th and 98th percentiles of earners in 2023.

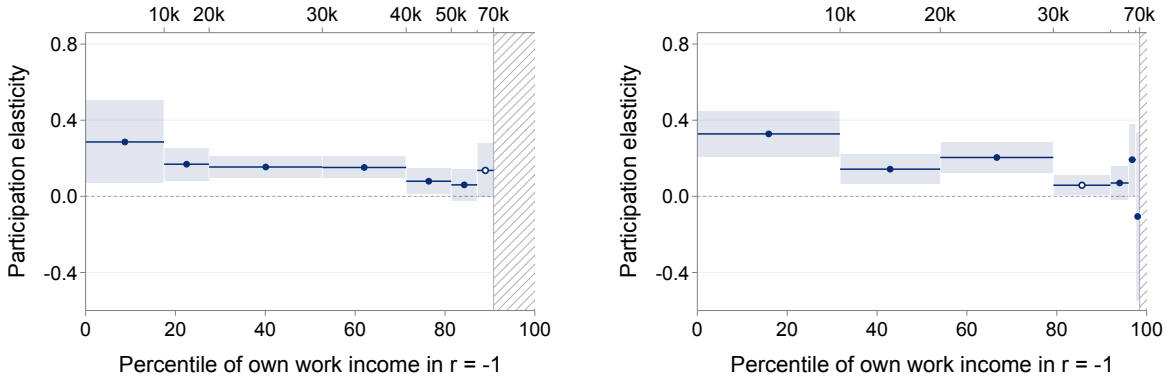
Figure F.18 draws the two schedules of Figure 10 against the percentile of earners. Because earnings respond to the schedule, an earner's income differs between the two schedules, and Figure 10 compares different earners at a given income. Because earners keep their rank from one schedule to the other, a percentile is the same earner on both lines. For about the bottom 56 percent of earners, the marginal rate with the fitted profile is lower than under the single elasticity of 0.229 (the grey dashed line of Figure 10). Figure F.19 repeats the calculation with two profiles that do not increase with income. Both fit the seven gender-pooled bin estimates behind Figure 10 by least squares and weight each estimate by its inverse squared standard error. The second fit also requires the profile to be convex. The estimates reject neither restriction ($p = 0.72$ and 0.57). The marginal rate is then lower than the single-elasticity rate for about the bottom 73 and 57 percent of earners. Compared at equal income (as in Section 5.1), it is lower below €10,000 by 13.4 and 13.8 percentage points on average in the unsmoothed solution, against 10.9 with the fitted profile. Above these percentiles the rate lies above the single-elasticity rate, as with the fitted profile. It does not decline towards €70,000 and reaches about 70 and 68 percent there, against about 60 percent with the fitted profile.



(a) Men (b) Women

Figure F.13: Earnings elasticities along the own-income distribution, in percentiles

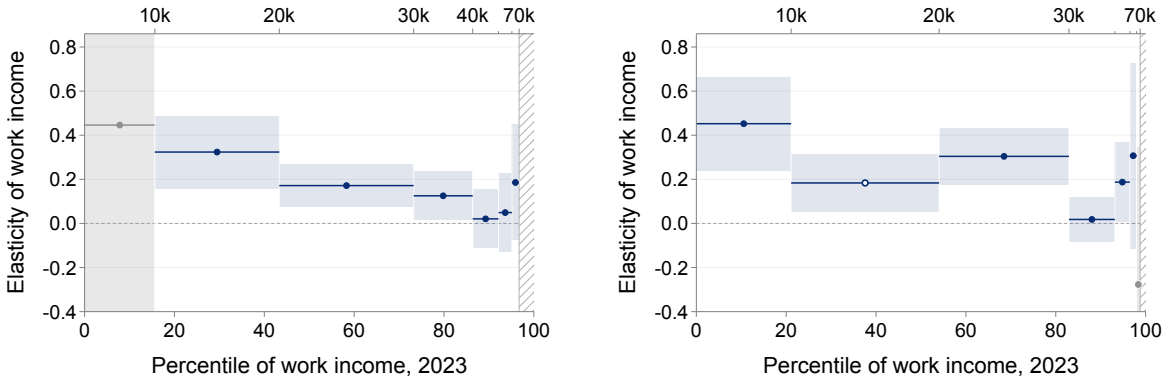
Notes: The bins of Figure F.4, each as wide as its share of the individuals with positive work income at $r = -1$ in the child's fiscal exit sample before its income cap of €70,000. A horizontal line with its marker is a bin's estimate and the shaded area its 95% confidence interval. Grey bins and hollow markers as in Figure F.4. The upper axis gives the bin edges in euros; its shorter ticks without a label are the €10,000 steps in between. The hatched area is the share above the cap, which the profiles do not cover.



(a) Men (b) Women

Figure F.14: Participation elasticities along the own-income distribution, in percentiles

Notes: As Figure F.13, for the bins of Figure F.5 and with percentiles among all individuals. The lowest bin includes those without work income at $r = -1$: 14 percent of the men and 20 percent of the women ranked, or 15 and 21 percent of those below the income cap (Table F.1).

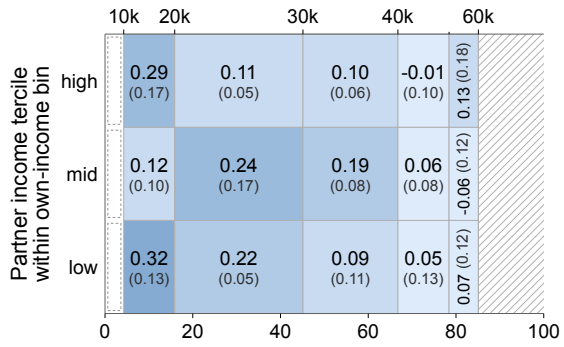


(a) Men

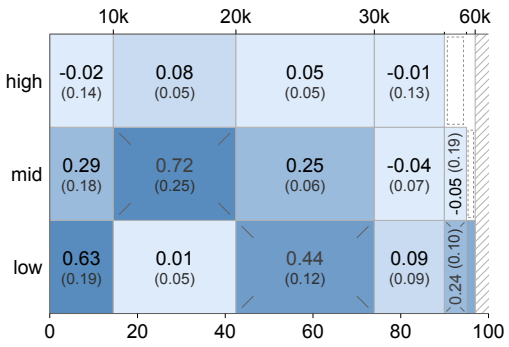
(b) Women

Figure F.15: Earnings elasticities along the own-income distribution, in percentiles of 2023 work income

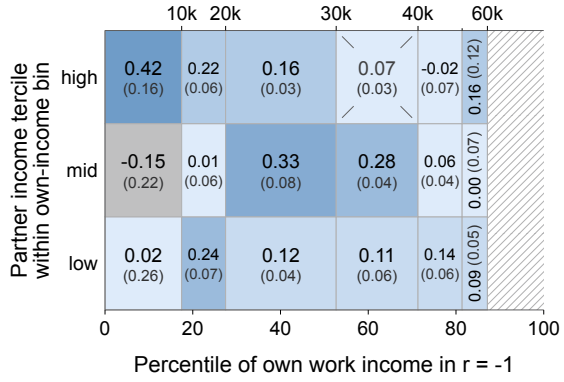
Notes: The bins of Figure F.4, placed in the 2023 work-income distribution of positive earners of the same gender; the bin edges are converted from 2015 to 2023 prices. The 2023 distribution is observed in nominal €10,000 bins up to €70,000. The edges are placed inside these bins by the fitted distribution of Figure 10, and the edges from €60,000 up lie in its assumed upper tail. Display as in Figure F.13.



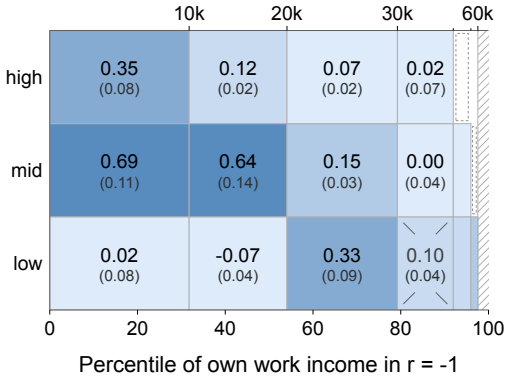
(a) Men, earnings elasticity



(b) Women, earnings elasticity



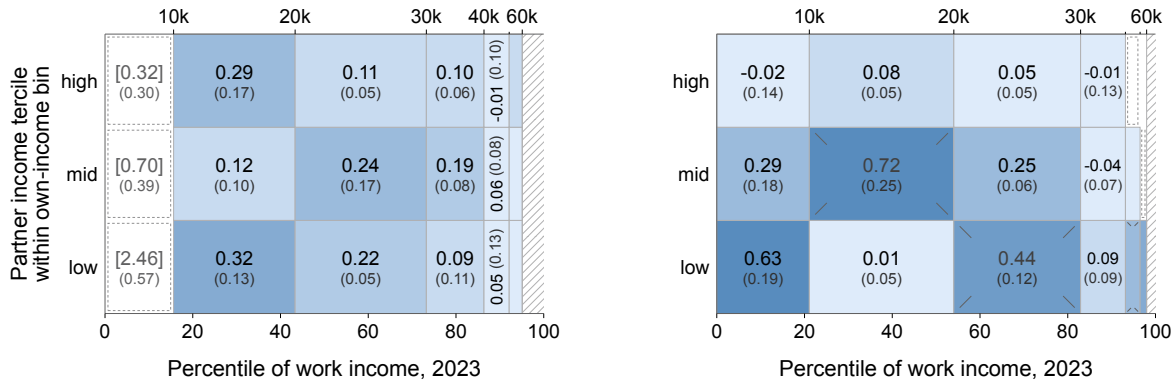
(c) Men, participation elasticity



(d) Women, participation elasticity

Figure F.16: Earnings and participation elasticities by own and partner income, own income in percentiles

Notes: The columns of Figure 8, each as wide as its share of the child's fiscal exit sample before its income cap of €70,000. Panels (a) and (b) rank individuals with positive work income at $r = -1$, and Panels (c) and (d) rank all individuals. The width of a column is not the precision of its estimates. Colours, dashed cells and corner ticks as in Figure 8. A narrow column has smaller type or its value and standard error on one rotated line, and the narrowest have no label. Type size and the length of the corner ticks have no meaning. Dashed cells omit the count of individuals. Figure 8 prints the labels and counts left out, and Appendix Tables F.7 and F.8 report the estimate and standard error of every tile. Upper axis as in Figure F.13; the hatched area is the share above €60,000, which the maps do not cover.



(a) Men (b) Women

Figure F.17: Earnings elasticities by own and partner income, own income in percentiles of 2023 work income

Notes: The columns of Panels (a) and (b) of Figure 8, placed as in Figure F.15. Display as in Figure F.16.

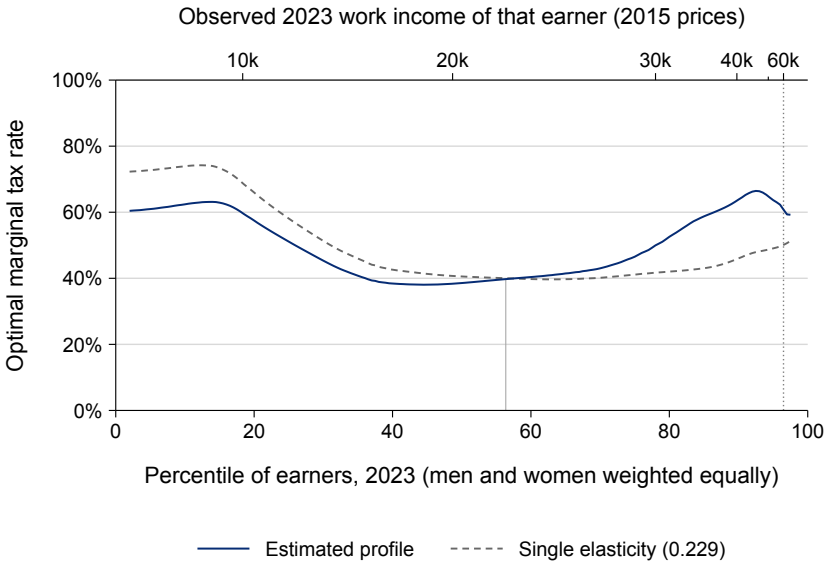
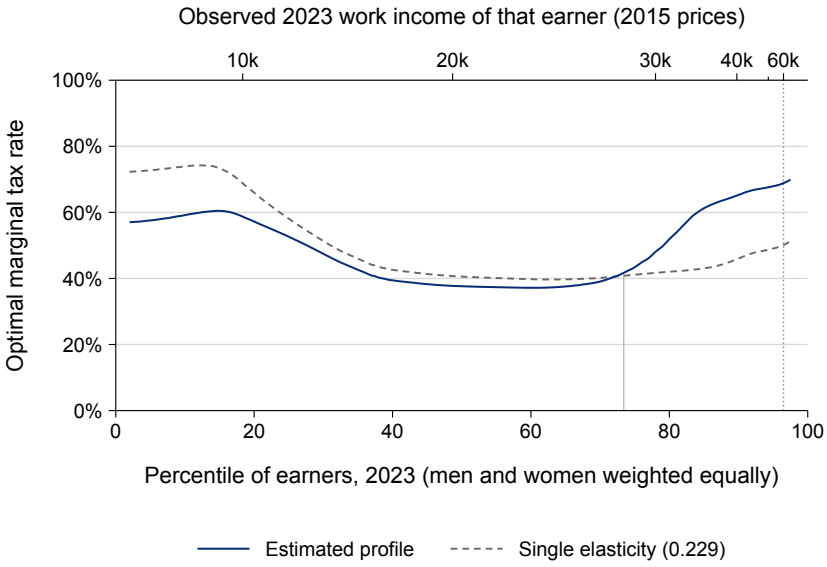
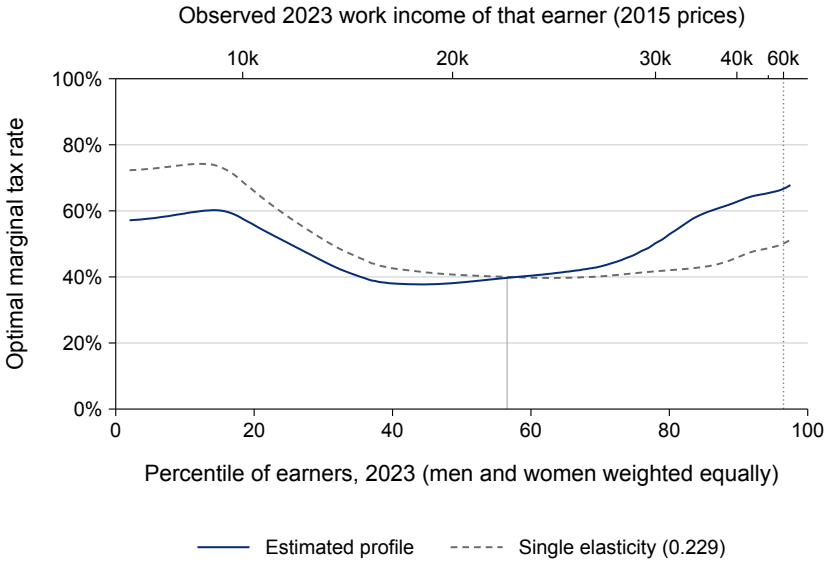


Figure F.18: Optimal marginal tax rates by percentile of earners

Notes: The two schedules of Figure 10, each line giving the marginal rate of the earner at a percentile. Because earners keep their rank from one schedule to the other, a percentile is the same earner on both lines. The upper axis gives the observed 2023 work income at a percentile in 2015 prices; its shorter tick without a label is €50,000. The thin vertical line marks the percentile at which the two schedules cross in the unsmoothed solution (56.4); the displayed lines run from the 2nd to the 97.5th percentile, are smoothed as in Figure 10 and cross within one and a half percentiles of it. Right of the dotted line at the 96.5th percentile, incomes exceed the last €10,000 bin of the income data (€70,000 in 2023 prices). The mass there is observed and its shape is the fitted Pareto tail of Section 5.1.



(a) Non-increasing profile



(b) Non-increasing and convex profile

Figure F.19: Optimal marginal tax rates by percentile of earners, non-increasing profiles

Notes: As Figure F.18, with the fitted profile replaced by the least-squares fit to the same seven gender-pooled bin estimates among profiles that do not increase with income (Panel a: 0.452, 0.254 and 0.238 for the three lowest bins and 0.093 for the four bins from €30,000 up) and among those that are also convex across the bins (Panel b: 0.452, 0.289, 0.197 and 0.104). Both fits weight each estimate by its inverse squared standard error. As for the fitted profile of Figure 10, the values are assigned at the bin midpoints, joined by a smooth curve between them and held fixed below the first and above the last midpoint. The weighted squared deviations of the seven estimates from the two fits are 1.36 and 2.95 ($p = 0.72$ and 0.57 on three and four degrees of freedom). The degrees of freedom are the seven bins less the free values of each fit, and the bins are treated as independent. The schedules cross at percentiles 73.4 and 56.6.

G The marriage design

The marriage design uses first marriages of spouses who were childless and separately taxed in the year before the wedding and file jointly the year after. It keeps the spouses for whom the merger predicts a rate increase and compares them within cells with the zero-dose spouses. Below €70,000 of own work income the estimation sample holds 1,249,585 women and 1,126,070 men, of whom 308,483 (25 percent) and 121,154 (11 percent) are predicted a rate increase. The predicted-increase share falls with own income, from 58 percent for women and 36 percent for men in the lowest bin to under 13 and 6 percent above €30,000. This is why the design identifies little above that level. In both genders 93 percent of spouses have positive work income at $r = -1$, and 42 and 40 percent of the weddings precede the 2011 reform. The design draws on the 2005 to 2017 cohorts, of 83,000 to 113,000 women each; later cohorts are much smaller.

The design follows the child's fiscal exit design of Sections 3.2 and 3.5, with the following differences. The instrument applies the pre-event formula to the two spouses' $r = -1$ incomes pooled into one unit with quotient 2. Unlike at the child's exit, it uses the statutory schedule, because the dose computed on the effective schedule of Section 2.1 removes the women's response at €10,000–20,000 and fails its placebo on the strict sample (both spouses also never married or in a civil union before). Only spouses whom the merger moves to a higher marginal rate (marrying up) are kept. After the residualisation described below, the predicted-decrease side is indistinguishable from zero for earnings and participation. As elsewhere, cells fix the own-income bin, the partner-income tercile within the bin and the pre-event bracket. A cell enters if it meets the floors of Appendix F.1 and holds at least 30 zero-dose spouses. Because the sample's screens fix civil status and the number of children, the controls replace them by the number of fiscal parts of each spouse before the wedding. Because the plain estimates show a post-event contrast of the same sign for predicted increases and predicted decreases, the outcomes are residualised before estimation. Within each cell and period the zero-dose group's outcome is regressed on the partner's income at $r = -1$ (capped at its within-cell 95th percentile), both spouses' ages and own work income at $r = -2$ and $r = -3$, with the fixed effects of the estimation. The fitted values are subtracted from every observation. For weddings before 2011 the wedding year was declared on separate returns before the wedding

and on a joint return after it. The $r = 0$ incomes of these spouses are imputed from the split-year returns and are not used, so $r = 0$ rests on the 2011 to 2017 cohorts alone. Only on the strict sample does the women's participation reduced form at $r = 0$ have the opposite sign to the later years. The exhibits pool $r = 0$ to 3 as elsewhere, and we report no event-time profile for this design. A participation build-up after the event would be read over $r = 1$ to 3.

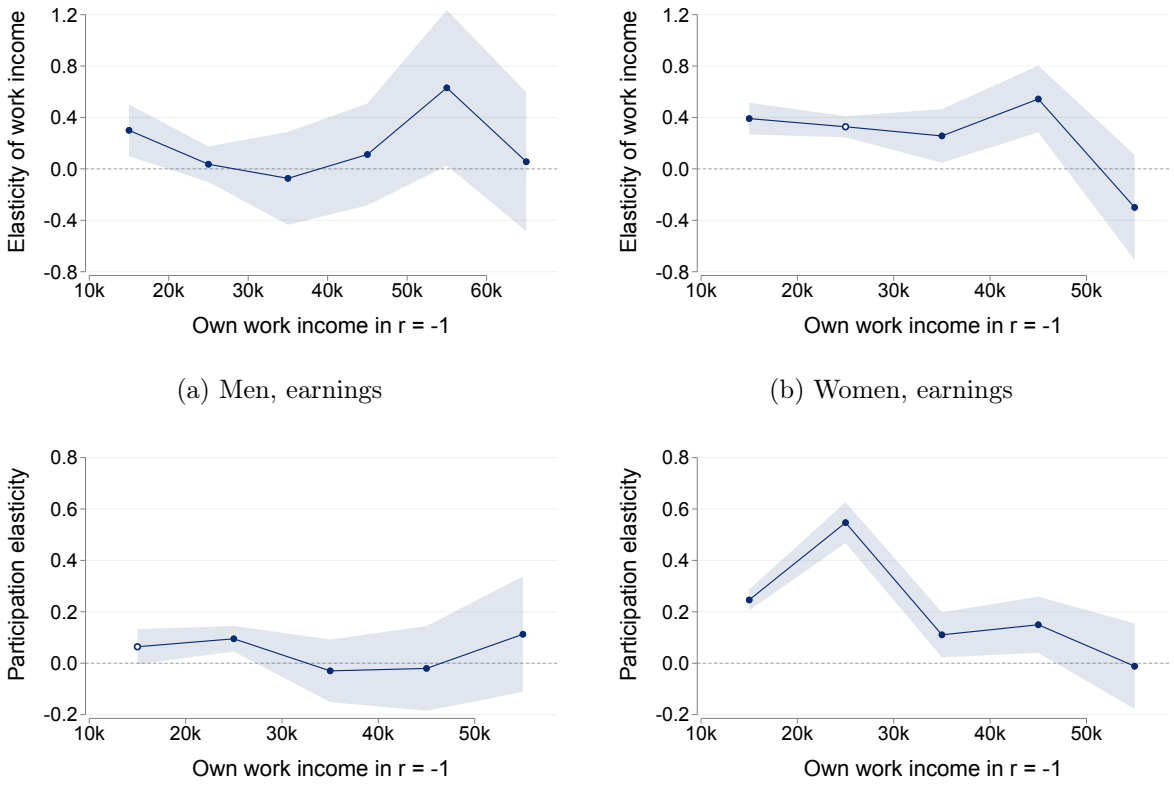
The estimates are pooled with the frequency weights of Section 3.4. In this design the weights matter more than at the child's exit. The marrying-up spouses are a small share of a cell (a median of 5.4 percent for women and 3.6 percent for men), the zero-dose spouses are shared by the doses of a cell, and a dose group with few marrying-up spouses receives the weight of the whole comparison pool. Where the two weightings differ by more than 0.02, we report both and prefer the treated-weighted value.

Within an own-income bin the marrying-up spouses face rate jumps of different sizes, and the mix changes from bin to bin, so the marriage response is one number per bracket jump rather than a profile. The jump from the 14 to the 30 percent bracket below €30,000 is made by 9,331 and 46,556 marrying-up women and 2,820 and 18,529 men in the €10,000–20,000 and €20,000–30,000 bins. For this jump the earnings elasticity is 0.14 and 0.18 for women and 0.32 and 0.16 for men (standard errors 0.02 to 0.04), and treated weights move these values by at most 0.02. The participation elasticity for the same jump is 0.11 and 0.08 for women and 0.09 and 0.09 for men under treated weights. Under frequency weights, cells with few marrying-up spouses and large comparison pools raise the women's €20,000–30,000 value to 0.75. No jump-level estimate of the 14 to 30 percent transition below €30,000 fails its pre-event placebo under frequency weights. Under treated weights, the women's €20,000–30,000 estimates fail it for earnings ($t = 4.61$) and for participation ($t = 2.29$). Above €30,000 no jump gives a number that is stable across weightings. The 14 to 30 percent jump is indistinguishable from zero for earnings under both weightings and changes sign or significance with them for participation. The 30 to 41 percent jump reaches 500 marrying-up spouses only for women (502 to 576), and its estimates move by 0.1 to 0.3 with the weights.

Figure G.1 shows the design in the paper's profile format along own income. Its notes state two differences from the display conventions of Appendix F.2. The profiles start at €10,000 of own income, because below it fewer than half of the spouses are in work two years before

the wedding and a participation response would measure entry into work. Above €10,000 nearly all spouses are in work, and the design measures exit from work. The profiles end at €60,000 for women and at €70,000 on earnings and €60,000 on participation for men. The floor of 1,000 individuals binds nowhere on the profiles, because every marrying-up cell has a large shared comparison pool. The profiles follow the paper's format and should not be read as a gradient. Their bins mix bracket jumps, and above €30,000 they rest on 114 to 4,961 marrying-up spouses. Below €30,000 they show how much the weighting matters. The women's earnings bins are 0.39 and 0.33 under frequency weights and 0.18 and 0.19 under treated weights, and the men's 0.30 and 0.04 against 0.31 and 0.17. Two of the 21 bins fail their pre-event placebo when it is pooled with the weights of the estimate (the women's earnings €20,000–30,000 bin and the men's participation €10,000–20,000 bin).

Inference in the marriage design is classical. Cells are screened on the first-stage rule of Appendix F.2 in its classical form, and no cell fails it. Standard errors and placebo tests treat the cells as independent. Because a cell pools four post-event years of the same people, these standard errors are too small. From €10,000 up, the person-clustered standard errors of the child's fiscal exit design exceed the classical ones by a median factor of 1.45 over the income-profile bins and by between 1.18 and 1.91 for four in five map tiles.



(a) Men, earnings

(b) Women, earnings

(c) Men, participation

(d) Women, participation

Figure G.1: Marriage design: elasticities along the own-income distribution, predicted rate increases

Notes: Sample: the marriage sample of Table F.1, restricted to spouses whom the merger moves to a higher marginal rate (marrying up) and compared within cells with the zero-dose spouses. Outcome: work-income growth relative to $r = -1$ in Panels (a) and (b), an indicator for positive work income in Panels (c) and (d), residualised on pre-event covariates fitted in the zero-dose group as described in the text, pooled over $r = 0$ to 3 with frequency weights; the participation estimates are divided by pre-event participation. Each marker pools the cell–dose elasticities of an own-income bin; the band is the 95 percent confidence interval. Display conventions as in Figure F.4 and Appendix F.2, with two differences. Own income runs from €10,000 to €60,000, or to €70,000 for men's earnings. The horizontal axis of each panel ends at its last bin. Every bin passes the cell screens, the standard-error cap of 0.5 and the floor of 1,000 individuals, and pools at least two cell–dose–period estimates; the vertical axis of an outcome is set by the shown bins and shared by the two genders. A hollow marker fails its pre-event placebo: the women's earnings bin at €20,000–30,000 and the men's participation bin at €10,000–20,000. Bins above €30,000 mix bracket jumps; the text reports one number per bracket jump. Cells are screened on the classical first-stage t-ratio, and no cell fails the screen. Standard errors and placebo tests are classical and understate the uncertainty because they treat the cells as independent.

H First-stage monotonicity

This appendix (i) states the monotonicity condition in the standard reform-based simulated-IV setting (Gruber & Saez, 2002; Jakobsen & Søgaard, 2022), (ii) characterises when the condition fails and illustrates a violation in a calibrated example, and (iii) argues that violations are less

likely under schedule *shifts* such as the one in our design than under reforms that change the schedule's gradient.

H.1 The condition in the reform-based setting

Consider a reform that raises statutory rates on part of the schedule. Let $g \in \{0, 1\}$ indicate whether the reform is in force, and let $\text{MTR}_i(g)$ denote the marginal tax rate individual i faces under regime g, inclusive of behavioural responses. The simulated instrument is the mechanical prediction of $\text{MTR}_i(1) - \text{MTR}_i(0)$, holding income fixed at its pre-reform value. The IV estimand averages individual responses with weights proportional to the realised first-stage responses $\pi_i \equiv \text{MTR}_i(1) - \text{MTR}_i(0)$ (Angrist & Imbens, 1995; Borusyak et al., 2022). The π_i must share a common sign for the weights to be non-negative and for the estimand to be a convex average that cannot reverse the sign of a uniformly signed effect (Blandhol et al., 2026). First-stage monotonicity requires

$$\text{MTR}_i(1) \;\geq\; \text{MTR}_i(0) \quad \forall\, i, \tag{H.1}$$

with the reverse inequality for a reform that cuts rates. A violation puts negative weight on the responses of individuals whose realised marginal tax rate moves *against* the direction of the reform.

H.2 When the condition fails

Under the strong assumptions standard in the literature of (i) a quasi-linear utility model without income effects, (ii) continuous labour supply, and (iii) no heterogeneity in elasticities, condition (H.1) is not violated. The intuition is that the utility ranking between the kink point and interior solutions in brackets unaffected by the reform remains unchanged, whether the reform increases or decreases the marginal tax rate in a given bracket. As a result, any new interior optimum in a bracket unaffected by the reform is dominated by the kink point at the reform-affected bracket.

When either of the first two assumptions is relaxed (allowing income effects or labour-supply rigidities) and elasticities or preferences are heterogeneous, first-stage monotonicity can be violated.

We illustrate a violation of first-stage monotonicity using a model with income effects, non-linear labour-supply responses, and heterogeneity in the curvature of the utility function.[66] Individuals derive utility from consumption c and hours worked h according to

$$u_i(c,h) = \frac{c^{1-\sigma_i}}{1-\sigma_i} - \chi_i \frac{h^{1+\varphi_i}}{1+\varphi_i},$$

where $\sigma_i > 0$ governs the curvature of utility from consumption and the strength of income effects, $\varphi_i > 0$ governs the curvature of the disutility from labour, and $\chi_i > 0$ scales the marginal disutility of hours worked. Labour income is given by $y = wh$. We assume that hours rigidities restrict individuals to four discrete choices: 0, 20, 30, or 40 hours per week.

For the tax schedule, we consider the French income tax system as of 2015. In the part of the schedule on which we focus, labour income up to €26,791 per year is taxed at a marginal rate of 14%, and income above this threshold at 30%.[67] Because utility is defined over hours worked, we can equivalently express the tax schedule as a function of hours, with a kink at $h = k$ corresponding to the income threshold. In the example, we set the hourly wage to €20.6, so that the threshold at €26,791 corresponds to approximately 25 hours of work per week.

We study a hypothetical reform ($g = 1$) that eliminates the 30% bracket by raising the marginal tax rate above the threshold from 30% to 41% (the next statutory rate in the French schedule).

We consider two preference types, Type A and Type B, and calibrate the model such that both share the same discrete optimum of 30 hours per week under no reform. The consumption-curvature parameter σ_i governs the relative strength of income and substitution effects. Type A is calibrated with $\sigma_A < 1$, so that the substitution effect dominates, and Type B with $\sigma_B \gg 1$, so that the income effect dominates. For each type, the scale χ_i is set to deliver this common

[66] Hours rigidities would be sufficient to illustrate a violation. To make the argument applicable more broadly, however, we consider a more general model with income effects.

[67] To simplify the exposition, the example uses the statutory schedule, without the tax reduction for small liabilities of Section 2.1; the reduction does not affect the example, and results including it are available on request.

optimum. The resulting preference parameters are

$$(\sigma_A,\, \varphi_A,\, \chi_A) = (0.696,\ 0.300,\ 0.272), \qquad (\sigma_B,\, \varphi_B,\, \chi_B) = (7.000,\ 0.300,\ 9.17 \times 10^{-30}).$$

Under no reform, both types choose 30 hours per week and are in the 30% bracket:

$$\mathrm{MTR}_A(0) = \mathrm{MTR}_B(0) = 0.30.$$

The simulated instrument holds both types' incomes fixed at their 30-hour values and predicts that both face a rise in the marginal tax rate from 30% to 41%. The predicted first-stage response is +0.11.

The optimal hours choices under the reform and under no reform are shown in Figure H.1.[68] Under the reform, the two types' optimal choices diverge:

$$h_A^* = 20\,\text{hours/week}, \qquad h_B^* = 40\,\text{hours/week}.$$

The realised treatments therefore respond to the same shock in opposite directions:

$$\pi_A = \mathrm{MTR}_A(1) - \mathrm{MTR}_A(0) = 0.14 - 0.30 = -0.16 < 0,$$
$$\pi_B = \mathrm{MTR}_B(1) - \mathrm{MTR}_B(0) = 0.41 - 0.30 = +0.11 > 0.$$

Type A's realised marginal tax rate moves against the direction of the reform. It *falls* although the reform raises statutory rates and the instrument predicts a tax increase for Type A. This violates condition (H.1). Type B's realised treatment moves with the reform.

The estimand places negative weight on Type A's response. It is not weakly causal in the sense of Blandhol et al. (2026), because with heterogeneous elasticities it can even reverse the sign of a uniformly signed response.

[68] The optimal hours choice under each regime is determined by maximising $u_i(c, h)$ subject to the piecewise-linear budget constraint. There is no closed-form solution in general, so the optimum is obtained numerically by comparing utility across all candidate solutions: interior first-order-condition roots on each linear segment, the kink, and the corner at zero hours.

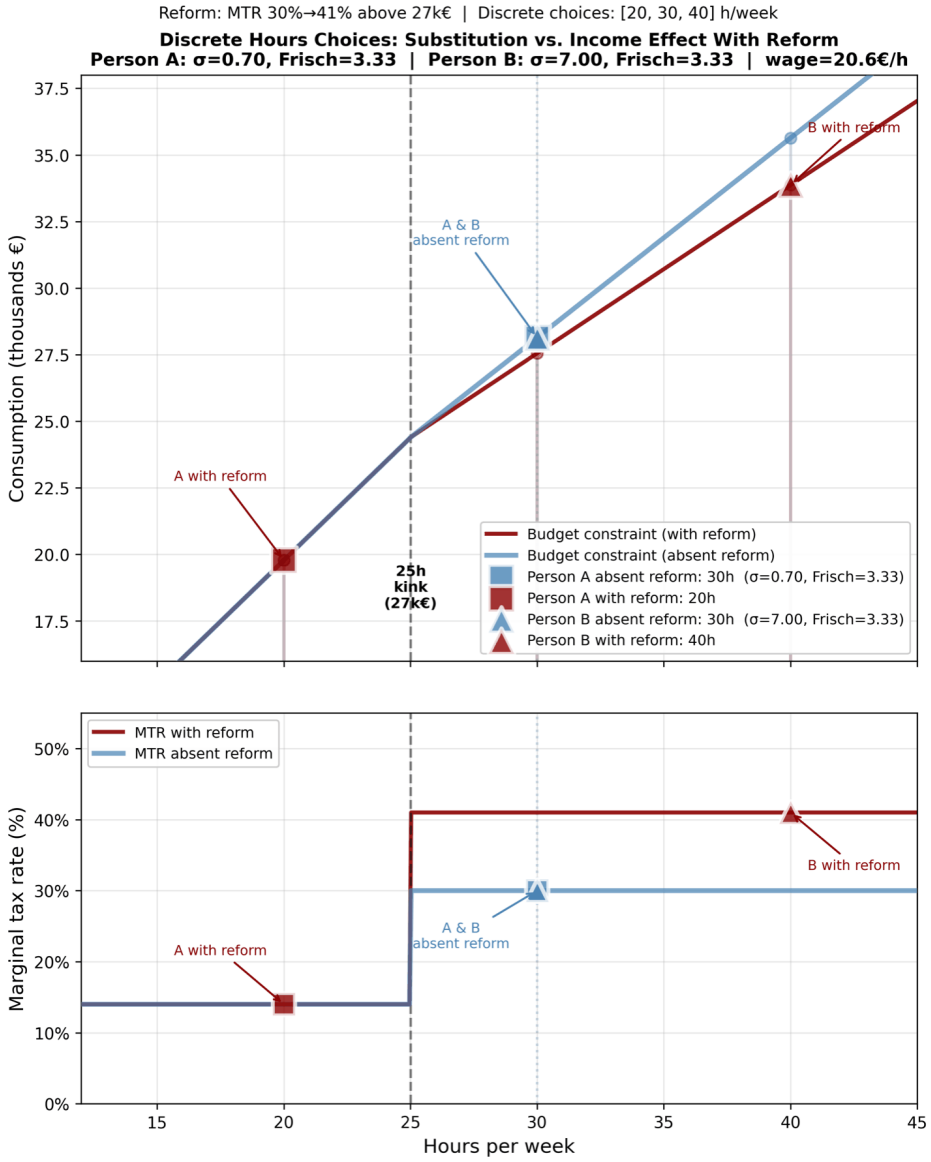


Figure H.1: Violation of first-stage monotonicity under heterogeneous labour-supply responses

Notes: Top panel. Budget constraints under the reform (red) and under no reform (blue). The reform raises the marginal tax rate above the second bracket threshold from 30% to 41%. Both Type A and Type B optimally choose 30 h/week under no reform. Under the reform, Type A chooses 20 h/week and is below the kink; Type B ($\sigma_B = 7$) chooses 40 h/week and is above the kink.
Bottom panel. Marginal tax rates under the reform (red) and under no reform (blue), as a function of hours worked, with the same optimal choices indicated.

A strong income effect for Type B is not necessary for the violation. If Type B simply stays at 30 hours under the reform ($\pi_B = +0.11$) while Type A moves to 20 ($\pi_A = -0.16$), the same violation arises.

Models with participation responses can also generate violations of first-stage monotonicity. For example, an individual who leaves work in response to a reform that increases tax rates moves out of the tax schedule altogether (in France, to a marginal tax rate of zero), so that $\pi_i < 0$. If other individuals j remain employed at the same time and face the higher rate under the reform ($\pi_j > 0$), the realised treatment responds to the shock in opposite directions across individuals. This violates condition (H.1). Because participation responses to tax increases are empirically common (as we also document in this paper), this is an important source of monotonicity violations in reform-based IV strategies.

The reform-based setting of this example is the canonical design from which our identification strategy departs. We now turn to our design.

H.3 The condition in the event-based setting

In the main design the schedule change is the fiscal exit of the first-born, which lowers the quotient from 2.5 to 2 with one dependent child and from 3 to 2.5 with two (Section 3.2). Because household income is divided by the quotient, the marginal rate at every level of household income y satisfies $T_t'(y/Q') \geq T_t'(y/Q)$ for $Q' < Q$: the exit weakly raises the marginal rate at every point of the budget set, so only the rate-increase direction of condition (H.1) applies, and a violation cannot arise from the shift itself. It requires a household to reduce its income by enough that its realised marginal rate falls strictly below the rate it would have faced under the old quotient. The required fall in income is much larger than under a gradient reform. An individual predicted into the next bracket has income per part under the new quotient at or above the upper threshold of the pre-event bracket, and a violation requires it to fall below the lower threshold. Household income must fall to less than the ratio of the two thresholds times its pre-event level. Partner income makes even this channel unlikely. Where partner income alone places the household above the tax-free threshold under the post-event quotient, a complete exit of the individual leaves the household in a positive bracket (a condition that can be checked cell by cell, since the cells fix the partner-income range).[69] The same floor arises at marriage, where the partner's income is added to the tax base.

[69] Two residual channels remain: (i) a simultaneous collapse of both spouses' incomes below the household tax-free threshold, and (ii) the partner's own exit from work, which lowers household income through the same mechanism and is common at the ages at which the event occurs.

Figure H.2 illustrates why such a monotonicity violation is unlikely with our instrument. The figure shows the marginal tax rate an individual faces as a function of own income, both when single (solid) and when married to a partner earning €40,000 (dashed). The marginal tax rate for a married individual remains strictly positive even when own income is zero, because the partner's income already places the household above the tax-free threshold. Even a complete exit leaves the individual facing a 14% rate. For an individual predicted to move from the 14% to the 30% bracket, a violation would require the realised marginal rate to fall strictly below 14%, which the partner's income prevents unless it collapses as well. The logic extends to all individuals facing predicted tax increases. Violations are also unlikely for predicted decreases, because they would require a large increase in labour supply.

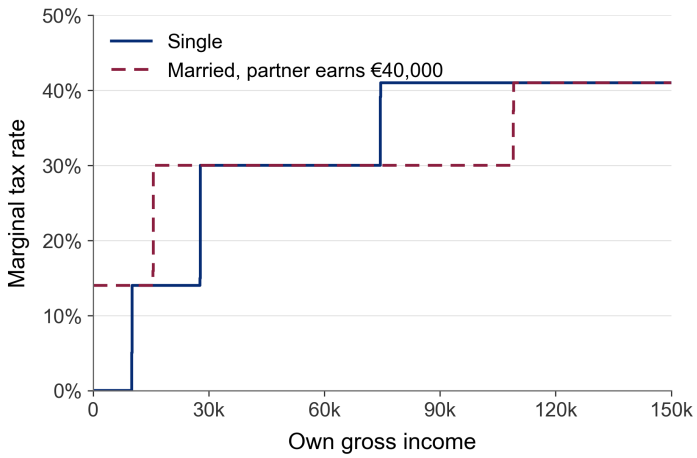


Figure H.2: Marginal tax rate under marriage, partner earns €40,000

Notes: The figure shows the marginal tax rate as a function of own gross income under individual taxation (blue solid) and under joint taxation when married to a partner earning €40,000 (red dashed), based on the statutory 2019 French income tax schedule (Section 2.1). Income is not pooled, so own consumption corresponds to own after-tax income only. The marginal tax rate for a married individual is 14% even at zero own income, because the partner's income of €40,000 already places the household above the tax-free threshold. A monotonicity violation is unlikely for individuals predicted to move from the 14% to the 30% bracket, since even a complete exit from the labour market cannot lower their marginal tax rate below 14%.

I Published estimates of work-income responses

Appendix Table I.1 records the estimates underlying Figure 1 and where they sit in the income distribution, together with each estimate's design, estimand, horizon and population; ranking universes and source versions are documented in the notes. It includes only studies represented in the figure and reports alternative estimates from those studies for context.

Table I.1: Evidence base for the literature map of work income responses

Study	Outcome	Design	Estimand	Horizon	Population	Estimate (SE)	Identifying variation
Saez (2010)	Work income	Bunching	Compensated	1995-2004 pooled	EITC filers with one child, wage earners only	**0.025 (0.036)**[a]	P13.4–P15.3 (C)
					EITC filers with two or more children, wage earners only	**0.003 (0.022)**[a]	P19.1–P21.1 (C)
	Earnings				All EITC filers with one child	0.213 (0.033)	P13.4–P15.3 (C)
					Self-employed EITC filers with one child	1.101 (0.092)	
					All EITC filers with two or more children	0.152 (0.02)	P19.1–P21.1 (C)
					Self-employed EITC filers with two or more children	0.755 (0.054)	
Chetty, Friedman and Saez (2013)	Work income	Reform DiD (EITC around first-child birth)	Frictionless-model elasticity matching observed EITC refund changes	Event study around first-child birth, 2000-2005	EITC-eligible parents around the birth of the first child, 2000-2005	**0.31 (0.018)**[b]	P0–P19.4 (C)
						0.14 (0.015)[b]	P24.8–P54.5 (C)
						0.19 (0.019)	P0–P54.5 (C)
						0.21 (0.012)	
Chetty et al. (2011)	Work income	Reform IV (small reforms)	Observed elasticity; the paper argues frictions attenuate it	1994-2001 pooled	All wage earners	**-0.005 (0.003)**	not locatable
	Work income (taxable base)	Bunching	Compensated (observed, frictions not netted out)			∼**0.01**[j]	P75 (A)
	Taxable income	Structural calibration	Structural lower bound (calibrated, frictions $\leq$ 5% of consumption)		Wage earners	$\geq$0.34	not locatable
	Work income (taxable base)	Bunching	Compensated (observed)		Married women, wage earners	∼0.02	P75 (A)
Bastani and Selin (2014)	Work income (taxable base)	Bunching	Compensated	1999-2005 pooled	Wage earners aged 15-70; anyone with active business income or closely-held-corporation ties removed	∼**0.00**[k]	P80 (A)
			Structural upper bound (tolerated loss = 1% of disposable income)		Wage earners aged 15-70	$\leq$0.70	
	Taxable income		Compensated	2000-2008 pooled	All self-employed	0.024	
	Work income (taxable base)			1998	Wage earners aged 15-70	∼0.00	
Lehmann et al. (2013)	Work income incl. contributions	Reform IV	Marginal-rate elasticity, average net-of-tax rates controlled	1-year, 2003-2006 pooled	Private-sector employees earning less than twice the minimum wage; public sector and self-employed excluded	**0.225 (0.085)**[g]	P10–P60 (C)
						0.016 (0.096)	
Saez (2003)	Work income	Reform IV (bracket creep, 1979-1981 inflation)	Compensated	1-year, 1979-1981	Tax returns in the NBER panel, 1979-1981; singles with AGI $3,000-40,000 and married $5,000-70,000 (1979 USD)	**0.084 (0.173)**[f]	P44–P94 (C)
						0.120 (0.196)	P55–P94 (C)

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Table I.1 continued

Study	Outcome	Design	Estimand	Horizon	Population	Estimate (SE)	Identifying variation
Kleven and Schultz (2014)	Earnings	Reform IV	No income effects	3-year, 1984-2005	All workers, top 10 percent earners	**0.085 (0.012)**[c]	P90–P100 (A)
					All workers, top 20 percent earners	**0.076 (0.008)**[c]	P80–P100 (A)
	Work income				Wage earners, top 10 percent earners	0.081 (0.012)	P90–P100 (A)
					Wage earners, top 20 percent earners	0.073 (0.009)	P80–P100 (A)
Jakobsen and Sogaard (2022)	Work income incl. contributions plus transfers	Local reform IV (2SLS local linear)	No income effects	4-year, 2008-2012 (pre-period 2004-2008)	Aged 15+, pre-reform personal income above DKK 250k; self-employed excluded	**∼0.038**	P74.9–P76.9 (C)
						∼0.051	P82.5–P83.8 (C)
						∼0.047	P87.7–P88.5 (C)
						∼0.031	P91.1–P91.5 (C)
						∼-0.004	P93.3–P93.6 (C)
						∼0	P94.9–P95 (C)
						∼0.028	P95.6–P95.7 (C)
						∼-0.001	P96.2–P96.3 (C)
		Reform IV		4-year, 2008-2012		0.016 (0.007)	P74.9–P100 (C)
	Taxable income					0.202 (0.007)	
Miao, Selin and Soderstrom (2025)	Earnings	Reform difference-in-differences (grouped)	No income effects (grouped Wald / DiD)	4-year, 2015 vs 2019	Percentile groups 96-100 (treatment) vs 88-95 (control), aged 20-65	**0.156 (0.023)**	P96–P100 (A)
					Percentile groups 96-100 vs 88-95, groups 94-97 excluded	0.171 (0.019)	
Kleven et al. (2025)	Work income incl. contributions	Reform IV, job switchers	Long-run population elasticity under orthogonality of switching timing	Four-year	Treated job switchers above the threshold	**0.487 (0.026)**[d]	P70–P98 (A)
					Treated job switchers in the top decile	**0.642 (0.029)**[d]	P90–P98 (A)
				Two-year	All treated job switchers	0.486 (0.022)	P70–P98 (A)
					Treated job switchers in the top decile	0.725 (0.029)	P90–P98 (A)
Gelber (2014)	Work income	Reform IV	Uncompensated	1-year, 1989-90 and 1990-91 pooled	Married men aged 18-65 with positive earnings in 1989 or 1990	**0.210 (0.02)**[e]	not locatable
					Married women aged 18-65 with positive earnings in 1989 or 1990	**0.230 (0.03)**[e]	
			Compensated		Married men aged 18-65 with positive earnings in 1989 or 1990	0.410 (0.02)	
					Married women aged 18-65 with positive earnings in 1989 or 1990	0.470 (0.02)	
Sigaard (2023)	Work income	Reform IV	No income effects	3-year	Danish wage earners with reported contract and performed hours, 2002-2016	**0.058 (0.0037)**	not locatable
	Hours worked					0.082 (0.0091)	
	Taxable income					0.095 (0.0042)	
Blomquist and Selin (2010)	Work income (taxable base)	Reform IV	Uncompensated	10-year, 1981 vs 1991	Married or cohabiting men; self-employed excluded	**0.210 (0.074)**[l]	not locatable
					Married or cohabiting women; self-employed excluded	**1.442 (0.832)**[l]	
He et al. (2021)	Work income (taxable base)	Reform IV	Compensated (income effects small and insignificant)	One-year	Wage earners aged 18-60 in one (unnamed) Chinese city	**2.423 (0.188)**[m]	not locatable
		Bunching		Pooled months		0.1 (0.02)	
						0.41 (0.07)	
						0.09 (0.03)	
		Reform IV (revised, long-run)		Long-run		∼4	

Continued on next page

Table I.1 continued

Study	Outcome	Design	Estimand	Horizon	Population	Estimate (SE)	Identifying variation
Kleven–Waseem (2013)	Work income (taxable base)	Bunching (notch)	Observed (friction-attenuated) elasticity	Annual, 2006-2007	Pakistani wage-earner filers 2006-07 (under 2% of the working-age population, over 99% male)	**∼0 (0.000-0.004 across notches)**[n]	not locatable
			Structural (frictionless) elasticity, bunching-hole lower bound			0.024 (0.009)	
						0.035 (0.010)	
						0.034 (0.019)	
						0.001 (0.013)	

Notes: One row per coded estimate; **bold** estimates are plotted in Figure 1, with superscripts matching the figure's label letters. Within a study block, repeated entries are shown once. (–) indicates that no standard error is reported.

Placement. Ranks in the identifying-variation column: (A) reported by the study; (B) computed by us from a closely matching official distribution; (C) computed from an official distribution with a related income concept, drawn dashed in the figure; (D) weak reconstruction; (Q) qualitative statement; (U) unrecoverable. Estimates marked (D), (Q) or (U) appear in the figure's right panel. The column gives the income range of the treated group only, never the estimation sample: control groups located elsewhere in the income distribution (Miao et al.: P88–95; Kleven et al.: below the reform threshold; Jakobsen–Søgaard: the validation region) are not part of the stated range, while for Kleven–Schultz the sample and the treated band coincide because identification uses within-sample variation in reform exposure.

Estimands. Not harmonised: bunching estimates are compensated elasticities; Gelber's plotted pair is uncompensated (the study's compensated estimates are 0.41 and 0.47); most register-based reform estimates are specified without income effects; and Chetty–Friedman–Saez report the elasticity a frictionless model requires to match observed EITC refund changes. Reported income effects are small throughout, so these differences are second order relative to design and horizon. Statutory-taxable-income estimates on wage-earner samples enter as work income (taxable base); population-wide taxable-income studies remain in the project archive.

Sources. Values are taken from the published articles, with exceptions: Chetty et al. (2011) from the revised NBER working paper; Sigurdsson (2025) from the November 2024 manuscript; Blomquist–Selin verified from CESifo Working Paper 2644. Jakobsen–Søgaard points and confidence bands are digitized from their Figure 3, Panel D; their local estimates use triangular weights within DKK 50,000 of each centre. Kleven–Waseem's observed elasticity is derived as the paper describes (structural lower bound times the responding share); no standard error applies.

Notes to the figure's letters. [a] One-child (left) and two-plus-children (right) EITC kinks, wage earners only. [b] EITC phase-in (left) and phase-out (right) regions. [c] Top-20 and nested top-10 percent samples. [d] All treated job switchers and, above, the top decile; long-run elasticities. The other reform estimates are one-to-four-year responses. [e] Husbands (left) and wives (right), both uncompensated. [f] Confidence interval truncated below; the segment covers the bracket-creep kinks holding roughly nine-tenths of treated taxpayers (full support about P21–P98). [g] Approximate: the treated interval of 0.3 to 2.1 minimum wages, which covers half of French employees. [h] Feldstein (1995) and Gruber and Saez (2002); surveyed in Saez et al. (2012). [i] From temporary tax holidays: 0.41 in Iceland (Sigurdsson, 2025), 0.025 in Switzerland (Martínez et al., 2021). [j] Top-kink bunching in taxable income among wage earners; no elasticity standard error is reported. The right-panel entry is the same study's schedule-wide wage-earnings reform estimate. [k] First central-government kink, taxable labour income; the paper reports 0.00 (rounded), without a standard error. [l] Taxable labour income, uncompensated; the marker is the male estimate, 0.21 (0.07); the female estimate, 1.44 (0.83), exceeds the axis and its confidence interval does not exclude zero. [m] Taxable wage income; the reform estimate, 2.42 (0.19), exceeds the axis; the same study's bunching estimates on the same schedule are 0.09–0.41. [n] Observed (friction-attenuated) elasticity of taxable income at the wage-earner notches, 0.000–0.004 across the four notches; no standard error applies. The friction-corrected structural bounds are 0.001–0.074. [o] The dashed line connects Jakobsen–Søgaard's local estimates at successive income thresholds (DKK 400,000–750,000) and does not denote interpolation or an estimated continuous profile.